Esercitazione N° 1: **La miostatina**

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Creiamo il nostro profilo su Ubuntu

questo è un commento!

lanciamo il terminale per aggiungere un nuovo profilo

ctrl+alt+t

```
$ sudo adduser nome_battesimo -uid 663
$ pw di labx

# per eliminare un profilo utente (in caso di errore)
$ sudo deluser --remove-home nome_utente
$ pw di labx

# per recuperare la passwd dimenticata
# sostituire <user name> con il proprio nome utente
$ sudo passwd user name
$ pw di labx
$ nuova pw
```

Creiamo il nostro profilo su Ubuntu

acquisiamo i privilegi dell'amministratore di sistema modificando il file di configurazione del Sistema Operativo (SO) `/etc/sudoers`

```
$ sudo visudo
```

questo comando lancia l'editor di testo **nano** per creare ed aprire il file `/etc/sudoers.tmp` in modalità scrittura (per poterlo modificare); tale file è una copia di backup di `/etc/sudoers`

Creiamo il nostro profilo su Ubuntu

```
GNU nano 7.2                                  /etc/sudoers.tmp
# This file MUST be edited with the 'visudo' command as root.
#
# Please consider adding local content in /etc/sudoers.d/ instead of
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file.
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults        use_pty

# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
#Defaults:%sudo env_keep += "http_proxy https_proxy ftp_proxy all_proxy no_proxy"

# This allows running arbitrary commands, but so does ALL, and it means
# different sudoers have their choice of editor respected.
#Defaults:%sudo env_keep += "EDITOR"

# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeeper
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

                                [ Lette 59 righe ]
^G Guida       ^O Salva       ^W Cerca       ^K Taglia      ^T Esegui      ^C Posizione   M-U Annulla    M-A Contrassegna M-] Parentesi  M-Q Precedente ^B Indietro
^X Esci        ^R Inserisci   ^\ Sostituisci ^U Incolla     ^J Giustifica  ^/ Vai a riga  M-E Ripeti     M-6 Copia        ^Q Cerca ind. M-W Successiva ^F Avanti
```

Creiamo il nostro profilo su Ubuntu

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# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
piero   ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

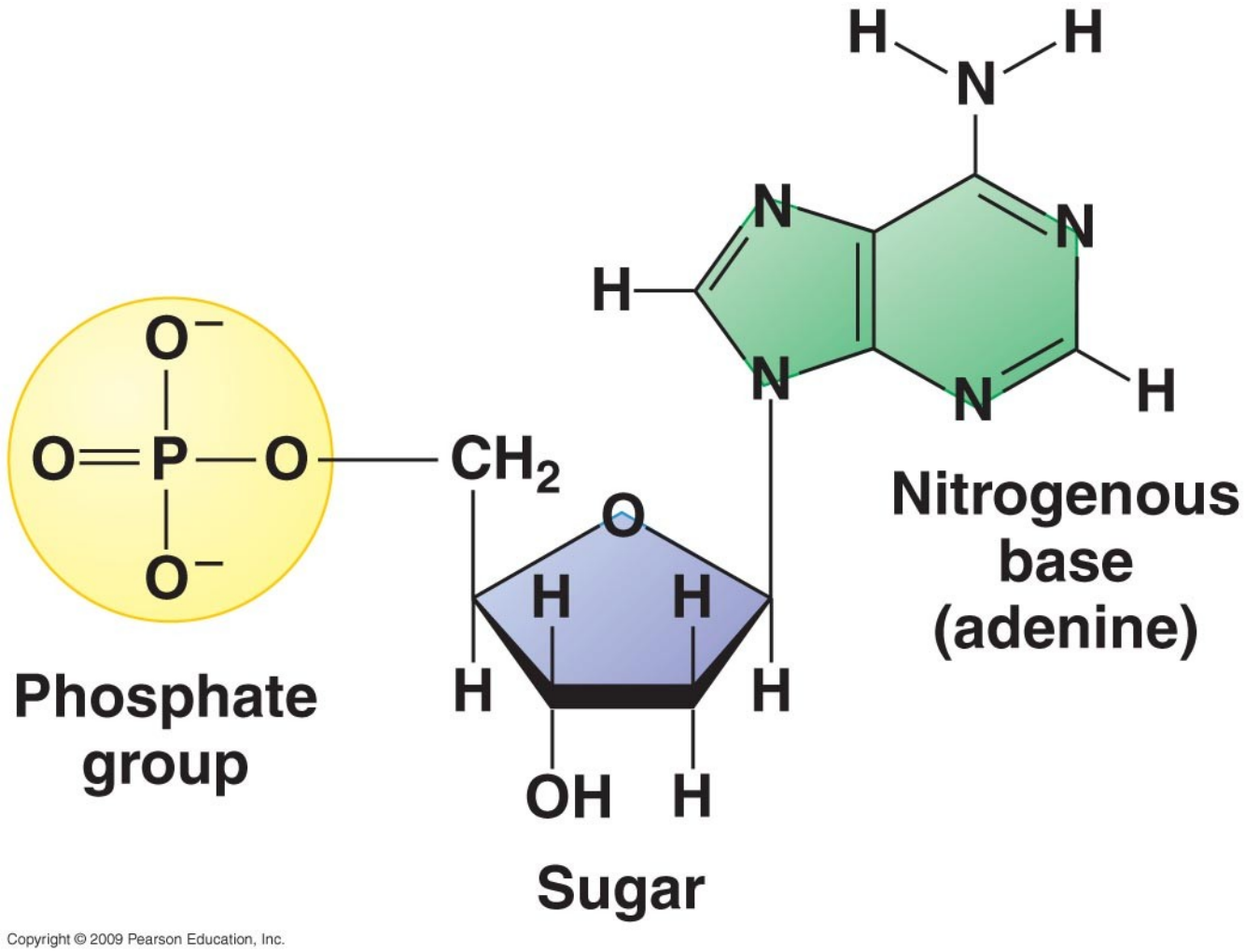
@includedir /etc/sudoers.d

^G Guida      ^O Salva       ^W Cerca       ^K Taglia      ^T Esegui      ^C Posizione   M-U Annulla    M-A Contrassegna  M-] Parentesi   M-Q Precedente  ^B Indietro
^X Esci       ^R Inserisci   ^\ Sostituisci ^U Incolla     ^J Giustifica  ^/ Vai a riga  M-E Ripeti     M-6 Copia         ^Q Cerca ind.   M-W Successiva  ^F Avanti
```

Elenco degli amministratori di sistema
portarsi con il cursore sull'ultima riga della lista
alt-6 # copia
ctrl-u # incolla l'intera riga
inserire il proprio nome utente
ctrl-o # per salvare
cancellare l'estensione .tmp
ctrl-x # per chiudere **nano**
alt-u # in caso di errore
(per annullare l'ultimo comando inserito)

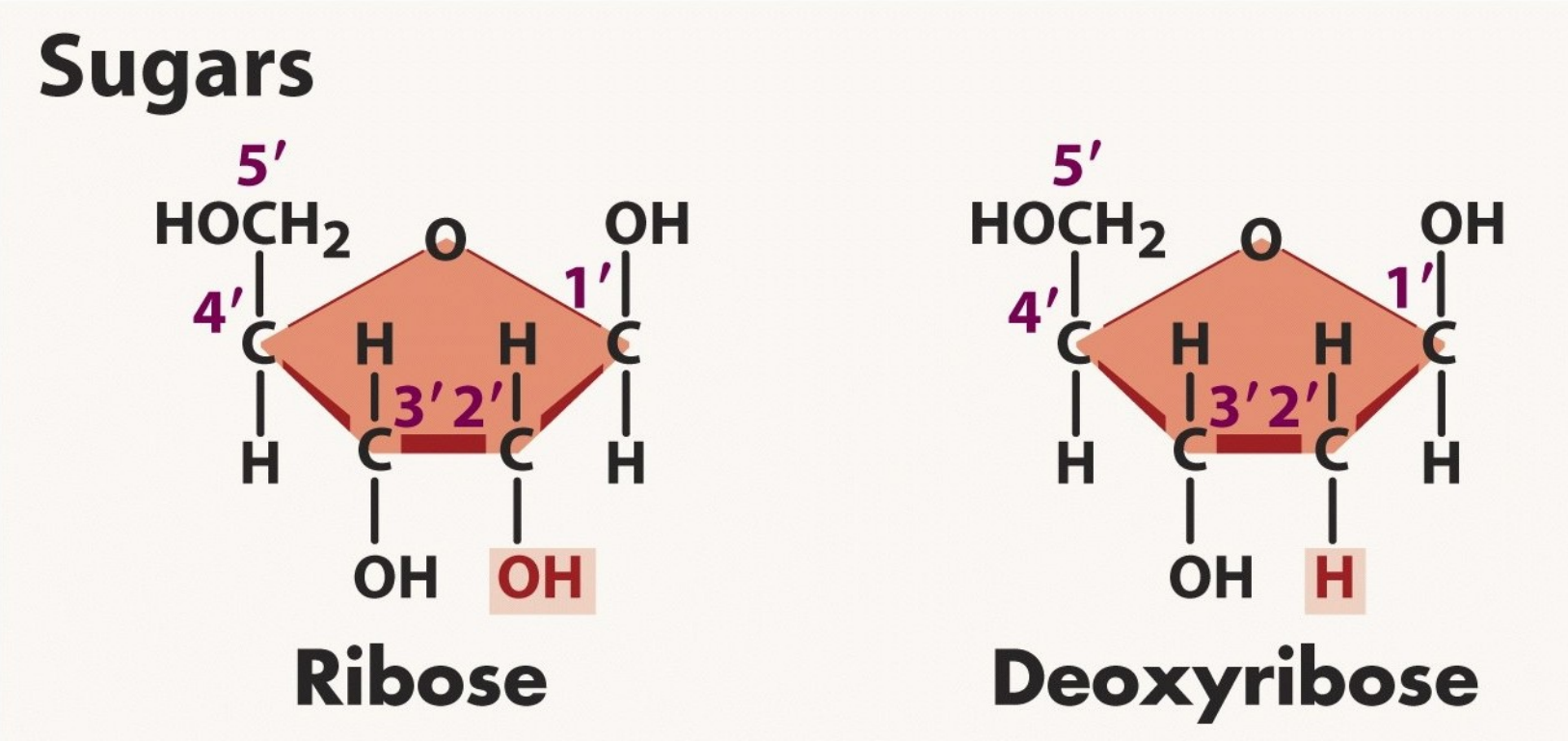
Elenco degli amministratori di sistema

Struttura di un nucleotide

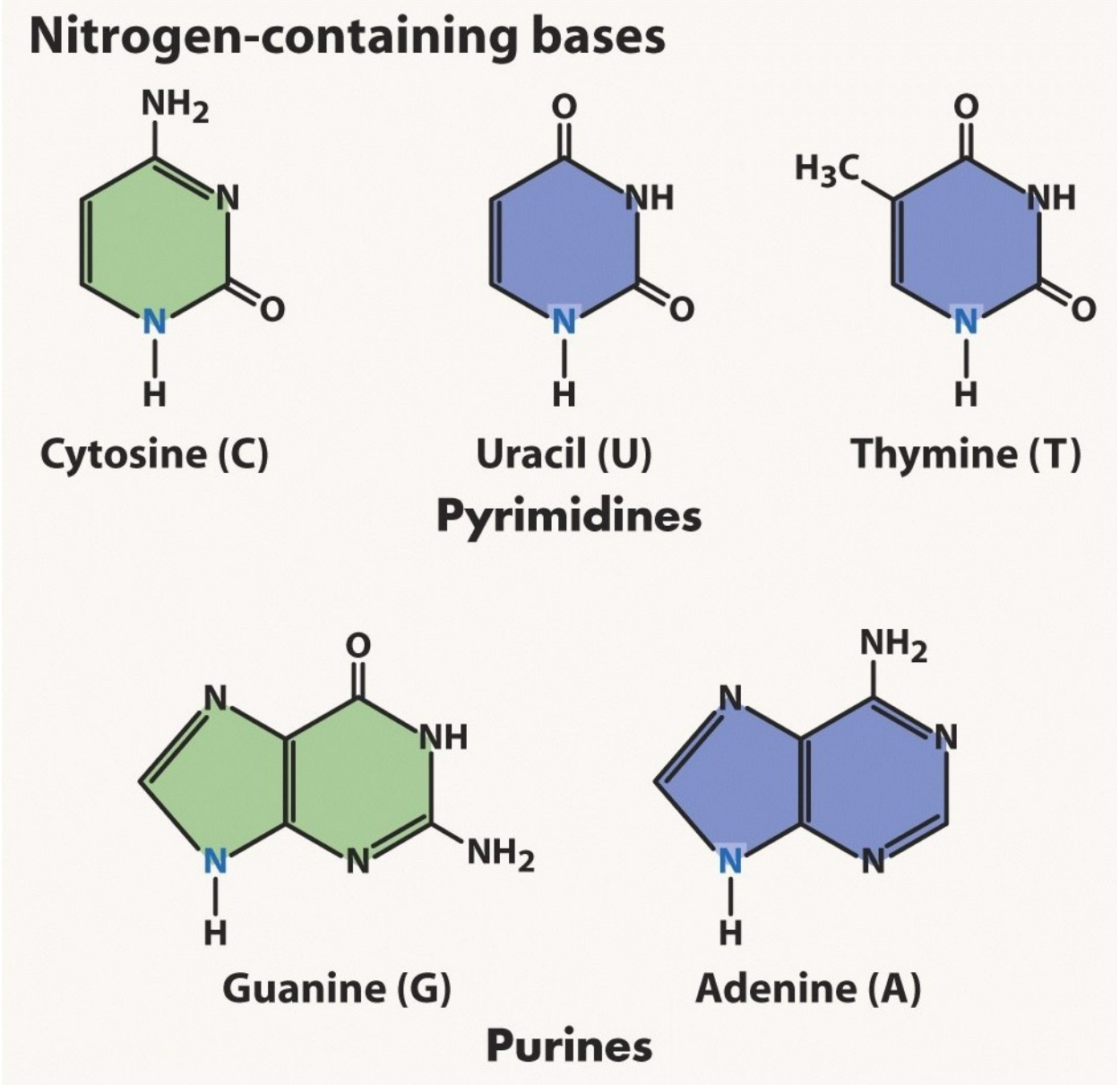


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Zuccheri

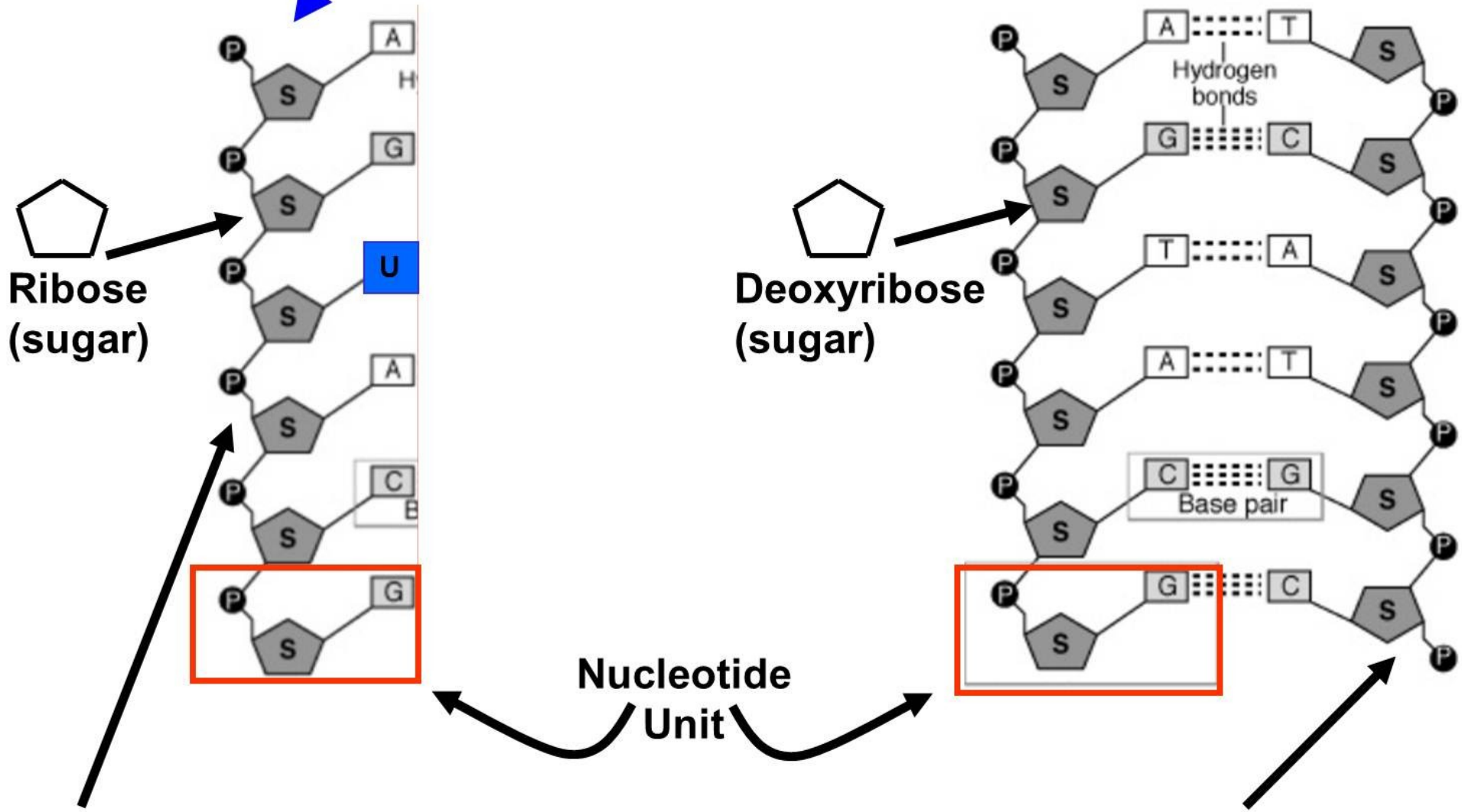

Sugars
5′
HOCH2
O
OH
4′
1′
H
H
3′ 2′
H
OH
OH
Ribose
5′
HOCH2
O
OH
4′
1′
H
H
3′ 2′
H
OH
H
Deoxyribose

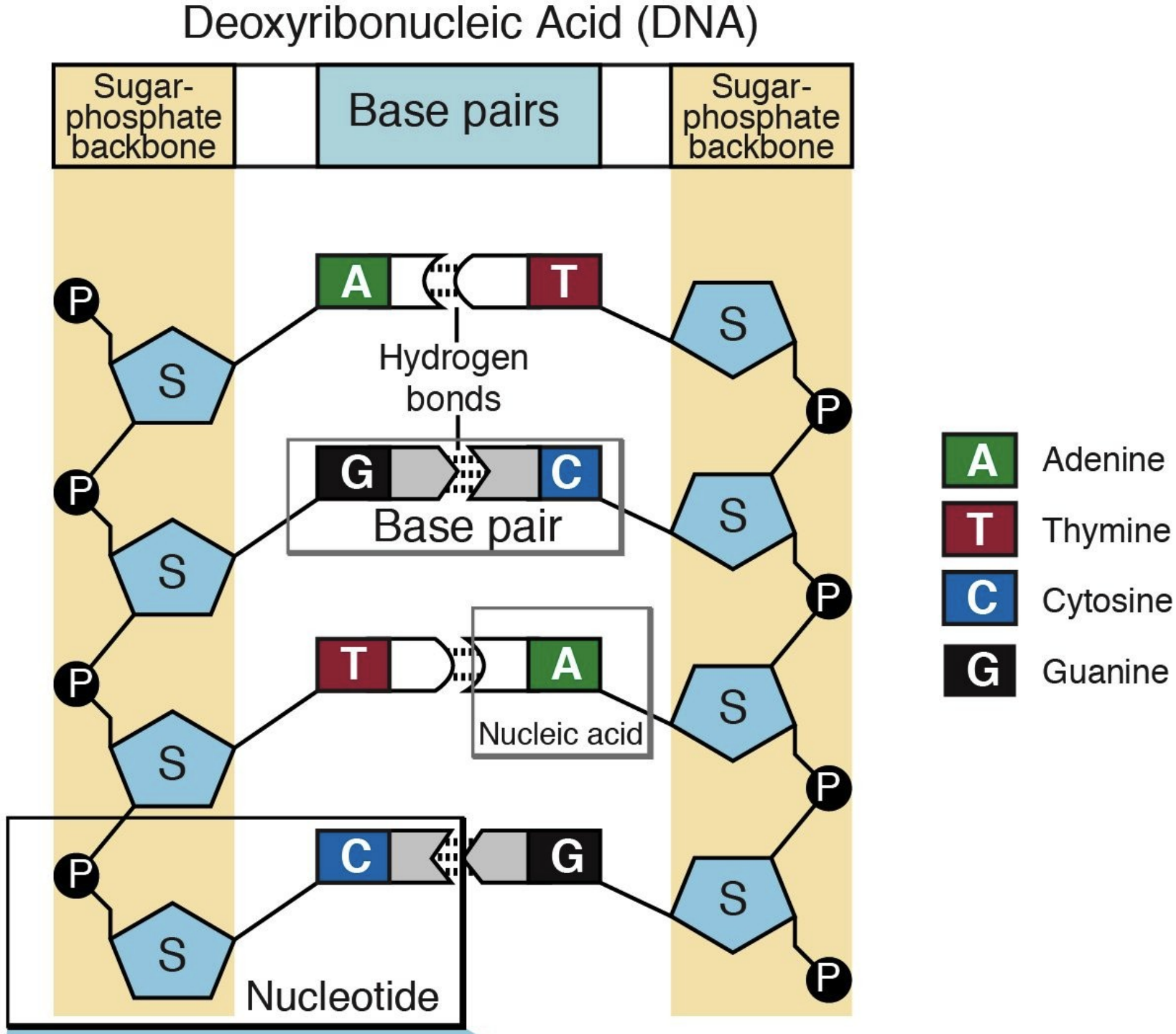
Basi pirimidiniche e puriniche

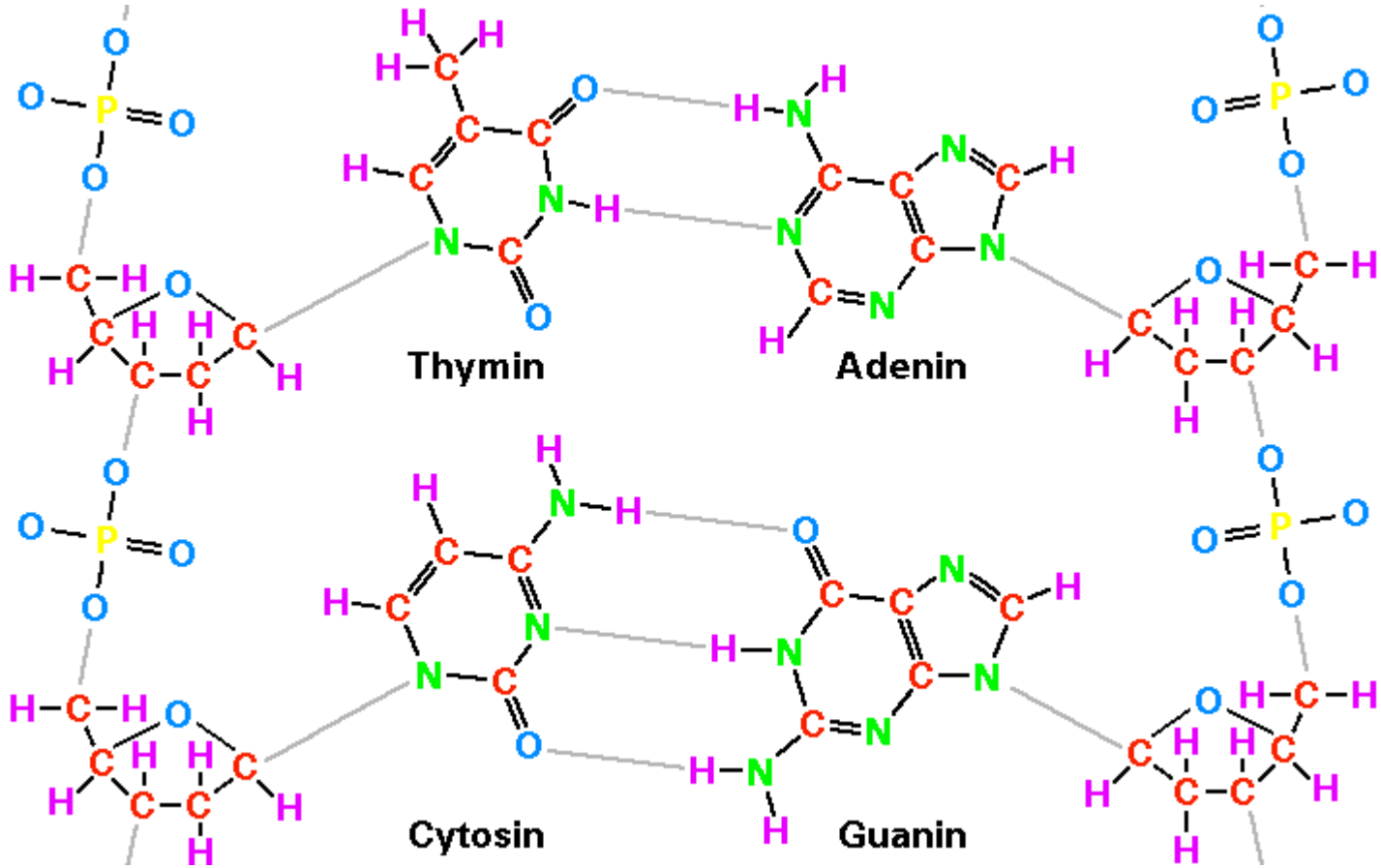


RNA & DNA

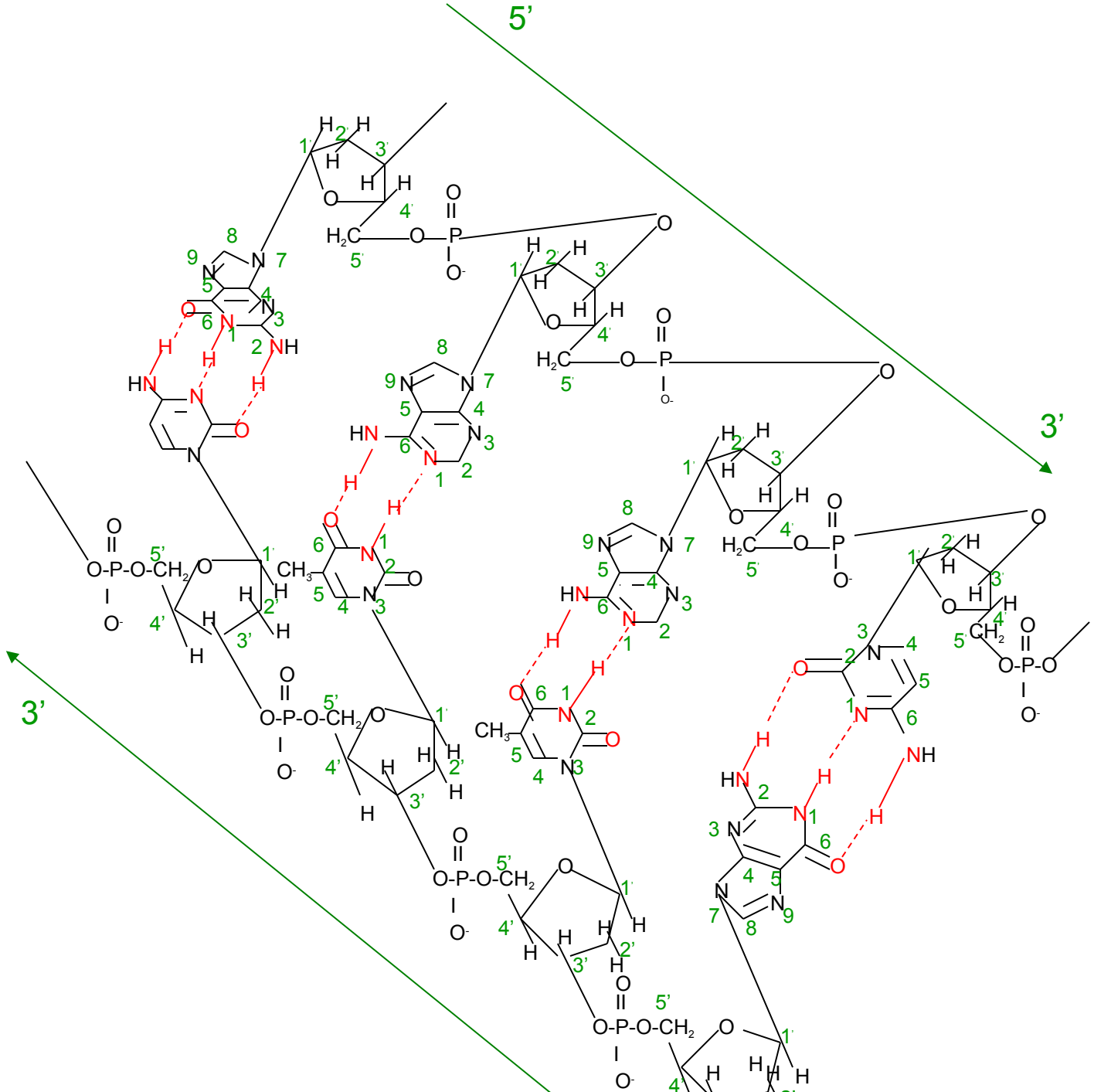
are chains of nucleotides




Deoxyribonucleic Acid (DNA)
Sugar-phosphate backbone
Base pairs
Sugar-phosphate backbone
A
T
Hydrogen bonds
G
C
Base pair
T
A
Nucleic acid
C
G
Nucleotide
P
S
Adenine
Thymine
Cytosine
Guanine

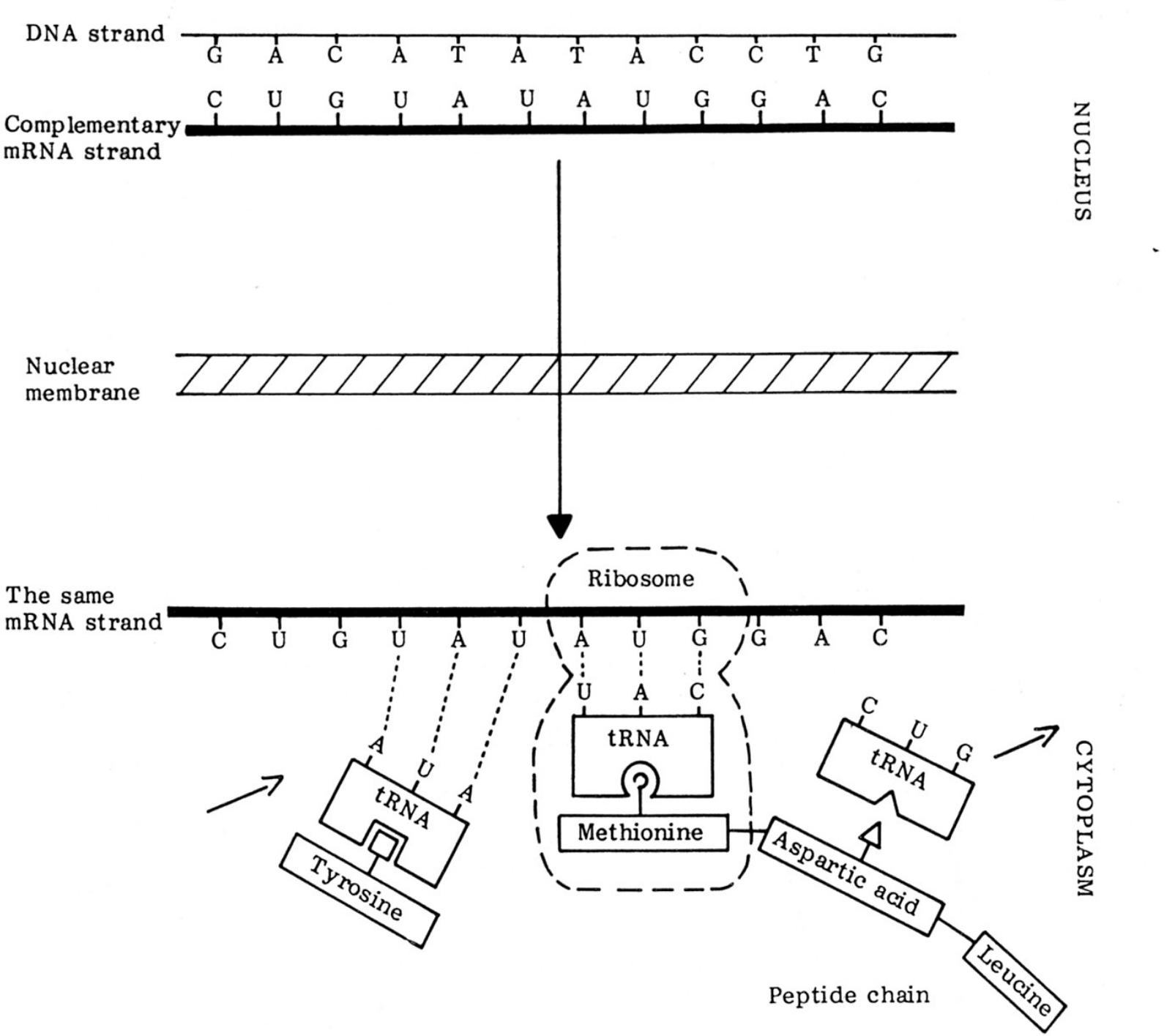

Thymin
Adenin
Cytosin
Guanin

Antiparallelismo

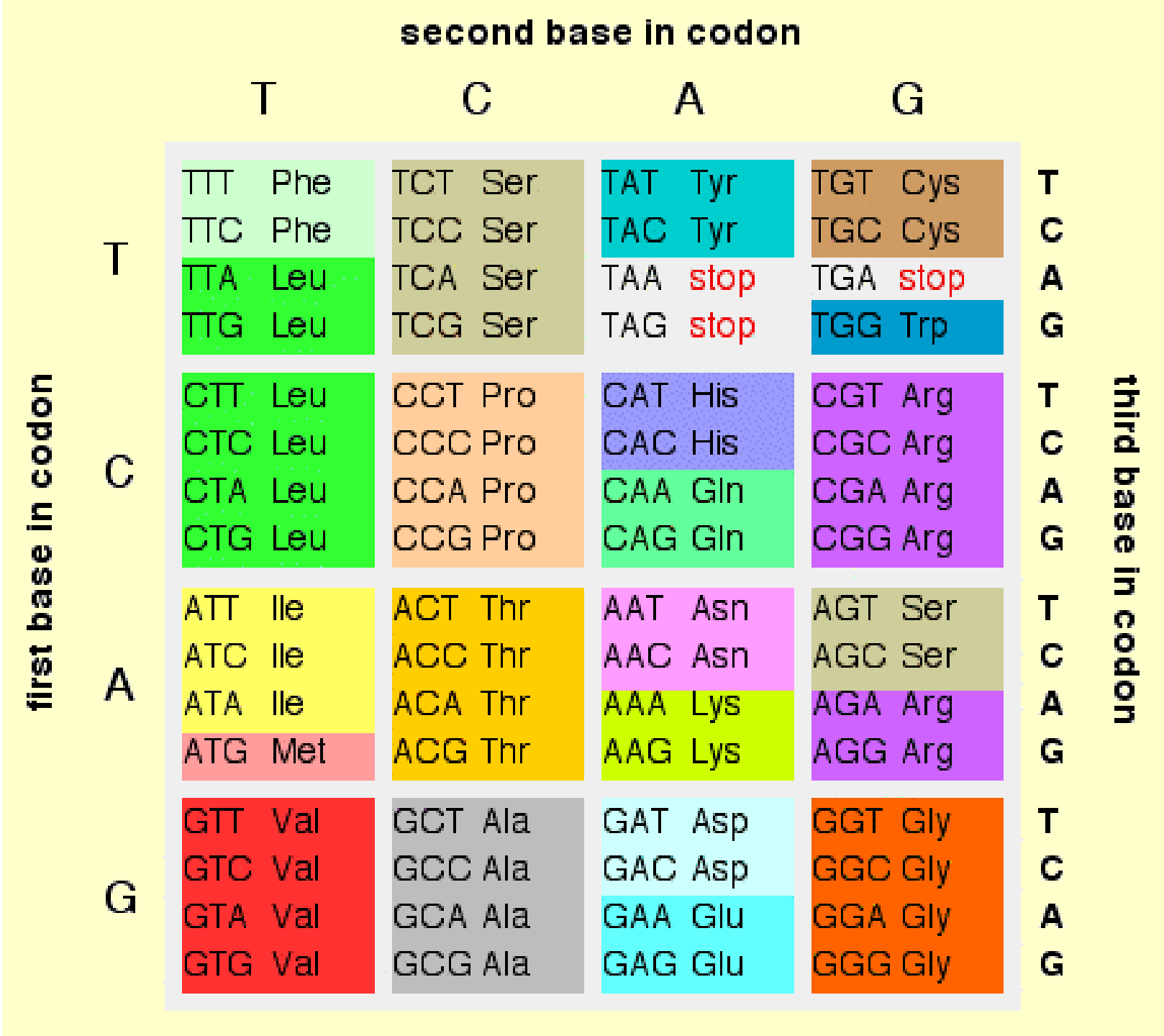


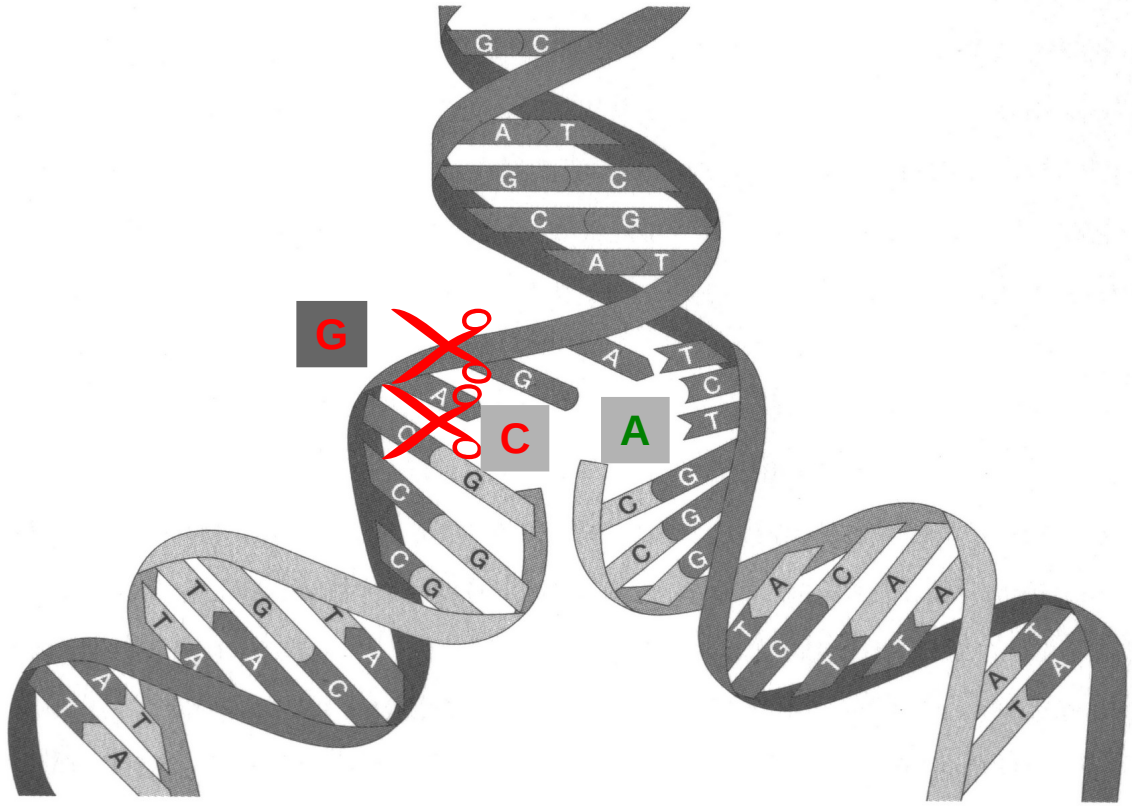
Tratto di DNA formato da 4 paia di basi (CTTG nella catena inferiore, GAAC nella catena antiparallela ad essa). Le frecce verdi indicano la direzione dei legami fra i nucleotidi, che è opposta nelle due catene.

Sintesi delle proteine



Sintesi delle proteine




G
C
A

L'incrocio industriale o F1 toro da carne X vacca da latte

- **Razza Belga** = razza incrociante più utilizzata in Piemonte, allevata in purezza solo in Belgio





- Vitelli ibridi spuntano prezzi ≤ 400 Euro
  [Piemontese puro ~ 500 E, frisone puro (*gaiot*) 50 E]

- Quante bovine da latte posso inseminare con il Belga senza compromettere la rimonta interna?

La razza Belga e la miostatina



Phenotype of mysotatin null vertebrates.
a) upper forelimb muscles of wild-type mouse;
B) upper forelimb of mouse that is myostatin null;
C) bovine with myostatin mutation;
D) and E) human with myostatin mutation as a neonate (D) and at 7 mo of age.









La razza Belga e la miostatina



La razza Belga e la miostatina

PNAS

RESEARCH ARTICLE | GENETICS |

Double muscling in cattle due to mutations in the myostatin gene

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November 11, 1997 | 94 (23) 12457-12461 | https://doi.org/10.1073/pnas.94.23.12457

La razza Belga e la miostatina

GenBank

the GenBank dat... 1

Proc. Natl. Acad. Sci. USA
Vol. 94, pp. 12457–12461, November 1997
Genetics

Check for updates

Double muscling in cattle due to mutations in the myostatin gene

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Communicated by Daniel Nathans, Johns Hopkins University School of Medicine, Baltimore, MD, August 26, 1997 (received for review August 12, 1997)

La razza Belga e la miostatina

Data deposition: The sequences reported in this paper have been deposited in the GenBank database [baboon (accession no. AF019619), bovine (accession no. AF019620), chicken (accession no. AF019621), ovine (accession no. AF019622), porcine (accession no. AF019623), rat (accession no. AF019624), turkey (accession no. AF019625), zebrafish (accession no. AF019626), and human (accession no. AF019627)].

A commentary on this article begins on page 12249.

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La razza Belga e la miostatina

lanciamo il terminale per usare **R**

ctrl+alt+t

$ **R**

```
> install.packages("ape")
> library(ape)
> x = paste("AF0196", 19:27, sep = "")
```

ctrl+MAIUSC+v per incollare nel terminale o in **R**

```
> myostatin.numbers = c('NM_001001525.3', 'XM_010862903.1', 'XM_005907427.1', 'GQ184147.1', 'AF019620.1',
'XM_061432883.1', 'BC134563.1', 'AY160688.1', 'XM_027555492.1', 'XM_019969486.1', 'AF019761.1',
'NM_001290967.1', 'XM_055574715.1', 'KP057613.1', 'AY629304.1', 'XM_055397770.1', 'XM_061135030.1',
'XM_043894881.1', 'XM_043488279.1', 'AY629307.1', 'XM_020900528.1', 'XM_052635569.1', 'EF629535.1',
'NM_001009428.3', 'XM_040227852.1', 'JN662463.1', 'AM992883.1', 'AY629305.1', 'AF019622.1', 'KM371731.1',
'JX840482.1', 'JN639006.1', 'HM462261.1', 'HM462259.1', 'NM_001285737.1', 'FM207636.1', 'AY629309.1',
'AY629306.1', 'AY436347.1', 'AM992884.1', 'GU377303.1', 'XM_007104442.2', 'XM_022568276.1', 'XM_061203308.1',
'XM_057551429.1', 'XM_036859085.1', 'AY629308.1', 'XM_029225345.1', 'XM_027114903.1', 'XM_024733638.1',
'XM_004323907.2', 'XM_030847723.2', 'XM_060151388.1', 'XM_060015612.1', 'XM_060106598.1', 'XM_004276934.2',
'XM_032637329.1', 'XM_059051852.1', 'XM_059928344.1', 'XM_057745280.1', 'XM_047770464.1', 'HM241657.1',
'AF019623.1', 'JN630464.1', 'EF612791.1', 'NM_214435.2', 'XM_006209983.3', 'XM_006189223.3', 'XM_028125258.1',
'XM_027590490.1', 'XM_025887402.1', 'XM_004426733.2', 'XM_010960036.2', 'XM_021703117.1', 'XM_004392403.2',
'XM_046659551.1', 'XM_046650639.1', 'XM_014837576.2', 'XM_032417983.1', 'XM_010991955.2', 'XM_058548651.1',
'XM_008516515.1', 'XM_036075506.1', 'NM_001081817.1', 'XM_006726961.2', 'XM_019626478.1', 'XM_054562700.1',
'XM_036253439.1', 'XM_019740713.1', 'MZ169554.1', 'XM_035007537.1', 'XM_026490450.4', 'XM_045868351.2',
'XM_045806119.1', 'XM_008687262.2', 'XM_029932956.1', 'XM_016215509.1', 'XM_036881965.2', 'XM_047086026.1',
'XM_045480514.1')
```

La razza Belga e la miostatina

```
> x = paste("AF0196", 19:27, sep = "")
> x = x[-c(1:8)]
> x
[1] "AF019627"
> x = x[9]
> x
[1] "AF019627"
>
```

perché l’output di questi due comandi è il medesimo?

La razza Belga e la miostatina

```
> y = c(x, myostatin.numbers)
> myostatin.seq = read.GenBank(x, as.character = TRUE)
> install.packages("seqinr")
> library(seqinr)
```

nel terminale

```
$ mkdir DNA_SEQUENCES
$ ls
$ pwd
```

in **R**

alt gr+ì → ~

```
> setwd("~/DNA_SEQUENCES")
> getwd()
> write.fasta(sequences = myostatin.seq, names = names(myostatin.seq), nbchar
=80, file.out = "myostatin.fas")
```

nel terminale

```
$ cd DNA_SEQUENCES
$ gedit myostatin.fas
```

La razza Belga e la miostatina

dopo
>AF019620
Aggiungere
>bovine_belgian
atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggcccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
ttgcccaaggctcctccactcctggaactgattgatcagttcgatgtccagagagatgccagcagtgacggctccttgga
agacgatgactaccacgccaggacggaaacggtcattaccatgcccacggagtctgatcttctaacgcaagtggaaggaa
aacccaaatgttgcttctttaaatttagctctaagatacaatacaataaactagtaaaggcccaactgtggatatatctg
aggcctgtcaagactcctgcgacagtgtttgtgcaaatcctgagactcatcaaacccatgaaagacggtacaaggtatac
tggaatccgatctctgaaacttgacatgaacccaggcactggtatttggcagagcattgatgtgaagacagtgttgcaga
actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
ttcccagaaccaggagaagatggactgactccttttttagaagtcaaggtaacagacacaccaaaaagatctaggagaga
ttttgggcttgattgtgacagaatctcgatgctgtcgttaccctctaactgtggattttgaagcttttggatgggattgg
attattgcacctaaaagatataaggccaattactgctctggagaatgtgaatttgtatttttgcaaaagtatcctcatac
ccatcttgtgcaccaagcaaaccccagaggttcagccggcccctgctgtactcctacaaagatgtctccaattaatatgc
tatattttaatggcgaaggacaaataatatacgggaagattccagccatggtagtagatcgctgtgggtgttcatga
>bovine_piedmontese
atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggcccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
ttgcccaaggctcctccactcctggaactgattgatcagttcgatgtccagagagatgccagcagtgacggctccttgga
agacgatgactaccacgccaggacggaaacggtcattaccatgcccacggagtctgatcttctaacgcaagtggaaggaa
aacccaaatgttgcttctttaaatttagctctaagatacaatacaataaactagtaaaggcccaactgtggatatatctg
aggcctgtcaagactcctgcgacagtgtttgtgcaaatcctgagactcatcaaacccatgaaagacggtacaaggtatac
tggaatccgatctctgaaacttgacatgaacccaggcactggtatttggcagagcattgatgtgaagacagtgttgcaga
actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
ttcccagaaccaggagaagatggactgactccttttttagaagtcaaggtaacagacacaccaaaaagatctaggagaga
ttttgggcttgattgtgatgaacactccacagaatctcgatgctgtcgttaccctctaactgtggattttgaagcttttg
gatgggattggattattgcacctaaaagatataaggccaattactgctctggagaatatgaatttgtatttttgcaaaag
tatcctcatacccatcttgtgcaccaagcaaaccccagaggttcagccggcccctgctgtactcctacaaagatgtctcc
aattaatatgctatattttaatggcgaaggacaaataatatacgggaagattccagccatggtagtagatcgctgtgggt
gttcatga

La razza Belga e la miostatina

ctrl+MAIUSC+s

```
# save as myostatin_mod.fas

$ clustalx
```

ClustalX 2.1

File Edit Alignment Trees Colors Quality Help

Multiple Alignment Mode Font: 10

Cancel **Load Sequence File** Open

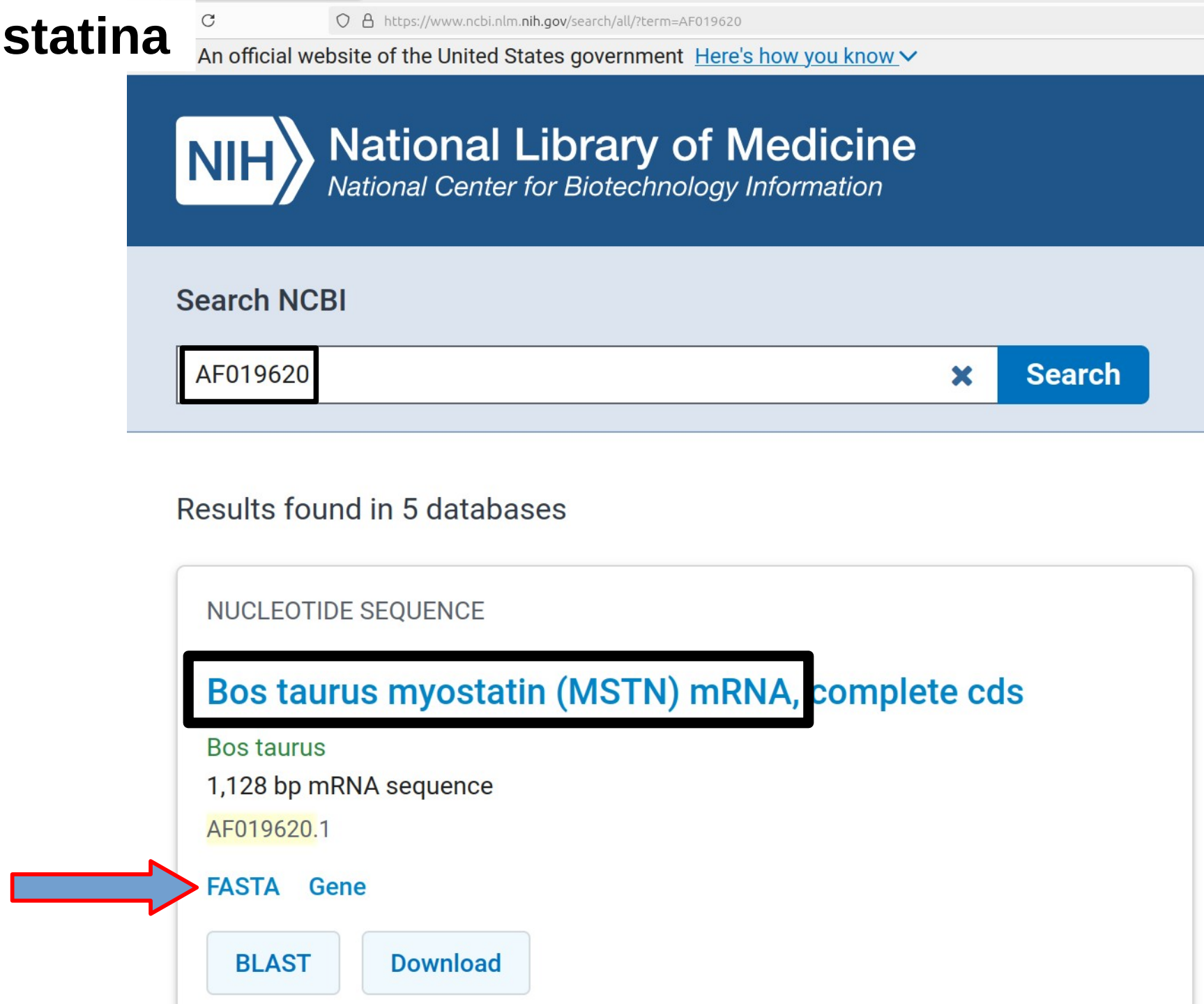
Recenti
Home
Documenti
Immagini
Musica
Scaricati
Video
Altre posizioni

piero DNA_SEQUENCES

Nome	Dimensione	Tipo	Data di modifica
allineamento_miostatina_blast.txt	480,1 kB	Testo	11 gen
database_completo_miostatina.txt	480,1 kB	Testo	13 gen
dexter.fasta	88,6 kB	Testo	3 set
link_Cervidae.txt	344 byte	Testo	10 gen
mcpherron-lee-1997-double-muscling-in-cattle-due-to-mutations-in-the-myostatin-ge…	537,5 kB	Documento	ven
MT_Cervidae	333,7 kB	Testo	10 gen
MT_Cervidae.aln	602,3 kB	Testo	11 gen
MT_Cervidae.dnd	627 byte	Testo	11 gen
MT_Cervidae.fasta	448,4 kB	Testo	11 gen
my_data.txt	98,9 kB	Testo	27 dic 2023
myostatin.fas	220,0 kB	Testo	10:36
myostatin.fasta	10,6 kB	Testo	Ieri
myostatin.RData	8,8 MB	R Data File	Ieri
MYOSTATIN_ACCESSION_NUMBERS.txt	1,7 kB	Testo	10:26
myostatin_all.fasta	215,1 kB	Testo	12 gen
MYOSTATIN_DATA.txt	17,3 kB	Testo	09:15
myostatin_mod.fas	222,9 kB	Testo	Ieri
myostatin_mod.fasta	740,2 kB	Testo	08:19
piero.RData	574,5 MB	R Data File	14 gen
wolves.csv	1,1 kB	Testo	25 dic 2023

All Files

La razza Belga e la miostatina



www.ncbi.nlm.nih.gov/nuccore/AF019620.1/?report=fasta 150% Accedi

FASTA Send to:

Bos taurus myostatin (MSTN) mRNA, complete cds

GenBank: AF019620.1

GenBank Graphics

>AF019620.1 Bos taurus myostatin (MSTN) mRNA, complete cds
ATGCAAAAACTGCAAATCTCTGTTTATATTTACCTATTTATGCTGATTGTTGCTGGCCCAGTGGATCTGA
ATGAGAACAGCGAGCAGAAGGAAAATGTGGAAAAAGAGGGGCTGTGTAATGCATGTTTGTGGAGGGAAAA
CACTACATCCTCAAGACTAGAAGCCATAAAAATCCAAATCCTCAGTAAACTTCGCCTGGAAACAGCTCCT
AACATCAGCAAAGATGCTATCAGACAACTTTTGCCCAAGGCTCCTCCACTCCTGGAACTGATTGATCAGT
TCGATGTCCAGAGAGATGCCAGCAGTGACGGCTCCTTGGAAGACGATGACTACCACGCCAGGACGGAAAC
GGTCATTACCATGCCCACGGAGTCTGATCTTCTAACGCAAGTGGAAGGAAAACCCAAATGTTGCTTCTTT
AAATTTAGCTCTAAGATACAATACAATAAACTAGTAAAGGCCCAACTGTGGATATATCTGAGGCCTGTCA
AGACTCCTGCGACAGTGTTTGTGCAAATCCTGAGACTCATCAAACCCATGAAAGACGGTACAAGGTATAC
TGGAATCCGATCTCTGAAACTTGACATGAACCCAGGCACTGGTATTTGGCAGAGCATTGATGTGAAGACA
GTGTTGCAGAACTGGCTCAAACAACCTGAATCCAACTTAGGCATTGAAATCAAAGCTTTAGATGAGAATG
GCCATGATCTTGCTGTAACCTTCCCAGAACCAGGAGAAGATGGACTGACTCCTTTTTTAGAAGTCAAGGT
AACAGACACACCAAAAAGATCTAGGAGAGATTTTGGGCTTGATTGTGATGAACACTCCACAGAATCTCGA
TGCTGTCGTTACCCTCTAACTGTGGATTTTGAAGCTTTTGGATGGGATTGGATTATTGCACCTAAAAGAT
ATAAGGCCAATTACTGCTCTGGAGAATGTGAATTTGTATTTTTGCAAAAGTATCCTCATACCCATCTTGT
GCACCAAGCAAACCCCAGAGGTTCAGCCGGCCCCTGCTGTACTCCTACAAAGATGTCTCCAATTAATATG
CTATATTTTAATGGCGAAGGACAAATAATATACGGGAAGATTCCAGCCATGGTAGTAGATCGCTGTGGGT
GTTCATGA

Change region shown

Customize view

Analyze this sequence

Run BLAST

Pick Primers

Articles about the MSTN gene

Combined Transcriptome and Metabolome Analysi [Int J Mol Sci. 2

Myostatin site-directed mutation and simultaneous PPAF [J Cell Physiol. 2

Association between polymorphisms Myf5, MSTN anc [Anim Biotechnol. 2

See

Reference sequence information

RefSeq mRNA

See reference mRNA sequence for t
MSTN gene (NM_001001525.3)

Miostatina: allineamento multi-sequenza

```
> x = paste("AF0196", 19:27, sep = "")
> setwd('~/DNA_SEQUENCES/')
> load('piero.RData')
> library(ape)
> library(seqinr)
> myostatin_x.seq = read.GenBank(x, as.character = TRUE)
```

usando la funzione <**read.GenBank**> del pacchetto APE (Analysis of Phylogenetics and Evolution), viene creata una lista delle sequenze nucleotidiche corrispondenti ai numeri di accesso a **GenBank** contenuti nel vettore **x**

www.ncbi.nlm.nih.gov/genbank/ 200% Accedi

GenBank Overview

What is GenBank?

GenBank ® is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences (*Nucleic Acids Research*, 2013 Jan;41(D1):D36-42). GenBank is part of the International Nucleotide Sequence Database Collaboration, which comprises the DNA DataBank of Japan (DDBJ), the European Nucleotide Archive (ENA), and GenBank at NCBI. These three organizations exchange data on a daily basis.

A GenBank release occurs every two months and is available from the ftp site. The release notes for the current version of GenBank provide detailed information about the release and notifications of upcoming changes to GenBank. Release notes for previous GenBank releases are also available. GenBank growth statistics for both the traditional GenBank divisions and the WGS division are available from each release.

An annotated sample GenBank record for a *Saccharomyces cerevisiae* gene demonstrates many of the features of the GenBank flat file format.

Access to GenBank

There are several ways to search and retrieve data from GenBank.

- Search GenBank for sequence identifiers and annotations with Entrez Nucleotide.

GenBank Resourc

GenBank Home
Submission Types
Submission Tools
Search GenBank
Update GenBank Rec

ape package 5.8-1

Analyses of Phylogenetics and Evolution

```
install.packages("ape")
```

Copy

Functions Readme Datasets Dependencies Imports Versions

subtreeplot

Zoom on a Portion of a Phylogeny by Successive Clicks

subtrees

All subtrees of a Phylogenetic Tree

summary.phylo

Print Summary of a Phylogeny

where

which.edge

write.dna

Download source package

Read PDF manual

Functions for reading, writing, plotting, and manipulating phylogenetic trees, analyses of comparative data in a phylogenetic framework, ancestral character analyses

Miostatina: allineamento multi-sequenza

```
> str(myostatin_x.seq)
List of 9
 $ AF019619: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019620: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019621: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019622: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019623: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019624: chr [1:1131] "a" "t" "g" "a" ...
 $ AF019625: chr [1:1128] "a" "t" "g" "c" ...
 $ AF019626: chr [1:1125] "a" "t" "g" "c" ...
 $ AF019627: chr [1:1128] "a" "t" "g" "c" ...
 - attr(*, "species")= chr [1:9] "Papio_hamadryas" "Bos_taurus"
"Gallus_gallus" "Ovis_aries" ...
> length(myostatin_x.seq)
[1] 9
```

Miostatina: allineamento multi-sequenza

```
> myostatin_x.seq[1]
$AF019619
   [1] "a" "t" "g" "c" "a" "a" "a" "a" "a" "c" "t" "g" "c" "a" "a" "c" "t" "c"
  [19] "t" "g" "t" "g" "t" "t" "t" "a" "t" "a" "t" "t" "t" "a" "c" "c" "t" "g"
  [37] "t" "t" "t" "a" "t" "g" "c" "t" "g" "a" "t" "t" "g" "t" "t" "g" "c" "t"
  ...
[1099] "a" "t" "g" "g" "t" "a" "g" "t" "a" "g" "a" "c" "c" "g" "c" "t" "g" "c"
[1117] "g" "g" "g" "t" "g" "c" "t" "c" "a" "t" "g" "a"

>
```

Miostatina: allineamento multi-sequenza

```
> myostatin_x.seq = read.GenBank(x, as.character = FALSE)
> str(myostatin_x.seq)
List of 9
 $ AF019619: raw [1:1128] 88 18 48 28 ...
 $ AF019620: raw [1:1128] 88 18 48 28 ...
 $ AF019621: raw [1:1128] 88 18 48 28 ...
 $ AF019622: raw [1:1128] 88 18 48 28 ...
 $ AF019623: raw [1:1128] 88 18 48 28 ...
 $ AF019624: raw [1:1131] 88 18 48 88 ...
 $ AF019625: raw [1:1128] 88 18 48 28 ...
 $ AF019626: raw [1:1125] 88 18 48 28 ...
 $ AF019627: raw [1:1128] 88 18 48 28 ...
 - attr(*, "class")= chr "DNAbin"
 - attr(*, "description")= chr [1:9] "AF019619.1 Papio hamadryas myostatin
(MSTN) mRNA, complete cds" "AF019620.1 Bos taurus myostatin (MSTN) mRNA,
complete cds" "AF019621.1 Gallus gallus myostatin (MSTN) mRNA, complete cds"
"AF019622.1 Ovis aries myostatin (MSTN) mRNA, complete cds" ...
 - attr(*, "species")= chr [1:9] "Papio_hamadryas" "Bos_taurus"
"Gallus_gallus" "Ovis_aries" ...
>
```

Miostatina: allineamento multi-sequenza

```
> attributes(myostatin_x.seq)
$names
[1] "AF019619" "AF019620" "AF019621" "AF019622" "AF019623" "AF019624" "AF019625"
[8] "AF019626" "AF019627"

$class
[1] "DNAbin"

$description
[1] "AF019619.1 Papio hamadryas myostatin (MSTN) mRNA, complete cds"
[2] "AF019620.1 Bos taurus myostatin (MSTN) mRNA, complete cds"
[3] "AF019621.1 Gallus gallus myostatin (MSTN) mRNA, complete cds"
[4] "AF019622.1 Ovis aries myostatin (MSTN) mRNA, complete cds"
[5] "AF019623.1 Sus scrofa myostatin (MSTN) mRNA, complete cds"
[6] "AF019624.1 Rattus norvegicus myostatin (MSTN) mRNA, complete cds"
[7] "AF019625.1 Meleagris gallopavo myostatin (MSTN) mRNA, complete cds"
[8] "AF019626.1 Danio rerio myostatin (MSTN) mRNA, complete cds"
[9] "AF019627.1 Homo sapiens myostatin (MSTN) mRNA, complete cds"

$species
[1] "Papio_hamadryas"     "Bos_taurus"          "Gallus_gallus"
[4] "Ovis_aries"          "Sus_scrofa"          "Rattus_norvegicus"
[7] "Meleagris_gallopavo" "Danio_rerio"         "Homo_sapiens"

>
```

Miostatina: allineamento multi-sequenza

```
> myostatin_x.seq = read.GenBank(x, as.character = TRUE)
> attributes(myostatin_x.seq)
$names
[1] "AF019619" "AF019620" "AF019621" "AF019622" "AF019623" "AF019624"
"AF019625"
[8] "AF019626" "AF019627"

$species
[1] "Papio_hamadryas"     "Bos_taurus"          "Gallus_gallus"
[4] "Ovis_aries"          "Sus_scrofa"          "Rattus_norvegicus"
[7] "Meleagris_gallopavo" "Danio_rerio"         "Homo_sapiens"

>
```

Miostatina: allineamento multi-sequenza

```
> library(seqinr)
> write.fasta(sequences = myostatin_x.seq, names = names(myostatin_x.seq),
nbchar =80, file.out = "myostatin_x.fas")
>
```

uso la funzione <**write.fasta**> del pacchetto SEQINR (Biological Sequences Retrieval and Analysis) per creare il file di testo in formato «fasta» <**myostatin_x.fas**>, contenente le sequenze elencate nella list <**myostatin_x.seq**>

seqinr package 4.2-36

Biological Sequences Retrieval and Analysis

```
install.packages("seqinr")
```

Copy

Functions Datasets Dependencies Imports Versions

a

Converts amino-acid three-letter code into the one-letter one

aaa

Converts amino-acid one-letter code into the three-letter one

AAstat

To Get Some Protein Statistics

acnucopen

al2bp

alllistranks

Download source package

Read PDF manual

Exploratory data analysis and data visualization for biological sequence (DNA and protein) data. Seqinr includes utilities for sequence data management under the

Miostatina: allineamento multi-sequenza

nel terminale

```
$ locate myostatin_x.fas
/home/piero/myostatin_x.fas
/home/piero/DNA_SEQUENCES/myostatin_x.fas
```

questo è il «percorso» (PATH) del file <**myostatin_x.fas**>

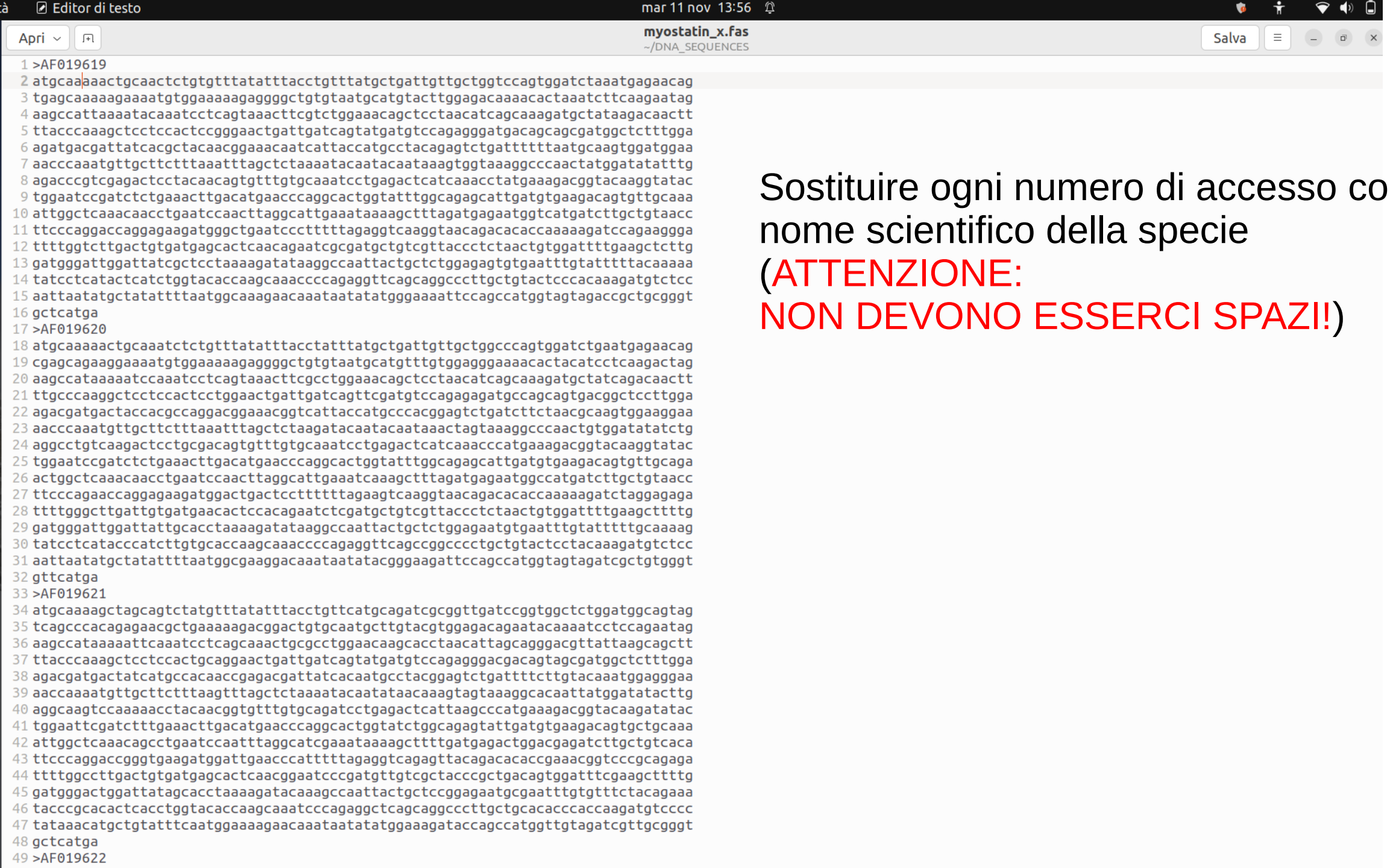
```
$ pwd
/home/piero
```

sono già nella home, quindi non è necessario specificare l'intero PATH

```
$ gedit DNA_SEQUENCES/myostatin_x.fas
```

oppure

```
piero@piero-XPS-9320:~$ cd DNA_SEQUENCES/
piero@piero-XPS-9320:~/DNA_SEQUENCES$ pwd
/home/piero/DNA_SEQUENCES
piero@piero-XPS-9320:~/DNA_SEQUENCES$ gedit myostatin_x.fas
```



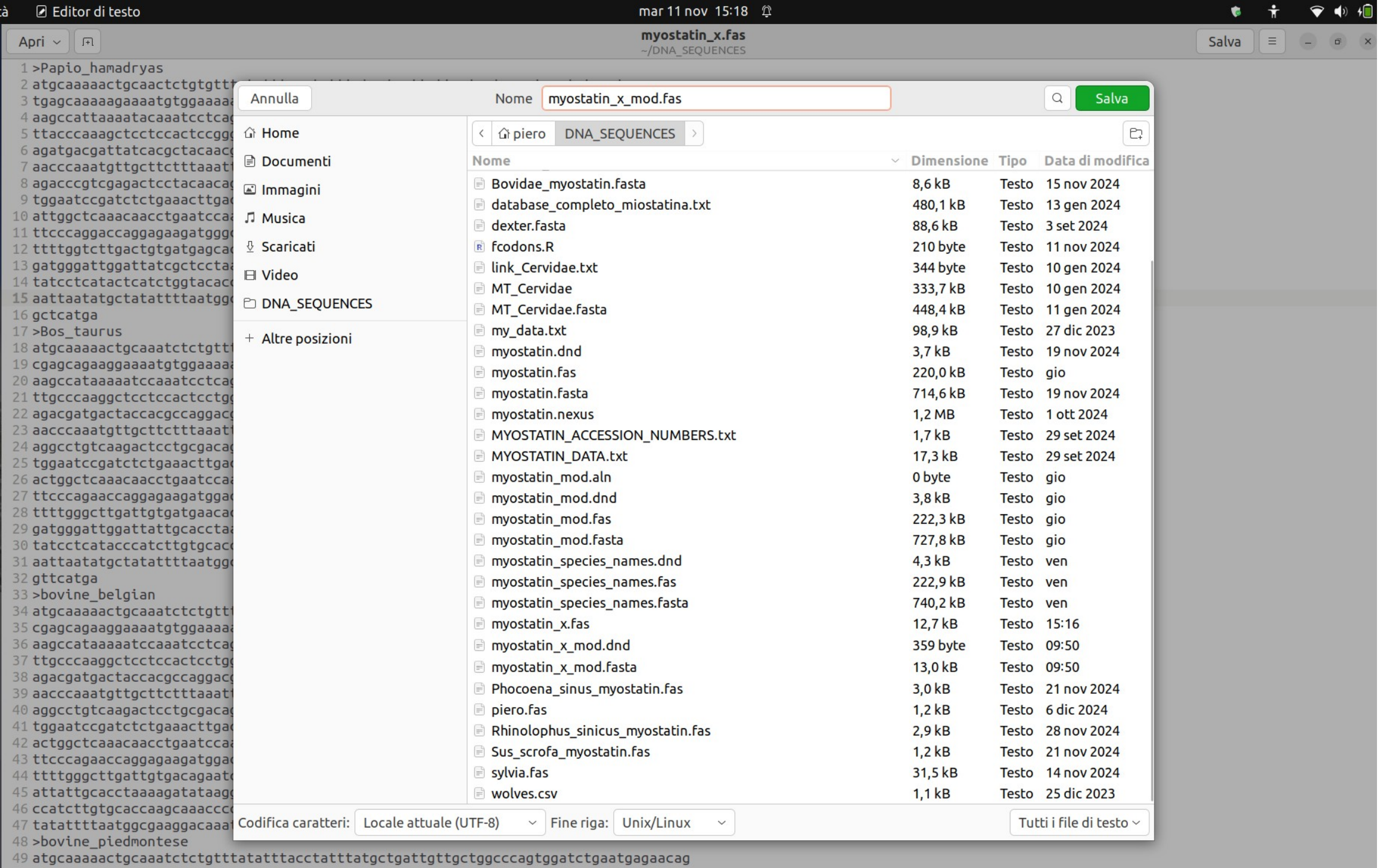
Sostituire ogni numero di accesso con il nome scientifico della specie (ATTENZIONE: NON DEVONO ESSERCI SPAZI!)

La razza Belga e la miostatina

dopo
>Bos_taurus
Aggiungere
>bovine_belgian
atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggcccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
ttgcccaaggctcctccactcctggaactgattgatcagttcgatgtccagagagatgccagcagtgacggctccttgga
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actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
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>bovine_piedmontese
atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggcccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
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agacgatgactaccacgccaggacggaaacggtcattaccatgcccacggagtctgatcttctaacgcaagtggaaggaa
aacccaaatgttgcttctttaaatttagctctaagatacaatacaataaactagtaaaggcccaactgtggatatatctg
aggcctgtcaagactcctgcgacagtgtttgtgcaaatcctgagactcatcaaacccatgaaagacggtacaaggtatac
tggaatccgatctctgaaacttgacatgaacccaggcactggtatttggcagagcattgatgtgaagacagtgttgcaga
actggctcaaacaacctgaatccaacttaggcattgaaatcaaagctttagatgagaatggccatgatcttgctgtaacc
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tatcctcatacccatcttgtgcaccaagcaaaccccagaggttcagccggcccctgctgtactcctacaaagatgtctcc
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gttcatga

La razza Belga e la miostatina

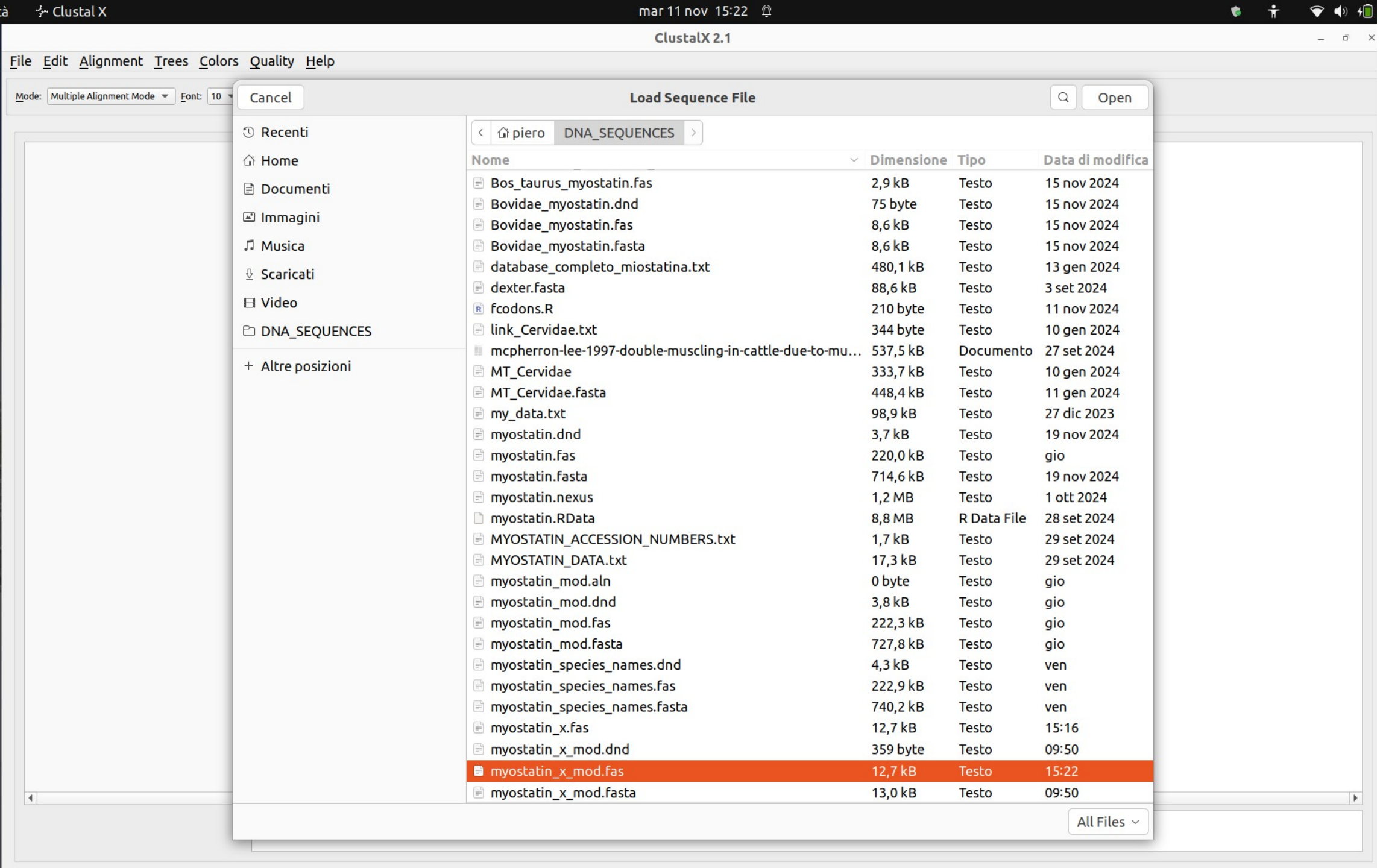
ctrl+MAIUSC+s
save as `myostatin_x_mod.fas`


Editor di testo
mar 11 nov 15:18
Apri
myostatin_x.fas
~/DNA_SEQUENCES
Salva
>Papio_hamadryas
atgcaaaaactgcaactctgtgttt
tgagcaaaaagaaaatgtggaaaa
aagccattaaaatacaaatcctcag
ttacccaaagctcctccactccggg
agatgacgattatcacgctacaacg
aacccaaatgttgcttctttaaatt
agacccgtcgagactcctacaacag
tggaatccgatctctgaaacttgac
attggctcaaacaacctgaatccaa
ttcccaggaccaggagaagatgggc
ttttggtcttgactgtgatgagcac
gatgggattggattatcgctcctaa
tatcctcatactcatctggtacacc
aattaatatgctatattttaatggc
gctcatga
>Bos_taurus
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cgagcagaaggaaaatgtggaaaaa
aagccataaaaatccaaatcctcag
ttgcccaaggctcctccactcctgg
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aggcctgtcaagactcctgcgacag
tggaatccgatctctgaaacttgac
actggctcaaacaacctgaatccaa
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tatcctcatacccatcttgtgcacc
aattaatatgctatattttaatggc
gttcatga
>bovine_belgian
atgcaaaaactgcaaatctctgttt
cgagcagaaggaaaatgtggaaaaa
aagccataaaaatccaaatcctcag
ttgcccaaggctcctccactcctgg
agacgatgactaccacgccaggacg
aacccaaatgttgcttctttaaatt
aggcctgtcaagactcctgcgacag
tggaatccgatctctgaaacttgac
actggctcaaacaacctgaatccaa
ttcccagaaccaggagaagatggac
ttttgggcttgattgtgacagaatc
attattgcacctaaaagatataagg
ccatcttgtgcaccaagcaaaccc
tatattttaatggcgaaggacaaat
>bovine_piedmontese
atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggcccagtggatctgaatgagaacag
Annulla
Nome
myostatin_x_mod.fas
Salva
Home
Documenti
Immagini
Musica
Scaricati
Video
DNA_SEQUENCES
Altre posizioni
piero
DNA_SEQUENCES
Nome
Dimensione
Tipo
Data di modifica
Bovidae_myostatin.fasta 8,6 kB Testo 15 nov 2024
database_completo_miostatina.txt 480,1 kB Testo 13 gen 2024
dexter.fasta 88,6 kB Testo 3 set 2024
fcodons.R 210 byte Testo 11 nov 2024
link_Cervidae.txt 344 byte Testo 10 gen 2024
MT_Cervidae 333,7 kB Testo 10 gen 2024
MT_Cervidae.fasta 448,4 kB Testo 11 gen 2024
my_data.txt 98,9 kB Testo 27 dic 2023
myostatin.dnd 3,7 kB Testo 19 nov 2024
myostatin.fas 220,0 kB Testo gio
myostatin.fasta 714,6 kB Testo 19 nov 2024
myostatin.nexus 1,2 MB Testo 1 ott 2024
MYOSTATIN_ACCESSION_NUMBERS.txt 1,7 kB Testo 29 set 2024
MYOSTATIN_DATA.txt 17,3 kB Testo 29 set 2024
myostatin_mod.aln 0 byte Testo gio
myostatin_mod.dnd 3,8 kB Testo gio
myostatin_mod.fas 222,3 kB Testo gio
myostatin_mod.fasta 727,8 kB Testo gio
myostatin_species_names.dnd 4,3 kB Testo ven
myostatin_species_names.fas 222,9 kB Testo ven
myostatin_species_names.fasta 740,2 kB Testo ven
myostatin_x.fas 12,7 kB Testo 15:16
myostatin_x_mod.dnd 359 byte Testo 09:50
myostatin_x_mod.fasta 13,0 kB Testo 09:50
Phocoena_sinus_myostatin.fas 3,0 kB Testo 21 nov 2024
piero.fas 1,2 kB Testo 6 dic 2024
Rhinolophus_sinicus_myostatin.fas 2,9 kB Testo 28 nov 2024
Sus_scrofa_myostatin.fas 1,2 kB Testo 21 nov 2024
sylvia.fas 31,5 kB Testo 14 nov 2024
wolves.csv 1,1 kB Testo 25 dic 2023
Codifica caratteri:
Locale attuale (UTF-8)
Fine riga:
Unix/Linux
Tutti i file di testo

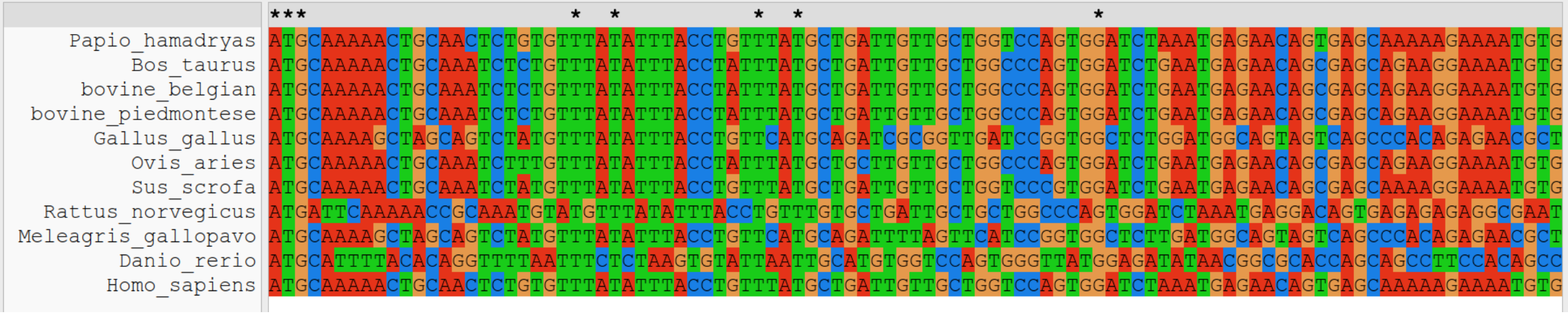
La razza Belga e la miostatina

nel terminale

```
$ clustalx
```

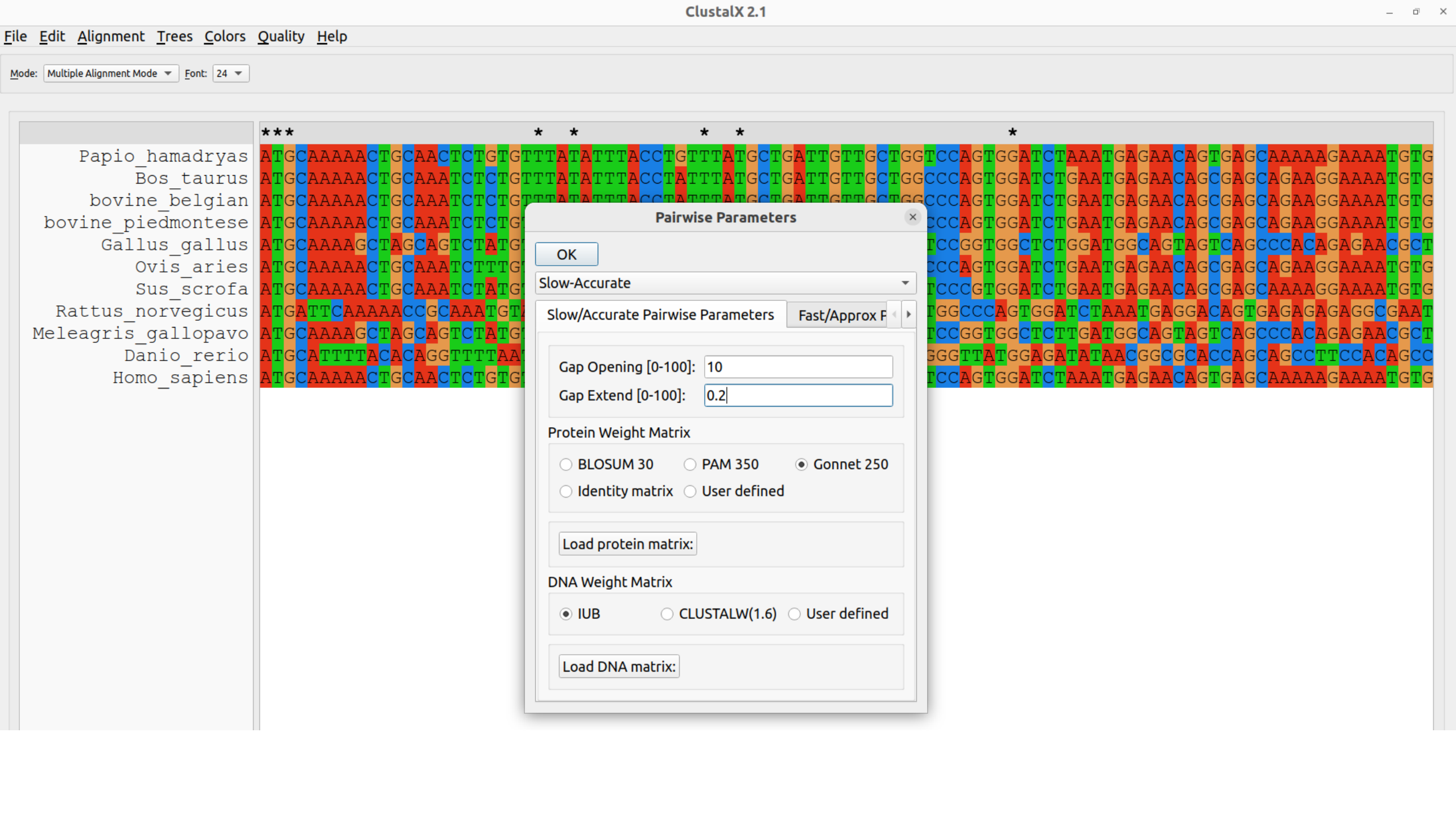

Clustal X
mar 11 nov 15:22
ClustalX 2.1
File Edit Alignment Trees Colors Quality Help
Mode: Multiple Alignment Mode
Font: 10
Cancel
Load Sequence File
Open
Recenti
Home
Documenti
Immagini
Musica
Scaricati
Video
DNA_SEQUENCES
Altre posizioni
piero
DNA_SEQUENCES
Nome
Dimensione
Tipo
Data di modifica
Bos_taurus_myostatin.fas 2,9 kB Testo 15 nov 2024
Bovidae_myostatin.dnd 75 byte Testo 15 nov 2024
Bovidae_myostatin.fas 8,6 kB Testo 15 nov 2024
Bovidae_myostatin.fasta 8,6 kB Testo 15 nov 2024
database_completo_miostatina.txt 480,1 kB Testo 13 gen 2024
dexter.fasta 88,6 kB Testo 3 set 2024
fcodons.R 210 byte Testo 11 nov 2024
link_Cervidae.txt 344 byte Testo 10 gen 2024
mcpherron-lee-1997-double-muscling-in-cattle-due-to-mu… 537,5 kB Documento 27 set 2024
MT_Cervidae 333,7 kB Testo 10 gen 2024
MT_Cervidae.fasta 448,4 kB Testo 11 gen 2024
my_data.txt 98,9 kB Testo 27 dic 2023
myostatin.dnd 3,7 kB Testo 19 nov 2024
myostatin.fas 220,0 kB Testo gio
myostatin.fasta 714,6 kB Testo 19 nov 2024
myostatin.nexus 1,2 MB Testo 1 ott 2024
myostatin.RData 8,8 MB R Data File 28 set 2024
MYOSTATIN_ACCESSION_NUMBERS.txt 1,7 kB Testo 29 set 2024
MYOSTATIN_DATA.txt 17,3 kB Testo 29 set 2024
myostatin_mod.aln 0 byte Testo gio
myostatin_mod.dnd 3,8 kB Testo gio
myostatin_mod.fas 222,3 kB Testo gio
myostatin_mod.fasta 727,8 kB Testo gio
myostatin_species_names.dnd 4,3 kB Testo ven
myostatin_species_names.fas 222,9 kB Testo ven
myostatin_species_names.fasta 740,2 kB Testo ven
myostatin_x.fas 12,7 kB Testo 15:16
myostatin_x_mod.dnd 359 byte Testo 09:50
myostatin_x_mod.fas 12,7 kB Testo 15:22
myostatin_x_mod.fasta 13,0 kB Testo 09:50
All Files

La razza Belga e la miostatina

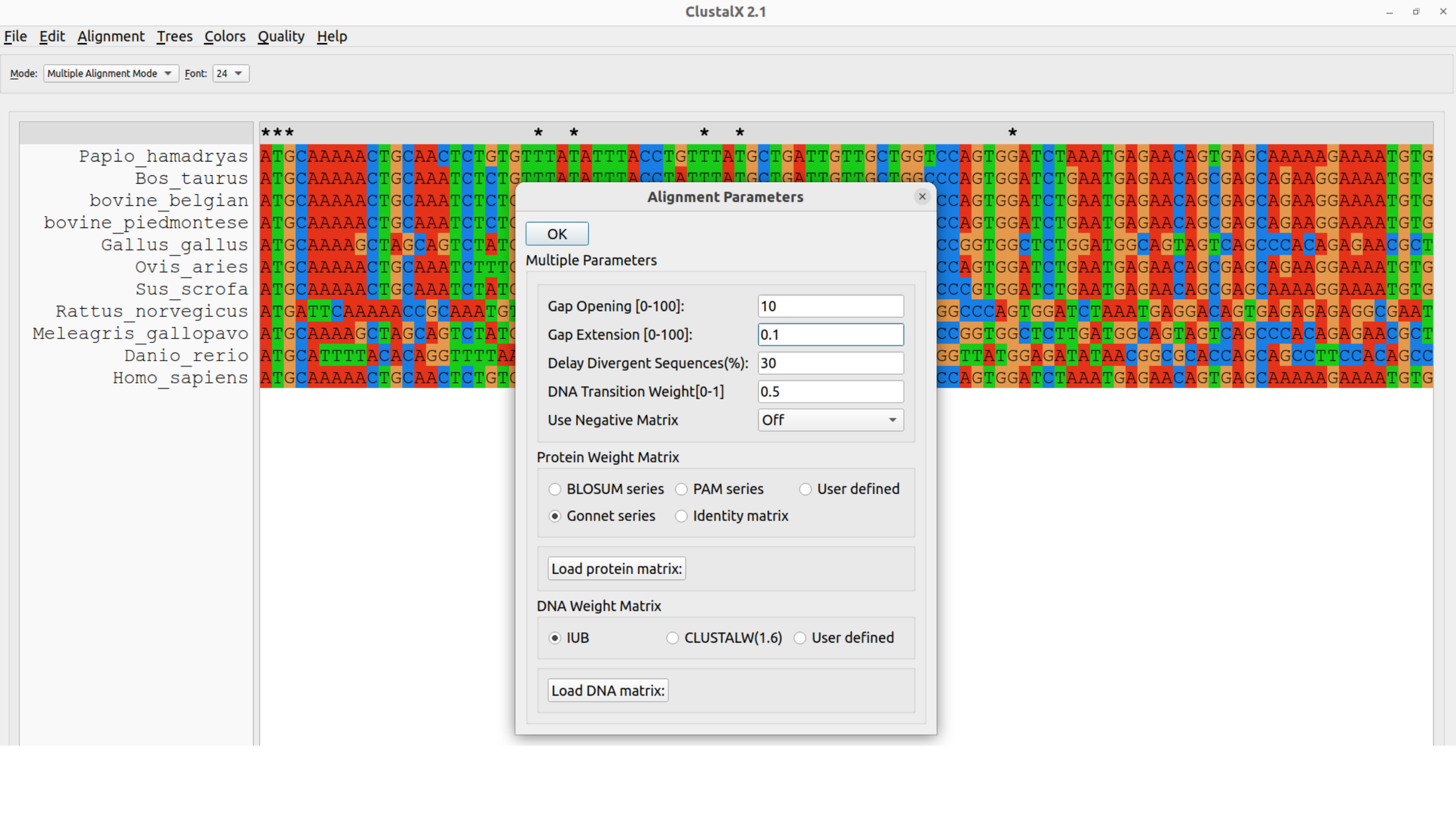


	GAPOPEN	GAPEXTENSION	DNA TRANSITION WEIGHT
PAIRWISE ALIGNMENT PARAMETERS	10.0	0.2	
MULTIPLE ALIGNMENT PARAMETERS	10.0	0.1	0.5


ClustalX 2.1
File
Edit
Alignment
Trees
Colors
Quality
Help
Mode:
Do Complete Alignment Ctrl+L
Do Guide Tree Only Ctrl+G
Do Alignment from Guide Tree
Realign Selected Sequences
Realign Selected Residue Range
Align Profile 2 to Profile 1
Align Profiles from Guide Trees
Align Sequences to Profile 1
Align Sequences to Profile 1 from Tree
Alignment Parameters
Iteration
Output Format Options
Set All Parameters to default
Reset New Gaps before Alignment
Reset All Gaps before Alignment
Pairwise Alignment Parameters
Multiple Alignment Parameters
Protein Gap Parameters
Secondary Structure Parameters
Danio_rerio
Homo_sapiens


ClustalX 2.1
File Edit Alignment Trees Colors Quality Help
Mode: Multiple Alignment Mode Font: 24
Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens
Pairwise Parameters
OK
Slow-Accurate
Slow/Accurate Pairwise Parameters
Fast/Approx P
Gap Opening [0-100]: 10
Gap Extend [0-100]: 0.2
Protein Weight Matrix
BLOSUM 30
PAM 350
Gonnet 250
Identity matrix
User defined
Load protein matrix:
DNA Weight Matrix
IUB
CLUSTALW(1.6)
User defined
Load DNA matrix:


ClustalX 2.1
File
Edit
Alignment
Trees
Colors
Quality
Help
Mode:
Do Complete Alignment Ctrl+L
Do Guide Tree Only Ctrl+G
Do Alignment from Guide Tree
Realign Selected Sequences
Realign Selected Residue Range
Align Profile 2 to Profile 1
Align Profiles from Guide Trees
Align Sequences to Profile 1
Align Sequences to Profile 1 from Tree
Alignment Parameters
Iteration
Output Format Options
Set All Parameters to default
Reset New Gaps before Alignment
Reset All Gaps before Alignment
Pairwise Alignment Parameters
Multiple Alignment Parameters
Protein Gap Parameters
Secondary Structure Parameters
Danio_rerio
Homo_sapiens

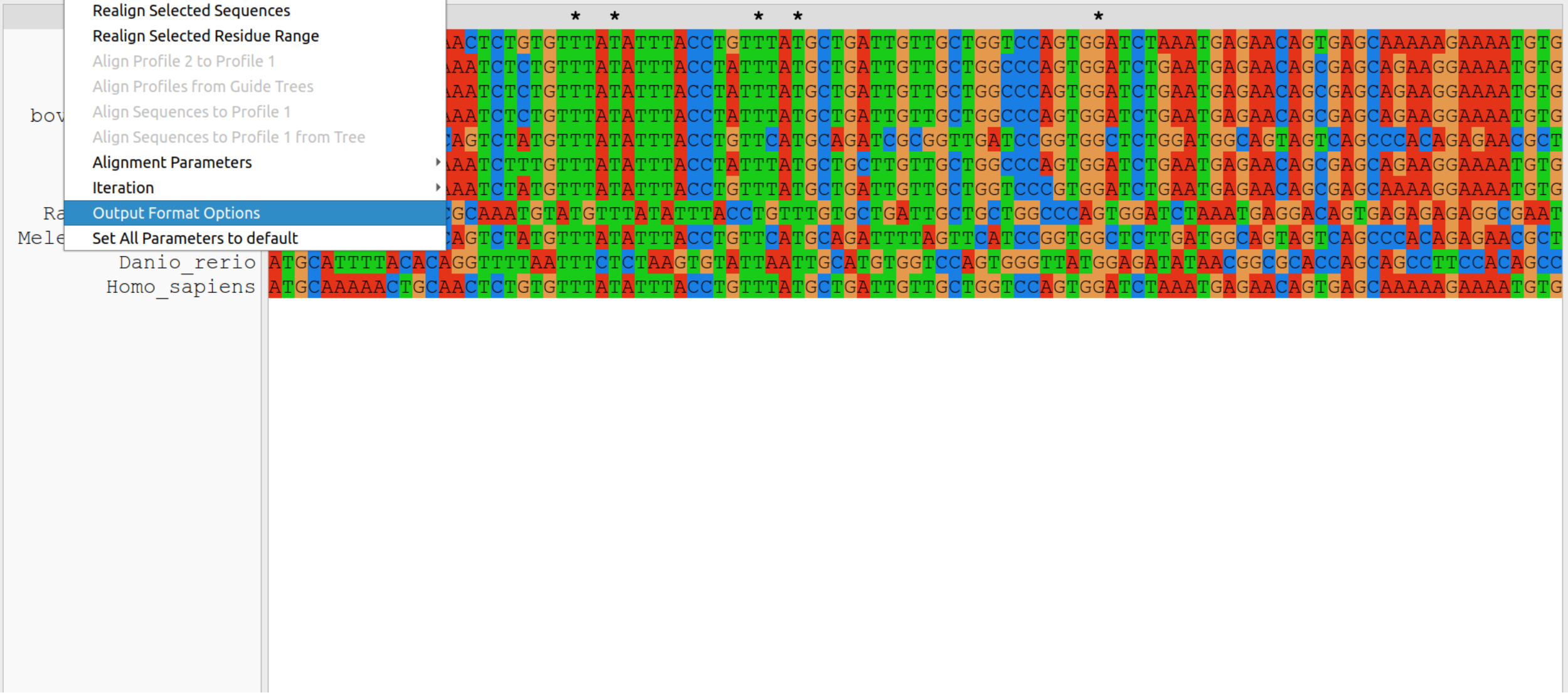

ClustalX 2.1
File Edit Alignment Trees Colors Quality Help
Mode: Multiple Alignment Mode Font: 24
Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens
Alignment Parameters
OK
Multiple Parameters
Gap Opening [0-100]: 10
Gap Extension [0-100]: 0.1
Delay Divergent Sequences(%): 30
DNA Transition Weight[0-1] 0.5
Use Negative Matrix Off
Protein Weight Matrix
BLOSUM series
PAM series
User defined
Gonnet series
Identity matrix
Load protein matrix:
DNA Weight Matrix
IUB
CLUSTALW(1.6)
User defined
Load DNA matrix:

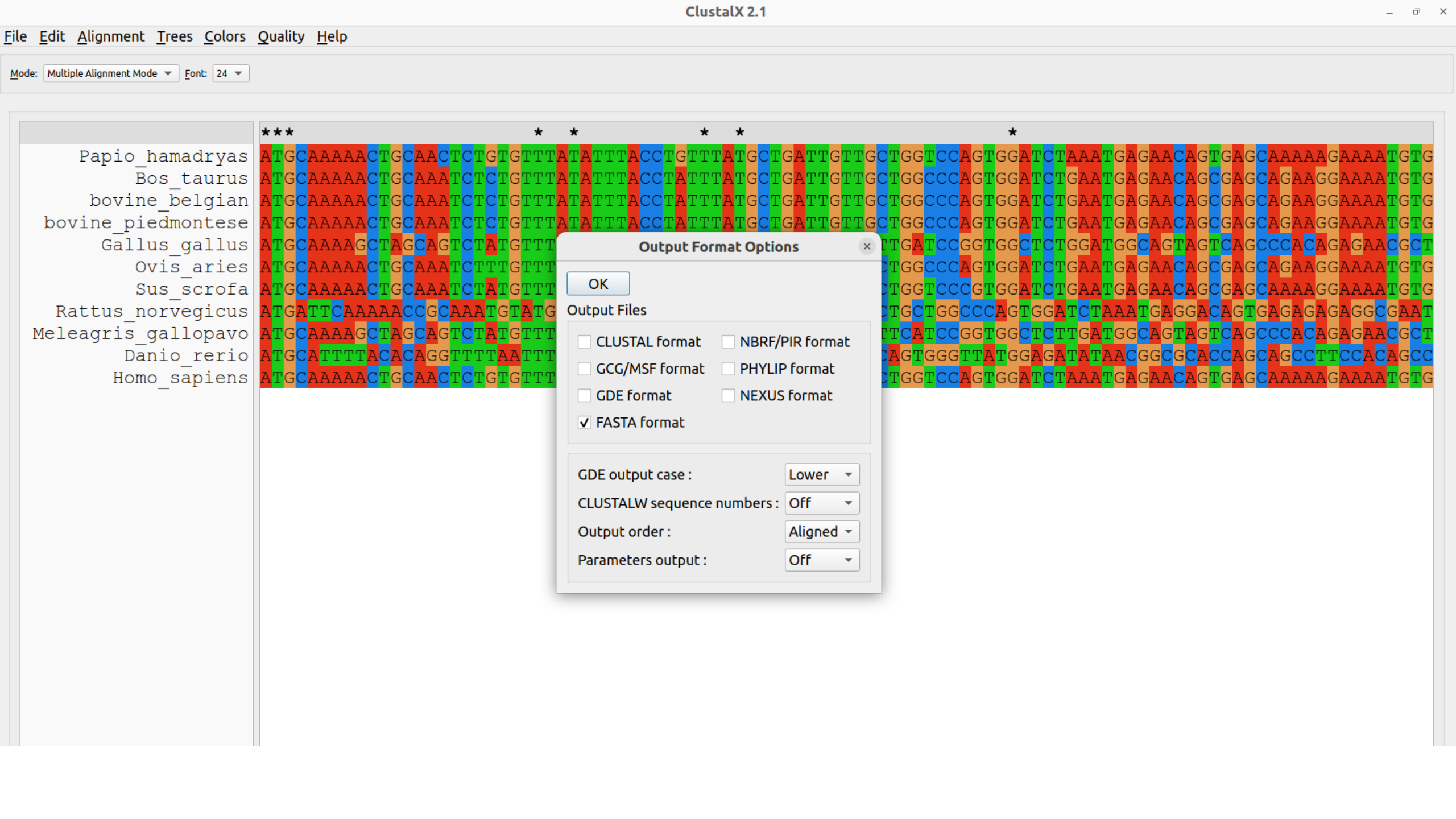
ClustalX 2.1

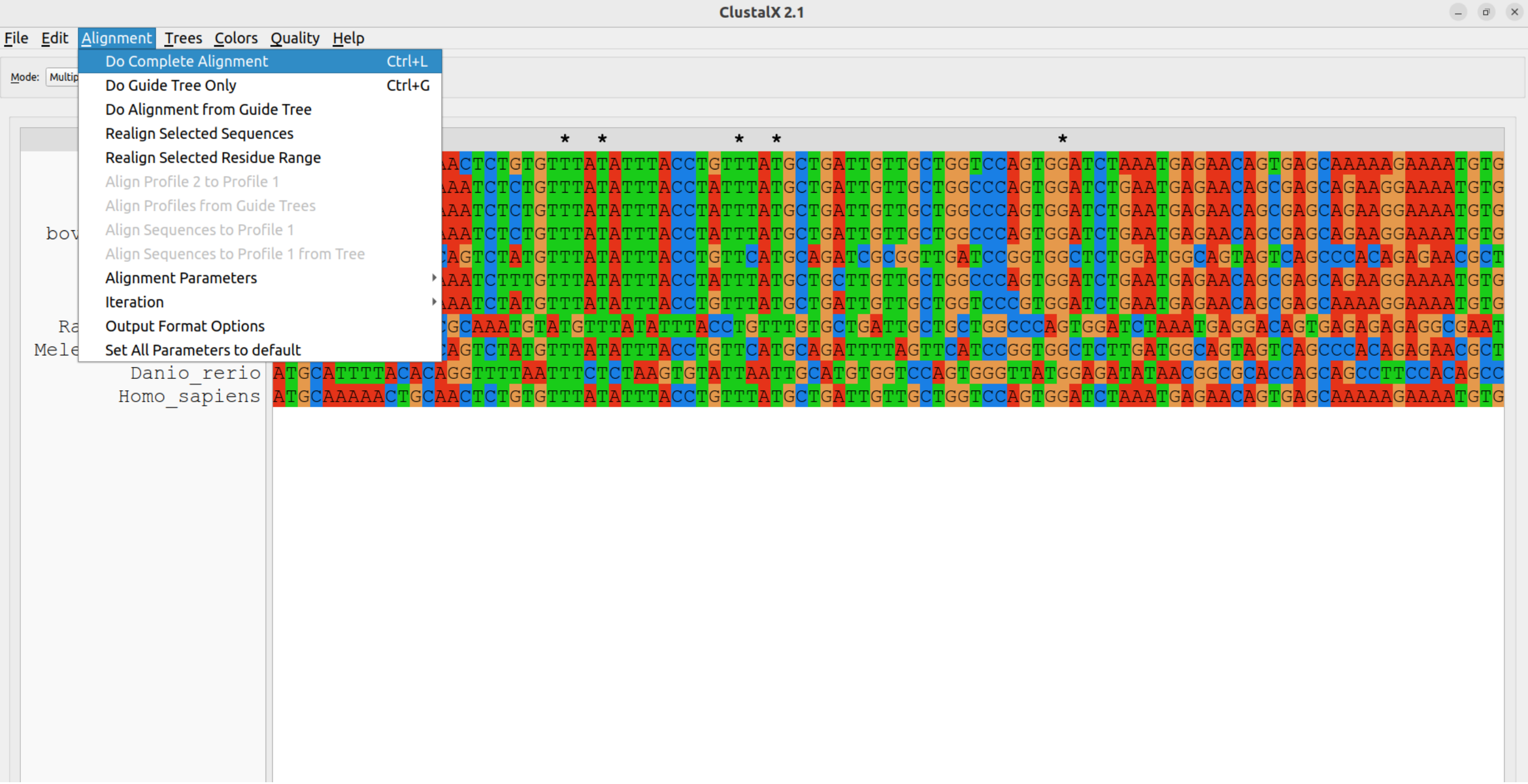
File Edit Alignment Trees Colors Quality Help

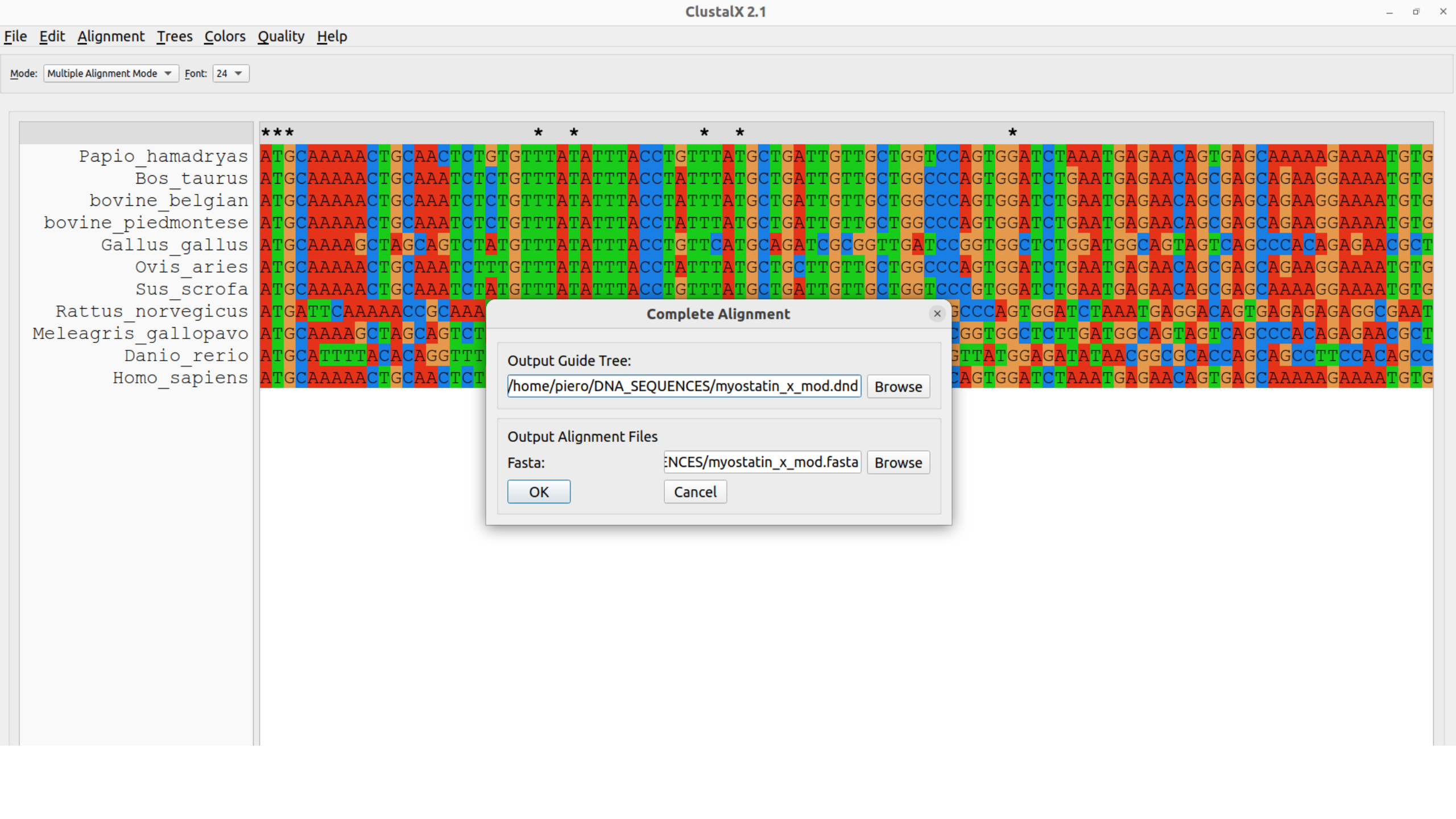
Mode: Multip

- Do Complete Alignment Ctrl+L
- Do Guide Tree Only Ctrl+G
- Do Alignment from Guide Tree
- Realign Selected Sequences
- Realign Selected Residue Range
- Align Profile 2 to Profile 1
- Align Profiles from Guide Trees
- Align Sequences to Profile 1
- Align Sequences to Profile 1 from Tree
- Alignment Parameters
- Iteration
- Output Format Options
- Set All Parameters to default



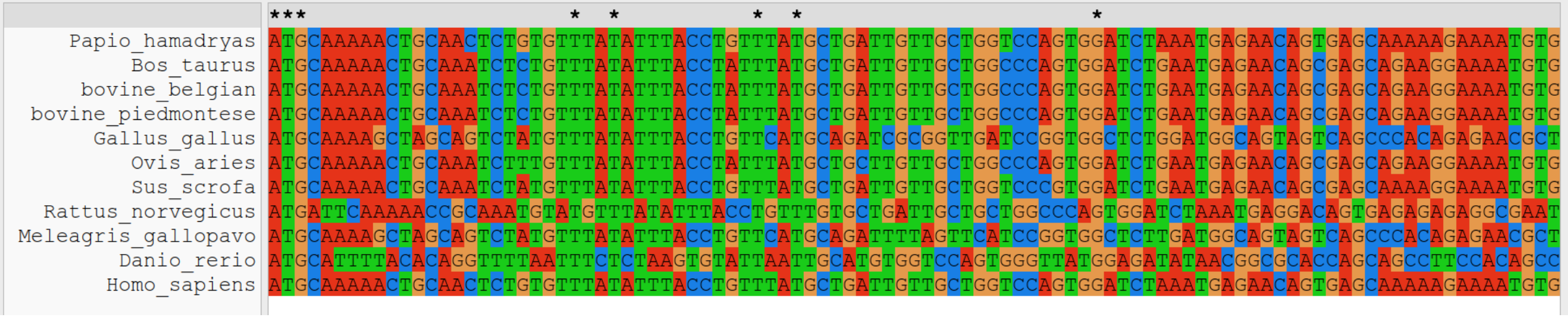

ClustalX 2.1
File Edit Alignment Trees Colors Quality Help
Mode: Multiple Alignment Mode Font: 24
Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens
Output Format Options
OK
Output Files
CLUSTAL format
NBRF/PIR format
GCG/MSF format
PHYLIP format
GDE format
NEXUS format
FASTA format
GDE output case : Lower
CLUSTALW sequence numbers : Off
Output order : Aligned
Parameters output : Off


ClustalX 2.1
File
Edit
Alignment
Trees
Colors
Quality
Help
Mode:
Do Complete Alignment Ctrl+L
Do Guide Tree Only Ctrl+G
Do Alignment from Guide Tree
Realign Selected Sequences
Realign Selected Residue Range
Align Profile 2 to Profile 1
Align Profiles from Guide Trees
Align Sequences to Profile 1
Align Sequences to Profile 1 from Tree
Alignment Parameters
Iteration
Output Format Options
Set All Parameters to default
Danio_rerio
Homo_sapiens

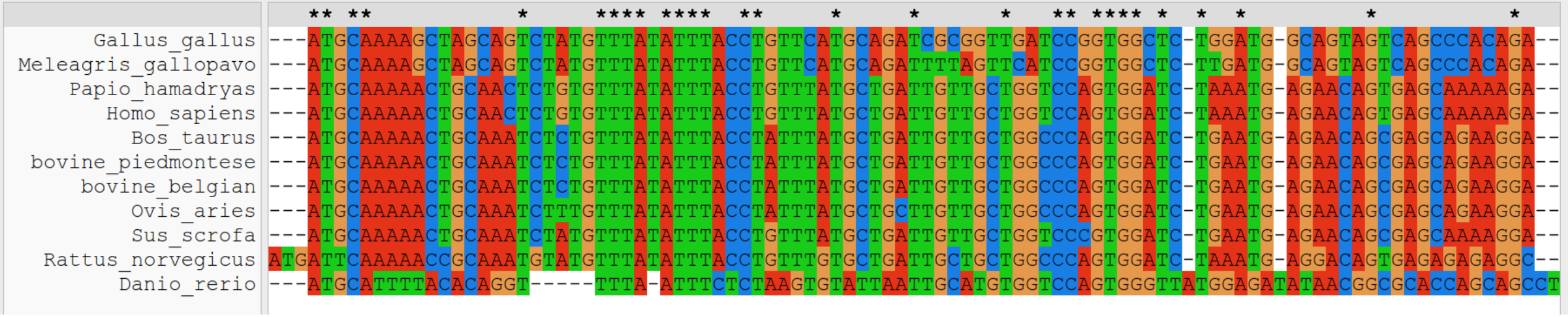

ClustalX 2.1
File Edit Alignment Trees Colors Quality Help
Mode: Multiple Alignment Mode Font: 24
Papio_hamadryas
Bos_taurus
bovine_belgian
bovine_piedmontese
Gallus_gallus
Ovis_aries
Sus_scrofa
Rattus_norvegicus
Meleagris_gallopavo
Danio_rerio
Homo_sapiens
Complete Alignment
Output Guide Tree:
/home/piero/DNA_SEQUENCES/myostatin_x_mod.dnd
Browse
Output Alignment Files
Fasta:
ENCES/myostatin_x_mod.fasta
Browse
OK
Cancel

La razza Belga e la miostatina

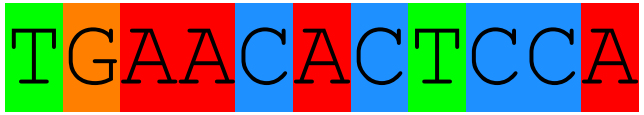
Prima

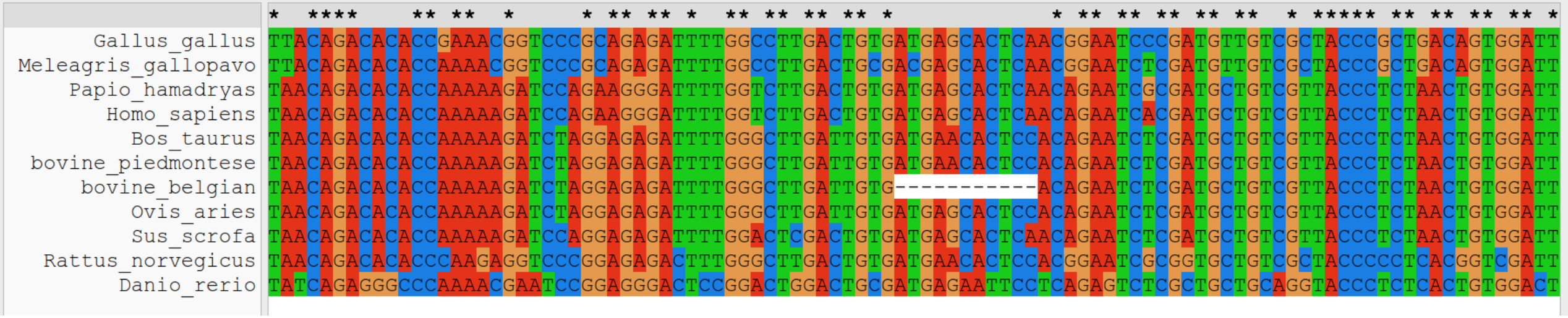


dopo

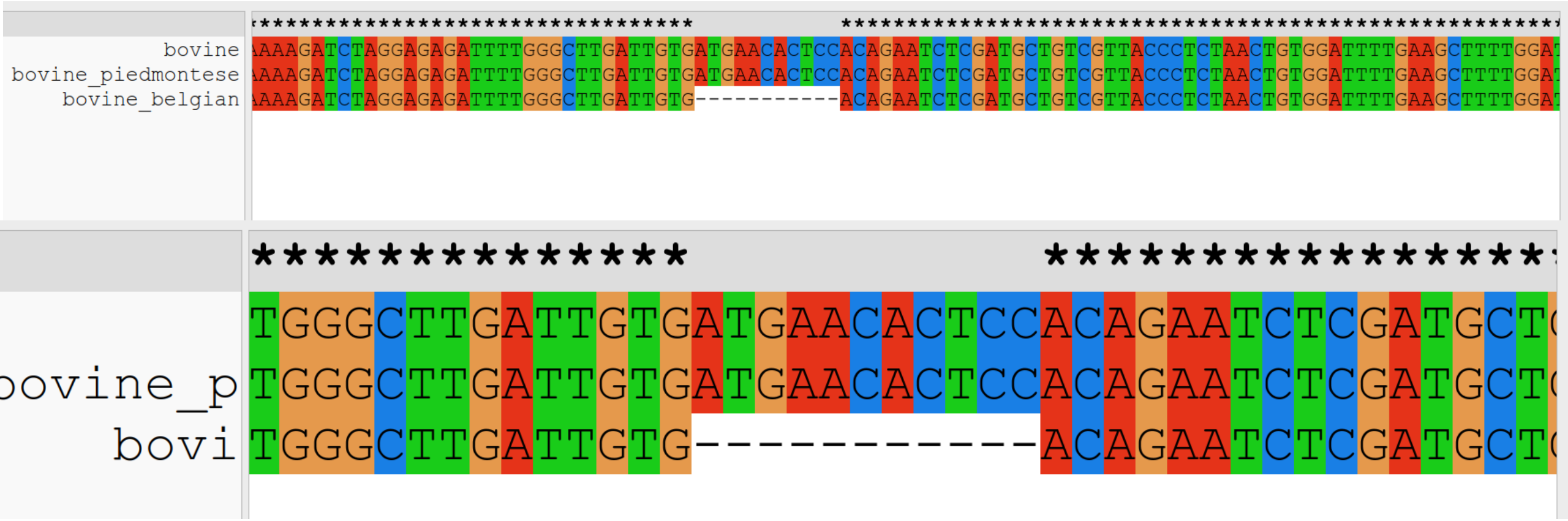


La razza Belga e la miostatina



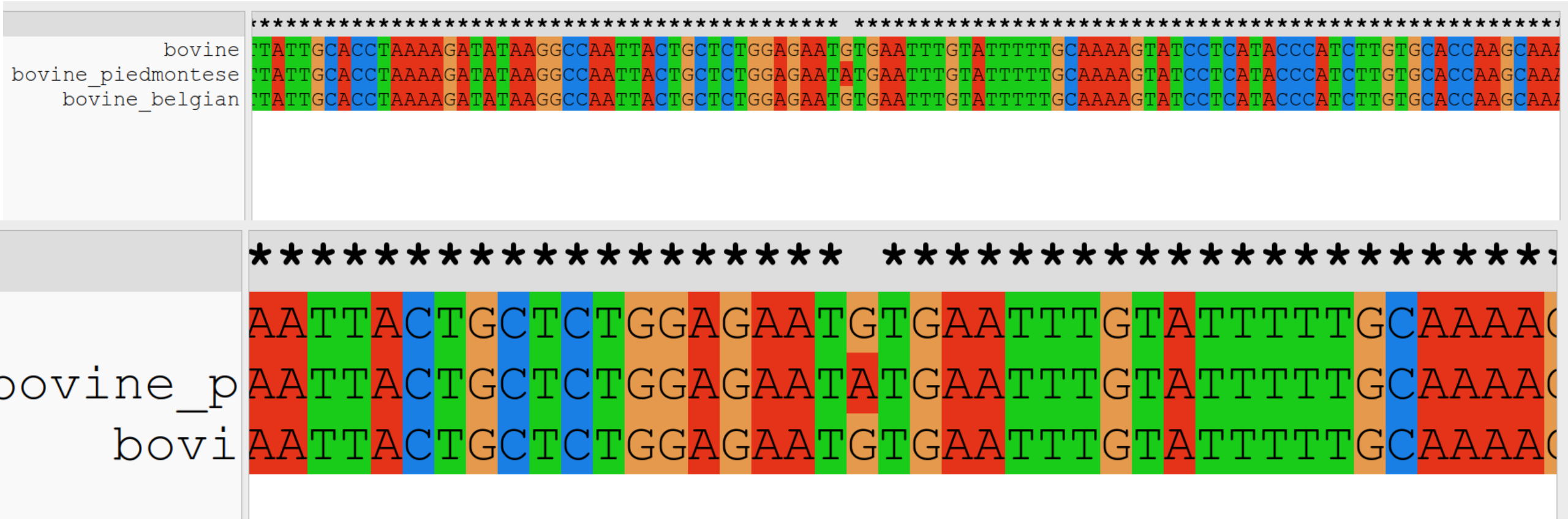


La razza Belga e la miostatina



La razza Piemontese e la miostatina

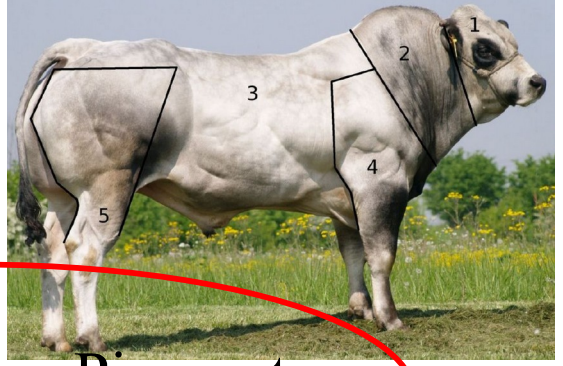
Transizione G → A

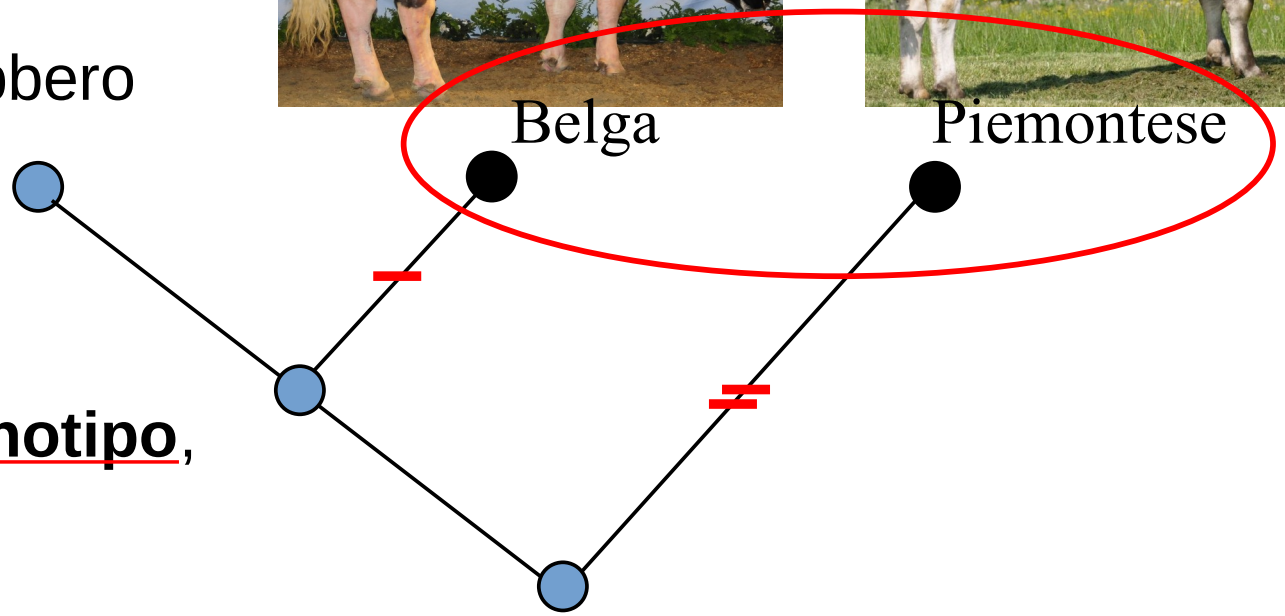


Le razze bovine Belga e Piemontese, se venissero classificate nella stessa categoria in base alla somiglianza fenotipica dovuta all'iperplasia muscolare congenita, formerebbero un **gruppo polifiletico**, ossia artificiale.

Ciò evidenzia i **limiti della classificazione basata sul fenotipo**, ossia sui caratteri morfologici.







github.com/privoira/ESERCITAZIONI/blob/main/myostatin_species_names.fas 120% Accedi

Files

main

Go to file t

- Apis_mellifera_mtDNA.fas
- BASAL_METABOLIC_RATE_DATA...
- E1_MIOSTATINA_10.12.2025.pdf
- E3_PESO_CORPOREO_VS_MET...
- E4_EARLY_HOLOCENE_SAHAR...
- E5_STUDI_DI_ASSOCIAZIONE_...
- INDICAZIONI_X_LA_STESURA_D...
- Papio_hamadryas_myostatin.fas
- README.md
- grafico_con_animali.jpg
- jane_1672_sm_supplementarymat...
- metabolismo_basale.xlsx
- myostatin_species_names.fas

ESERCITAZIONI / myostatin_species_names.fas

privoira Add files via upload — cdecaa6 · last month — History

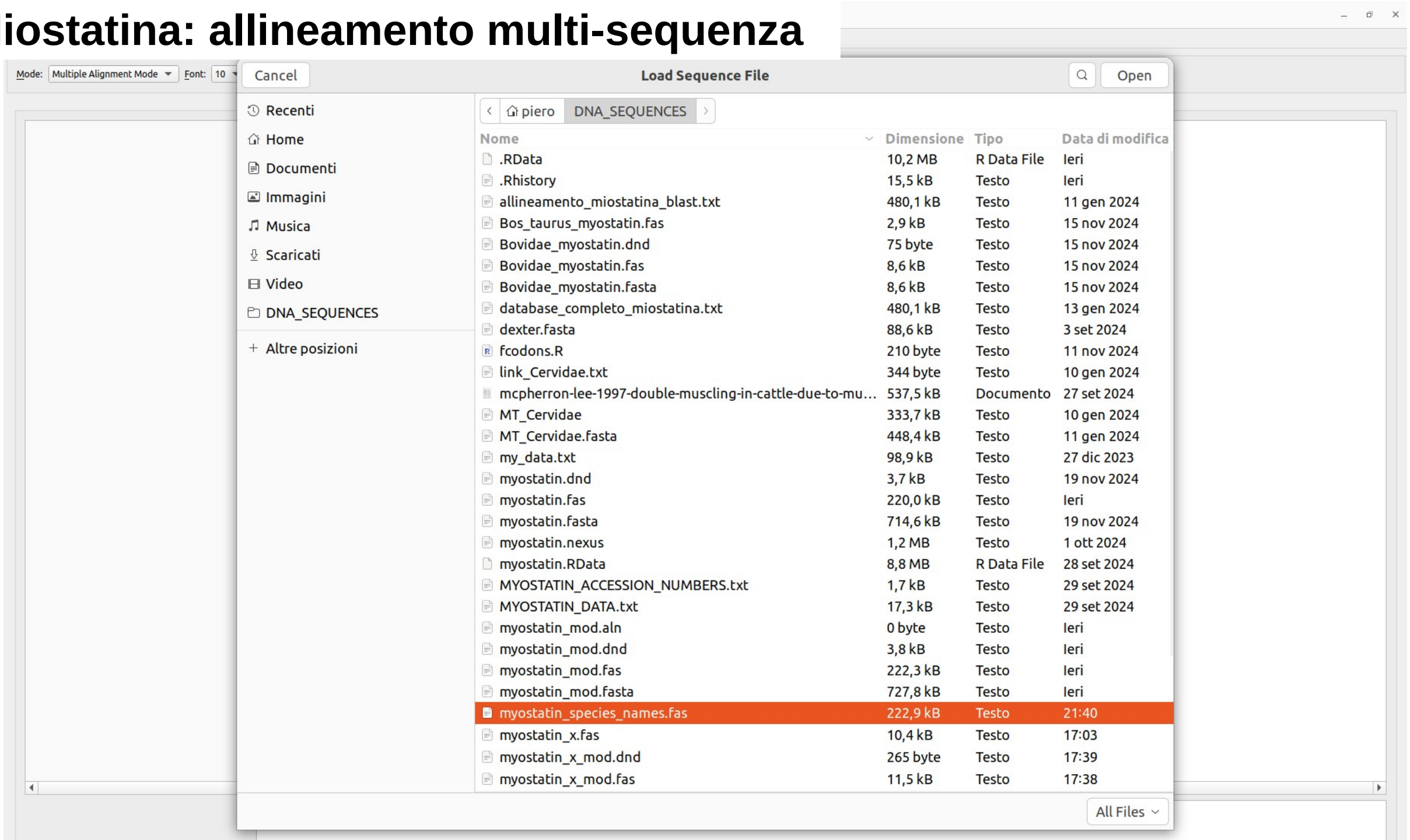
Download raw file

Code Blame — Raw

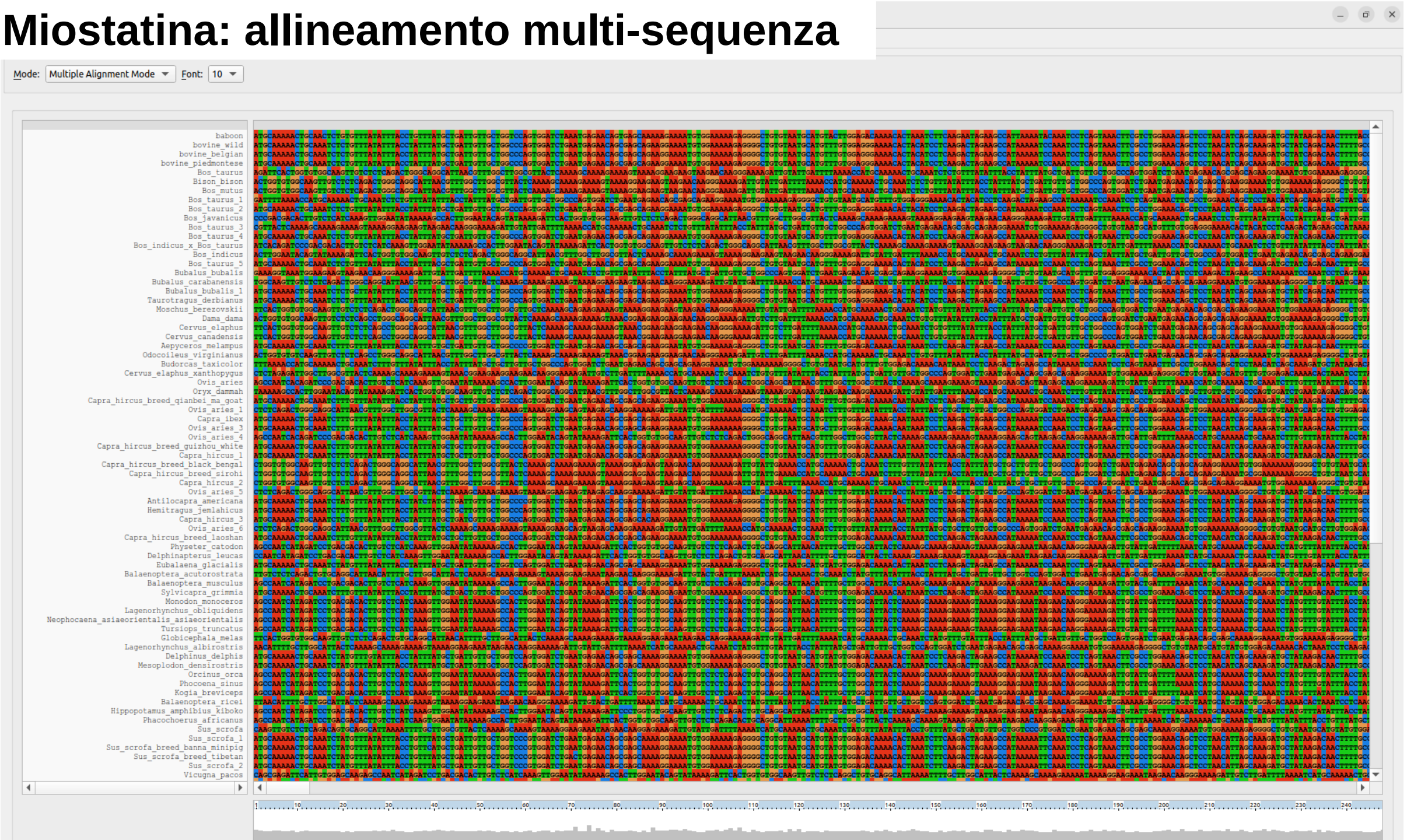


```
>baboon
atgcaaaaactgcaactctgtgtttatatttacctgtttatgctgattgttgctggtccagtggatctaaatgagaacag
tgagcaaaaagaaaatgtggaaaaagaggggctgtgtaatgcatgtacttggagacaaaacactaaatcttcaagaatag
aagccattaaaatacaaatcctcagtaaacttcgtctggaaacagctcctaacatcagcaaagatgctataagacaactt
ttacccaaagctcctccactccgggaactgattgatcagtatgatgtccagagggatgacagcagcgatggctctttgga
agatgacgattatcacgctacaacggaaacaatcattaccatgcctacagagtctgatttttaatgcaagtggatggaa
aacccaaatgttgcttctttaaatttagctctaaaatacaatacaataaagtggtaaaggcccaactatggatatatttg
agacccgtcgagactcctacaacagtgtttgtgcaaatcctgagactcatcaaacctatgaaagacggtacaaggtatac
tggaatccgatctctgaaacttgacatgaacccaggcactggtatttggcagagcattgatgtgaagacagtgttgcaaa
attggctcaaacaacctgaatccaacttaggcattgaaataaaagctttagatgagaatggtcatgatcttgctgtaacc
ttcccaggaccaggagaagatgggctgaatccctttttagaggtcaaggtaacagacacaccaaaaagatccagaaggga
ttttggtcttgactgtgatgagcactcaacagaatcgcgatgctgtcgttaccctctaactgtggattttgaagctcttg
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tatcctcatactcatctggtacaccaagcaaaccccagaggttcagcaggcccttgctgtactcccacaaagatgtctcc
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gctcatga
>bovine_wild
atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggcccagtggatctgaatgagaacag
cgagcagaaggaaaatgtggaaaaagaggggctgtgtaatgcatgtttgtggagggaaaacactacatcctcaagactag
aagccataaaaatccaaatcctcagtaaacttcgcctggaaacagctcctaacatcagcaaagatgctatcagacaactt
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```

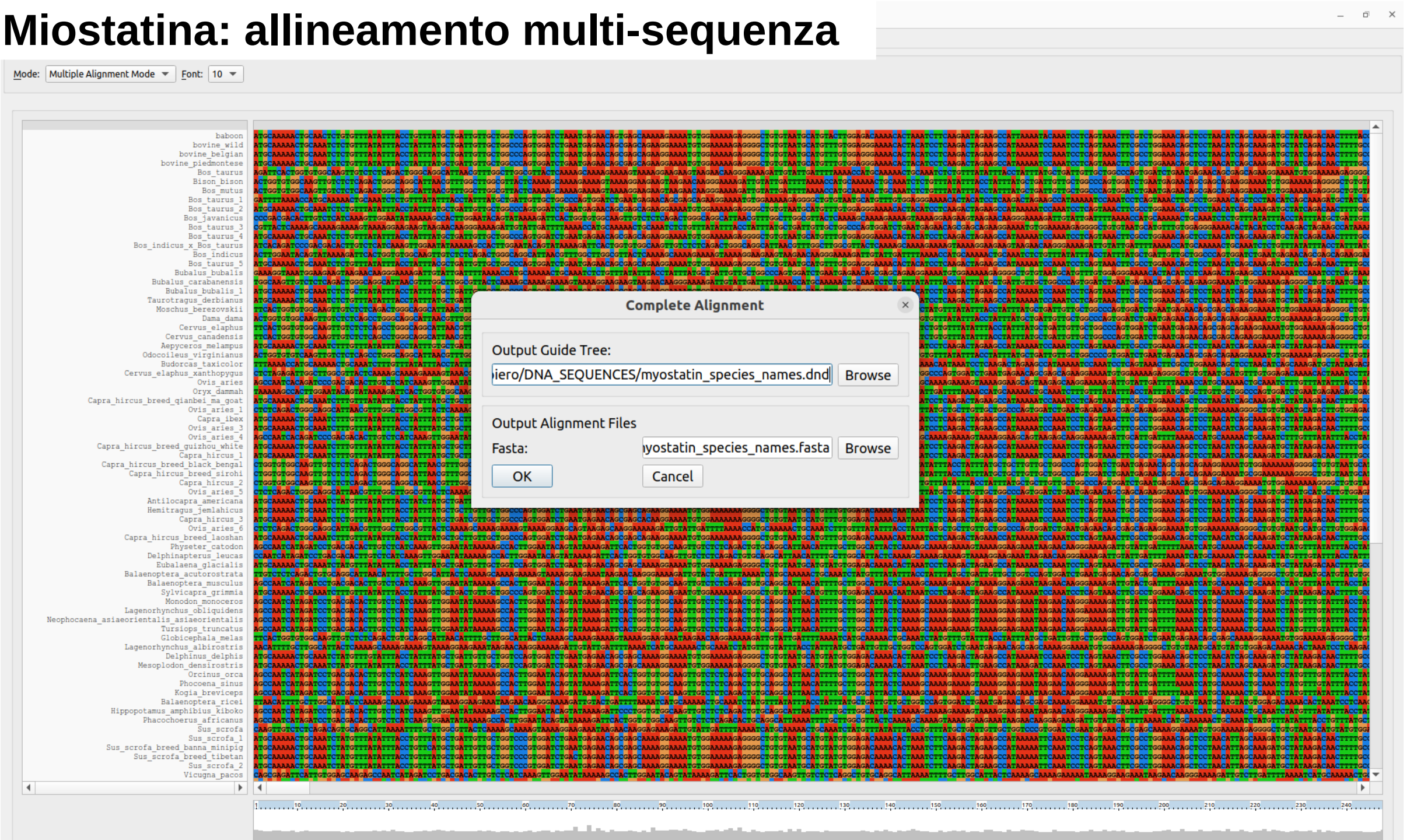
Miostatina: allineamento multi-sequenza



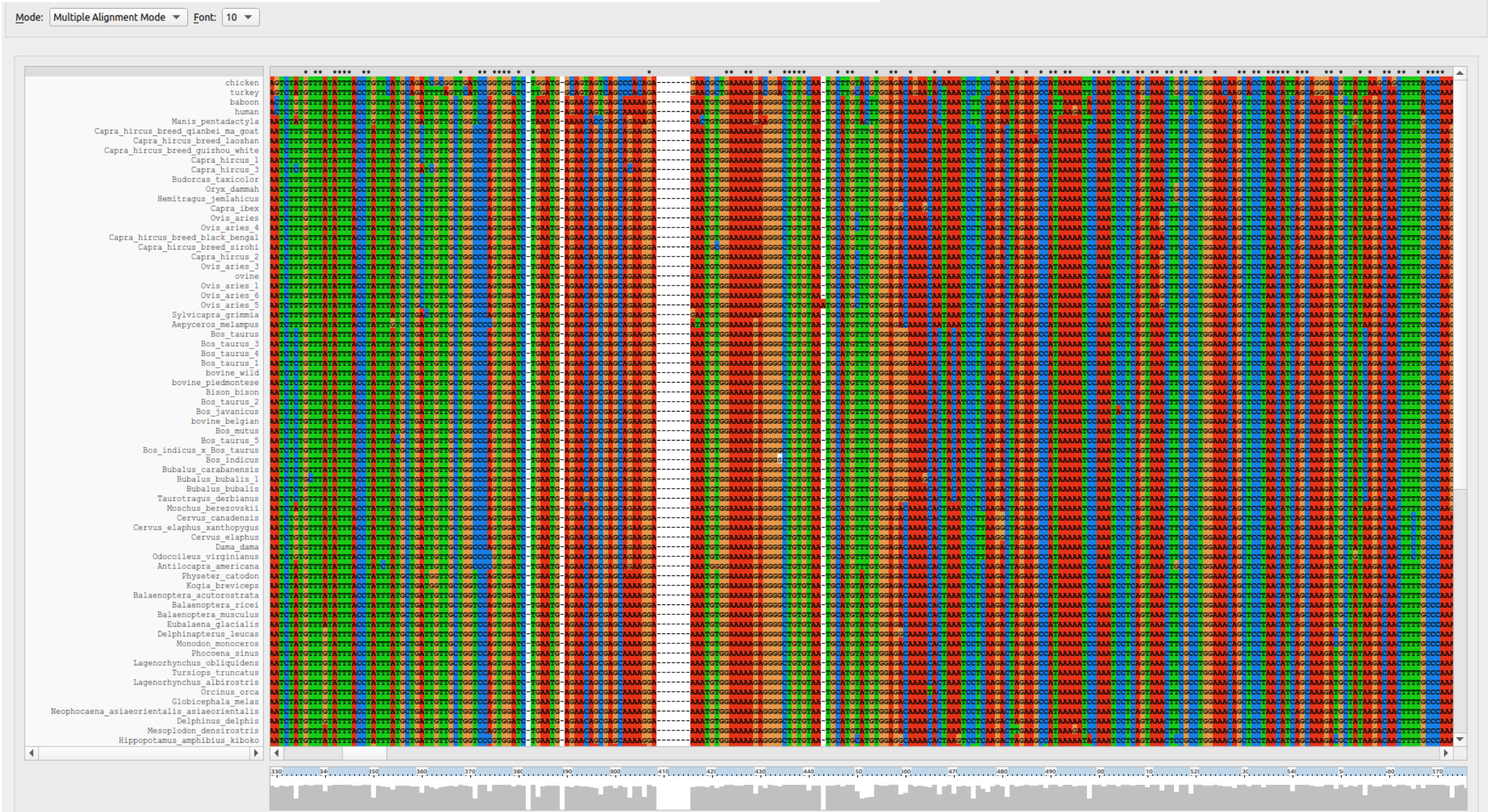
Miostatina: allineamento multi-sequenza



Miostatina: allineamento multi-sequenza



Miostatina: allineamento multi-sequenza



La razza Belga e la miostatina

```
> setwd('~/DNA_SEQUENCES/')
> load('myostatin.RData')
```
[sostituire **<myostatin.RData>** con il nome del file che avete creato (es.: **<piero.Rdata>**)]

```
> library(ape)
> library(seqinr)

> myostatin.fas = read.dna("myostatin_species_names.fas", format="fasta")
> myostatin.fasta = read.dna("myostatin_species_names.fasta", format="fasta")

> myostatin.fas

> myostatin.fasta
```

La razza Belga e la miostatina

```
> myostatin.fasta
111 DNa sequences in binary format stored in a matrix.

all sequences of same length: 6434

Labels:
chicken
turkey
baboon
human
Manis_pentadactyla
Capra_hircus_breed_qianbei_ma_goat
...

Base composition:
    a     c     g     t
a.337 a.181 a.189 a.293
(Total: 714.17 kb)
```

La razza Belga e la miostatina

```
> str(myostatin.fasta)
 'DNabin' raw [1:111, 1:6434] - - - - ...
 - attr(*, "dimnames")=List of 2
  ..$ : chr [1:111] "chicken" "turkey" "baboon" "human" ...
  ..$ : NULL

> myostatin.dist=dist.dna(myostatin.fasta,model="raw")
> myostatin.dist
```

	chicken	turkey	baboon
turkey	a.a26484a18		
baboon	a.15a684932	a.143378995	
human	a.149771689	a.142465753	a.aa8219178
Manis_pentadactyla	a.1652968a4	a.153424658	a.a54794521
Capra_hircus_breed_qianbei_ma_goat	a.175342466	a.168a3653a	a.a76712329
Capra_hircus_breed_laoshan	a.1762557a8	a.169863a14	a.a78538813
Capra_hircus_breed_guizhou_white	a.1762557a8	a.168949772	a.a77625571
Capra_hircus_1	a.174429224	a.167123288	a.a75799a87
Capra_hircus_3	a.173515982	a.168a3653a	a.a76712329
Budorcas_taxicolor	a.1762557a8	a.167123288	a.a739726a3

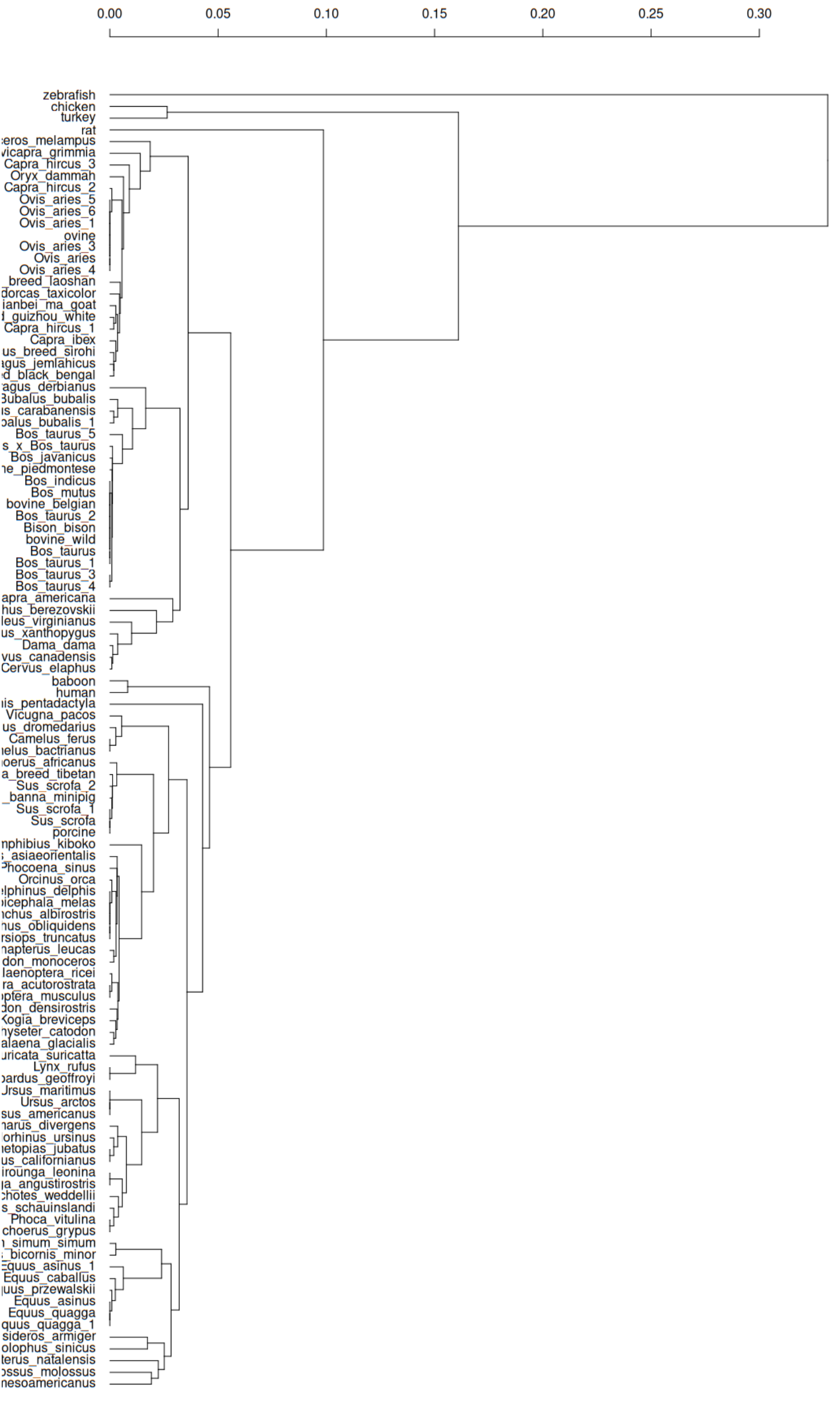
La razza Belga e la miostatina

```
> myostatin.tree = hclust(myostatin.dist,"average")

> plot(as.dendrogram(myostatin.tree))
# (orientamento verticale)

# oppure

> plot(as.phylo(myostatin.tree))
# (orientamento orizzontale)
```


0.00
0.05
0.10
0.15
0.20
0.25
0.30
zebrafish
chicken
turkey
rat
eros_melampus
vicapra_grimmia
Capra_hircus_3
Oryx_dammah
Capra_hircus_2
Ovis_aries_5
Ovis_aries_6
Ovis_aries_1
ovine
Ovis_aries_3
Ovis_aries
Ovis_aries_4
_breed_laoshan
dorcas_taxicolor
ianbei_ma_goat
_guizhou_white
Capra_hircus_1
Capra_ibex
us_breed_sirohi
agus_jemlahicus
d_black_bengal
agus_derbianus
Bubalus_bubalis
is_carabanensis
balus_bubalis_1
Bos_taurus_5
s_x_Bos_taurus
Bos_javanicus
ne_piedmontese
Bos_indicus
Bos_mutus
bovine_belgian
Bos_taurus_2
Bison_bison
bovine_wild
Bos_taurus
Bos_taurus_1
Bos_taurus_3
Bos_taurus_4
apra_americana
hus_berezovskii
leus_virginianus
us_xanthopygus
Dama_dama
vus_canadensis
Cervus_elaphus
baboon
human
is_pentadactyla
Vicugna_pacos
us_dromedarius
Camelus_ferus
elus_bactrianus
oerus_africanus
a_breed_tibetan
Sus_scrofa_2
_banna_minipig
Sus_scrofa_1
Sus_scrofa
porcine
nphibius_kiboko
_asiaeorientalis
Phocoena_sinus
Orcinus_orca
lphinus_delphis
icephala_melas
chus_albirostris
nus_obliquidens
rsiops_truncatus
napterus_leucas
don_monoceros
laenoptera_ricei
ra_acutorostrata
ptera_musculus
don_densirostris
Kogia_breviceps
nyseter_catodon
alaena_glacialis
uricata_suricatta
Lynx_rufus
ardus_geoffroyi
Ursus_maritimus
Ursus_arctos
sus_americanus
narus_divergens
orhinus_ursinus
etopias_jubatus
us_californianus
irounga_leonina
ia_angustirostris
chotes_weddellii
s_schauinslandi
Phoca_vitulina
choerus_grypus
_simum_simum
_bicornis_minor
Equus_asinus_1
Equus_caballus
uus_przewalskii
Equus_asinus
Equus_quagga
quus_quagga_1
sideros_armiger
olophus_sinicus
terus_natalensis
ossus_molossus
nesoamericanus

La razza Belga e la miostatina

- Scegliere 20 specie di Mammiferi, max 2 della stessa famiglia
- Uno dei taxa dev’essere ***Bos taurus***, un altro **Zebra fish**
- Costruire l'albero UPGMA basato sull’**analisi del DNA mitocondriale** (mtDNA)
- Come si distribuiscono i vari taxa sui rami dell'albero? Perchè?
  (rispondere sul quaderno)

Esempio

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NLM Catalog	0
PubMed	5
PubMed Central	277

Genes

Gene	1
GEO DataSets	0
GEO Profiles	0

Proteins

Conserved Domains	0
Identical Protein Groups	0
Protein	200
Protein Family Models	0
Structure	0

Genomes

Assembly / Genome NCBI Datasets	
BioCollections	0
BioProject	1
BioSample	0
Nucleotide	37
SRA	0

Clinical

ClinicalTrials.gov	0
ClinVar	2,786
dbGaP	0
dbSNP	0
dbVar	0
GTR	0

PubChem

BioAssays	0
Compounds	0
Pathways	0
Substances	0

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genomic DNA/RNA (37)
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INSDC (GenBank) (36)
RefSeq (1)
Customize ...

Sequence Type
Nucleotide (37)

Genetic compartments
Mitochondrion (1)

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Release date
Custom range...

Revision date
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Summary | 20 per page | Sort by Default order | Send to:

See ATPQ (**MITOCHONDRIAL**) ATP synthase D chain in the Gene database
mitochondrial reference sequences Transcript (2) Protein (2)

Items: 1 to 20 of 37

<< First < Prev Page 1 of 2 Next > Last >>

1. **Felis silvestris** mitochondrion, complete **genome**
16,437 bp circular DNA
Accession: OR095103.1 GI: 2531342991
Protein Taxonomy
GenBank FASTA Graphics

2. Fonsecaea multimorphosa strain CBS 980.96 Scaff3, whole **genome** shotgun sequence
444,857 bp linear DNA
Accession: LVCI01000003.1 GI: 1029155904
Assembly BioProject BioSample Protein Taxonomy
GenBank FASTA Graphics

3. Fonsecaea multimorphosa strain CBS 980.96 Scaff7, whole **genome** shotgun sequence
926,531 bp linear DNA
Accession: LVCI01000007.1 GI: 1029155199
Assembly BioProject BioSample Protein Taxonomy
GenBank FASTA Graphics

4. Fonsecaea multimorphosa strain CBS 980.96 Scaff12, whole **genome** shotgun sequence
688,162 bp linear DNA
Accession: LVCI01000012.1 GI: 1029154316
Assembly BioProject BioSample Protein Taxonomy
GenBank FASTA Graphics

Filters: Manage Filters

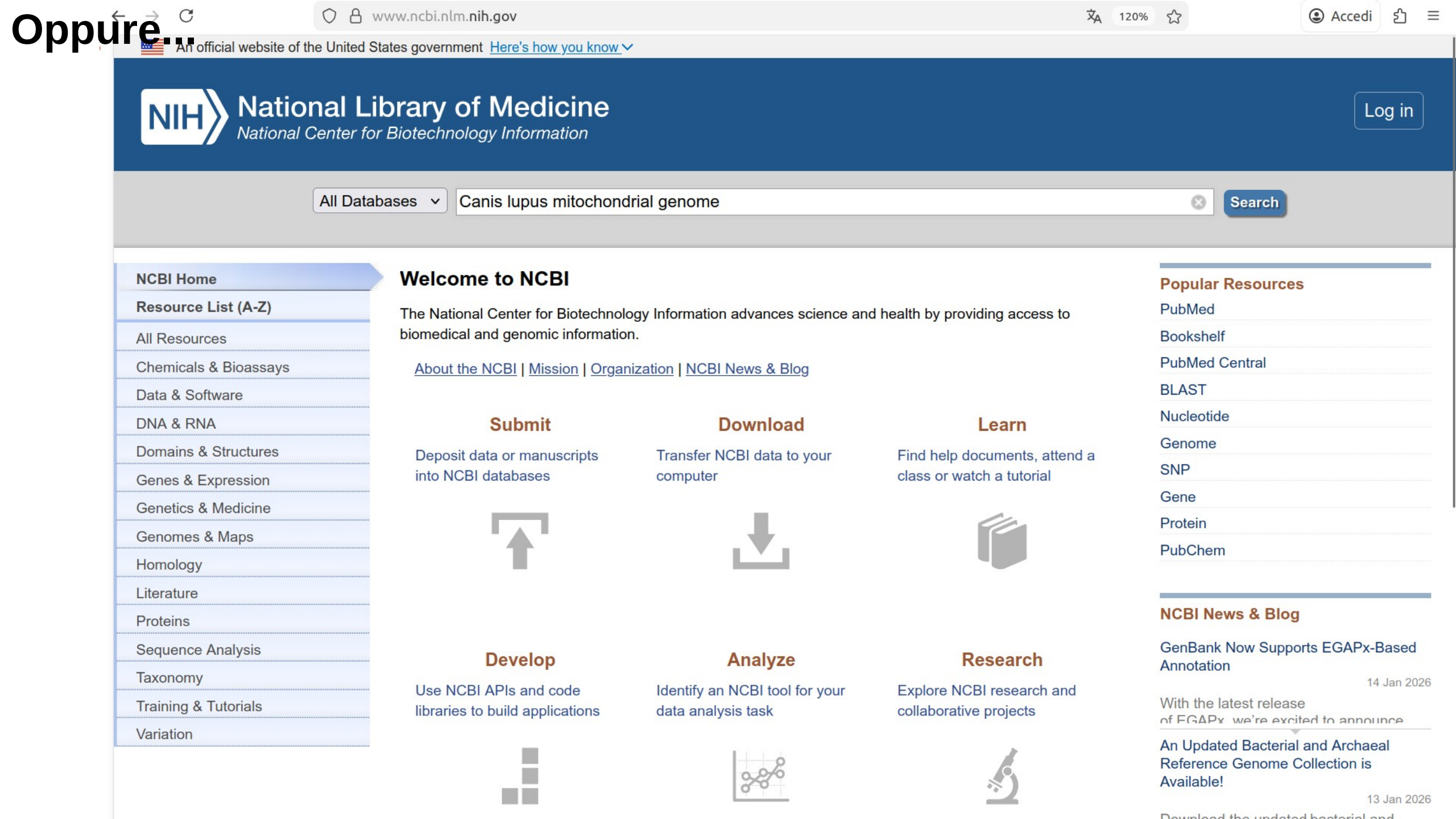
Results by taxon
Top Organisms [Tree]
Fonsecaea multimorphosa *(30)*
Aspergillus niger *(2)*
Aspergillus luchuensis *(2)*
Penicillium rubens Wisconsin 54-1255 *(1)*
Felis silvestris *(1)*
All other taxa *(1)*
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Find items

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("Felis silvestris"[Organism] OR Felis silvestris[All Fields]) AND mitochondrial[All Fields] AND genome[All Fields]
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Limited historical admixture between

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Wolf mitochondrial reference genome

Canis lupus (wolf)

RefSeq: NC_008092.1 Length: 16,729 bp circular

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Canis lupus mitochondrion, complete genome

NCBI Reference Sequence: NC_008092.1

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```
LOCUS       NC_008092              16729 bp    DNA     circular MAM 03-APR-2023
DEFINITION  Canis lupus mitochondrion, complete genome.
ACCESSION   NC_008092
VERSION     NC_008092.1
DBLINK      BioProject: PRJNA927338
KEYWORDS    RefSeq.
SOURCE      mitochondrion Canis lupus (wolf)
  ORGANISM  Canis lupus
            Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi;
            Mammalia; Eutheria; Laurasiatheria; Carnivora; Caniformia; Canidae;
            Canis.
REFERENCE   1  (bases 1 to 16729)
  AUTHORS   Bjornerfeldt,S., Webster,M.T. and Vila,C.
  TITLE     Relaxation of selective constraint on dog mitochondrial DNA
            following domestication
  JOURNAL   Genome Res. 16 (8), 990-994 (2006)
   PUBMED   16809672
REFERENCE   2  (bases 1 to 16729)
  CONSRTM   NCBI Genome Project
```

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Un po' scomodo...
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```
>NC_008092.1 Canis lupus mitochondrion, complete genome
GTTAATGTAGCTTAACTAATAAAGCAAGGCACTGAAAATGCCAAGATGAGTCGCACGACTCCATAAACAT
AAAGGTTTGGTCCTAGCCTTCCTATTAGTTTTTAGTAGACTTACACATGCAAGCCTCCACGCCCCAGTGA
GAATGCCCTTAAAATCACCAGTGATCTAAAGGAGCAGGTATCAAGCACACTCTTAAGTAGCTCATAACAC
CTTGCTAAGCCACACCCCCACGGGATACAGCAGTGATAGAAATTAAGCCATAAACGAAAGTTTGACTAAG
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CTACGGCGTAAAGCGTGTTCAAGATACTTTAACACTAAAGTTAAAACTTAACTAAGCCGTAAAAAGCTAC
AGTTATCATAAAATAAACCACGAAAGTGACTTTATAATAATCTGACTACACGATAGCTAAGACCCAAACT
GGGATTAGATACCCCACTATGCTTAGCCCTAAACATAGATAATTTTACAACAAAATAATTCGCCAGAGGA
CTACTAGCAATAGCTTAAAACTCAAAGGACTTGGCGGTGCTTTATATCCCTCTAGAGGAGCCTGTTCTAT
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CCTCAAAAGGTAGAACAGTAAGCACAATCATTTTACATAAAAAAGTTAGGTCAAGGTGTAACTTATGAGG
TGGGAAGAAATGGGCTACATTTTCTACCCAAGAACATTTCACGAATGTTTTTATGAAATTAAAAACTGAA
GGAGGATTTAGTAGTAAATTAAGAATAGAGAGCTTAATTGAATAGGGCCATGAAGCACGCACACACCGCC
CGTCACCCTCCTCAAGTAATAAGACACAACCATAACCATATTAACTTAACTAAAACACAAGAGGAGACAA
GTCGTAACAAGGTAAGCATACCGGAAGGTGTGCTTGGATTAATCAAAGTGTAGCTTAACTAAAGCGTCTG
GCCTACACCCAGAAGATTTCATTACTTATGGCCACTTTGAACAAAAGCTAGCCCAACTAACCCCAAACTT
AAGTATTACAGACACATAAAATAAAACATTTAGTTAAACAATAAAAGTATAGGAGATAGAAATTTTAATT
GGAGCGATAGAGATAGTACCGTAAGGGAATGATGAAAGACATCTTAACAGTATTAAACAGCAAAGATTAC
CCCTTCTACCTTTTGCATAATGAACTAGCCAGAAACAACTTAACAAAGAGAACTTAAGCTAAGCTCCCCG
AAACCAGACGAGCTACCCATAAACAATCTAAAAGGATCAACTCATCTATGTAGCAAAATAGTGAGAAGAT
TTGTGGGTAGAGGTGAAAAGCCTAACGAGCCTGGTGATAGCTGGTTACCCACAAACAGAATTTTAGTTCA
```

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Canis lupus mitochondrion, complete genome

NCBI Reference Sequence: NC_008092.1

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```
>NC_008092.1 Canis lupus mitochondrion, complete genome
GTTAATGTAGCTTAACTAATAAAGCAAGGCACTGAAAATGCCAAGATGAGTCGCACGACTCCATAAACAT
AAAGGTTTGGTCCTAGCCTTCCTATTAGTTTTTAGTAGACTTACACATGCAAGCCTCCACGCCCCAGTGA
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CTTGCTAAGCCACACCCCCACGGGATACAGCAGTGATAGAAATTAAGCCATAAACGAAAGTTTGACTAAG
CCATACTAAATAGGGTTGGTAAATTTCGTGCCAGCCACCGCGGTCATACGATTAACCCAAACTAATAGGC
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GTCGTAACAAGGTAAGCATACCGGAAGGTGTGCTTGGATTAATCAAAGTGTAGCTTAACTAAAGCGTCTG
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AAGTATTACAGACACATAAAATAAAACATTTAGTTAAACAATAAAAGTATAGGAGATAGAAATTTTAATT
GGAGCGATAGAGATAGTACCGTAAGGGAATGATGAAAGACATCTTAACAGTATTAAACAGCAAAGATTAC
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AAACCAGACGAGCTACCCATAAACAATCTAAAAGGATCAACTCATCTATGTAGCAAAATAGTGAGAAGAT
TTGTGGGTAGAGGTGAAAAGCCTAACGAGCCTGGTGATAGCTGGTTACCCACAAACAGAATTTTAGTTCA
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Canis lupus mitochondrion, complete genome

NCBI Reference Sequence: NC_008092.1

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```
>NC_008092.1 Canis lupus mitochondrion, complete genome
GTTAATGTAGCTTAACTAATAAAGCAAGGCACTGAAAATGCCAAGATGAGTCGCACGACTCCATAAACAT
AAAGGTTTGGTCCTAGCCTTCCTATTAGTTTTTAGTAGACTTACACATGCAAGCCTCCACGCCCCAGTGA
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AGTTATCATAAAATAAACCACGAAAGTGACTTTATAATAATCTGACTACACGATAGCTAAGACCCAAACT
GGGATTAGATACCCCACTATGCTTAGCCCTAAACATAGATAATTTTACAACAAAATAATTCGCCAGAGGA
CTACTAGCAATAGCTTAAAACTCAAAGGACTTGGCGGTGCTTTATATCCCTCTAGAGGAGCCTGTTCTAT
AATCGATAAACCCCGATAAACCTCACCACCTTTCGCTAATTCAGTCTATATACCGCCATCTTCAGCAAAC
CCTCAAAAGGTAGAACAGTAAGCACAATCATTTTACATAAAAAAGTTAGGTCAAGGTGTAACTTATGAGG
TGGGAAGAAATGGGCTACATTTTCTACCCAAGAACATTTCACGAATGTTTTTATGAAATTAAAAACTGAA
GGAGGATTTAGTAGTAAATTAAGAATAGAGAGCTTAATTGAATAGGGCCATGAAGCACGCACACACCGCC
CGTCACCCTCCTCAAGTAATAAGACACAACCATAACCATATTAACTTAACTAAAACACAAGAGGAGACAA
GTCGTAACAAGGTAAGCATACCGGAAGGTGTGCTTGGATTAATCAAAGTGTAGCTTAACTAAAGCGTCTG
GCCTACACCCAGAAGATTTCATTACTTATGGCCACTTTGAACAAAAGCTAGCCCAACTAACCCCAAACTT
AAGTATTACAGACACATAAAATAAAACATTTAGTTAAACAATAAAAGTATAGGAGATAGAAATTTTAATT
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CCCTTCTACCTTTTGCATAATGAACTAGCCAGAAACAACTTAACAAAGAGAACTTAAGCTAAGCTCCCCG
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TTGTGGGTAGAGGTGAAAAGCCTAACGAGCCTGGTGATAGCTGGTTACCCACAAACAGAATTTTAGTTCA
```

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```
>NC_008092.1 Canis lupus mitochondrion, complete genome
GTTAATGTAGCTTAACTAATAAAGCAAGGCACTGAAAATGCCAAGATGAGTCGCACGACTCCATAAACAT
AAAGGTTTGGTCCTAGCCTTCCTATTAGTTTTTAGTAGACTTACACATGCAAGCCTCCACGCCCCAGTGA
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AGTTATCATAAAATAAACCACGAAAGTGACTTTATAATAATCTGACTACACGATAGCTAAGACCCAAACT
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AATCGATAAACCCCGATAAACCTCACCACCTTTCGCTAATTCAGTCTATATACCGCCATCTTCAGCAAAC
CCTCAAAAGGTAGAACAGTAAGCACAATCATTTTACATAAAAAAGTTAGGTCAAGGTGTAACTTATGAGG
TGGGAAGAAATGGGCTACATTTTCTACCCAAGAACATTTCACGAATGTTTTTATGAAATTAAAAACTGAA
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TTGTGGGTAGAGGTGAAAAGCCTAACGAGCCTGGTGATAGCTGGTTACCCACAAACAGAATTTTAGTTCA
```

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LinkOut to external resources

```
> library(ape)
> setwd('~/MATERIALE_DIDATTICO_A.S._2025-2026/ESERCITAZIONI/MIOSTATINA/')
# sostituire il PATH
# <~/MATERIALE_DIDATTICO_A.S._2025-2026/ESERCITAZIONI/MIOSTATINA/>
# con un percorso compatibile con il proprio albero di cartelle
# Es. <~/DNA_SEQUENCES/>

> myostatin.fasta = read.dna("myostatin_5_species_100_bp.fasta",
format="fasta")
> X = myostatin.fasta
> myostatin.dist = dist.dna(X, model="raw")
```

raw: This is simply the proportion or the number of sites that differ between each pair of sequences.

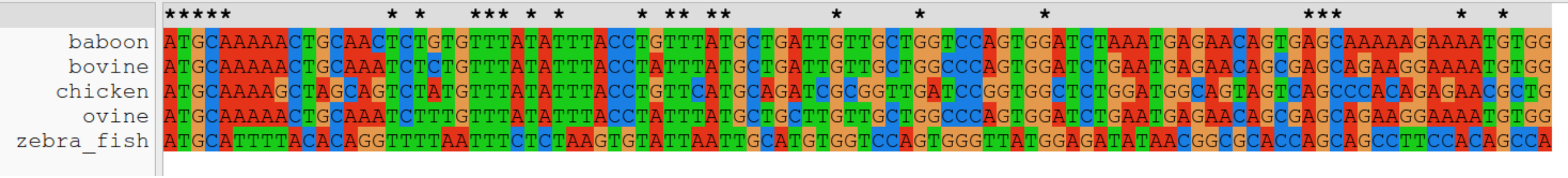
```
> library(ape)
> setwd('~/MATERIALE_DIDATTICO_A.S._2024-2025/ESERCITAZIONI/MIOSTATINA/')
> X = read.dna("myostatin_5_species_100_bp.fasta", format="fasta",
as.character = FALSE)
> str(X)
 'DNAbin' raw [1:5, 1:100] a a a a ...
 - attr(*, "dimnames")=List of 2
  ..$ : chr [1:5] "baboon" "bovine" "chicken" "ovine" ...
  ..$ : NULL
> X
5 DNA sequences in binary format stored in a matrix.

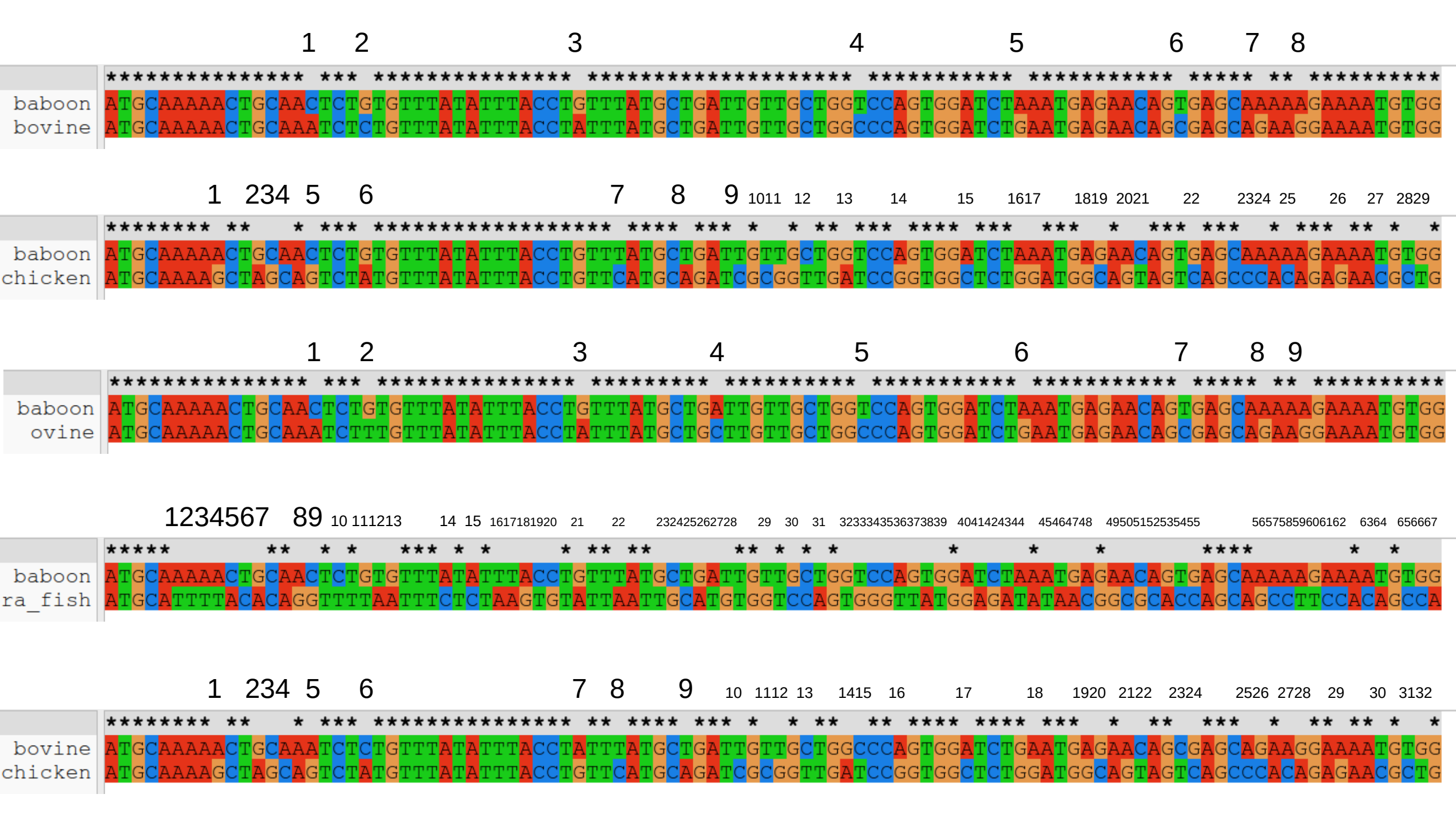
All sequences of same length: 100

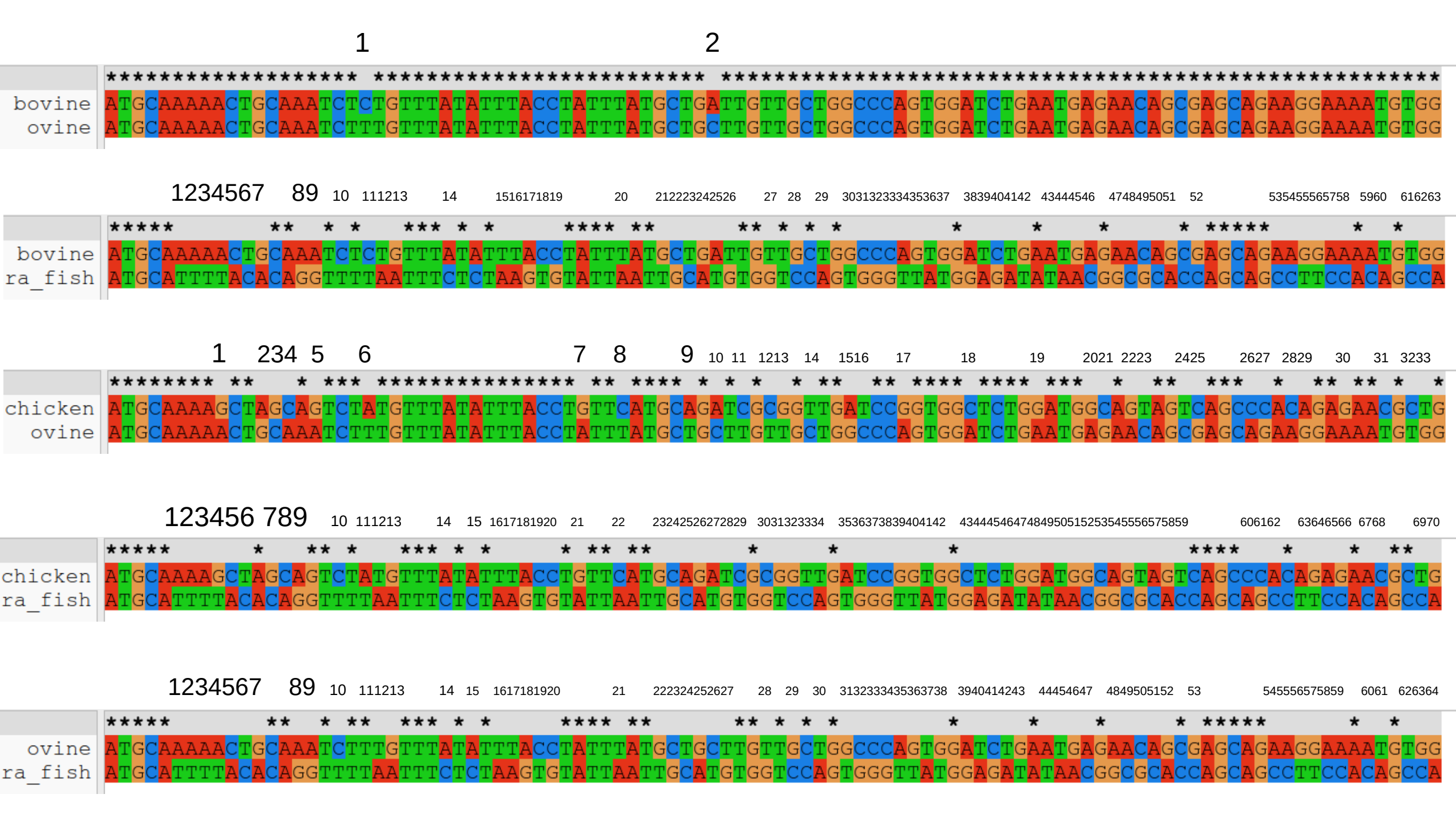
Labels:
baboon
bovine
chicken
ovine
zebra_fish

Base composition:
    a     c     g     t
0.286 0.178 0.240 0.296
(Total: 500 bases)
```

```
> myostatin.dist = dist.dna(X, model="raw")
> myostatin.dist
           baboon bovine chicken ovine
bovine       0.08
chicken      0.29   0.32
ovine        0.09   0.02    0.33
zebra_fish   0.67   0.64    0.70  0.63
```




baboon
bovine
baboon
chicken
baboon
ovine
baboon
ra_fish
bovine
chicken


1
2
bovine
ovine
bovine
ra_fish
chicken
ovine
chicken
ra_fish
ovine
ra_fish

```
> myostatin.UPGMA = hclust(myostatin.dist, "average")
```
simple UPGMA

```
> myostatin.UPGMA

Call:
hclust(d = myostatin.dist, method = "average")

Cluster method   : average
Distance         : raw
Number of objects: 5
```

La funzione **`hclust`** effettua una *cluster analysis* gerarchica utilizzando, in questo caso, l'algoritmo **UPGMA** (*Unweighted Pair Group Method with Arithmetic Mean*).

L'idea alla base della *cluster analysis* gerarchica è di rappresentare graficamente quali elementi (singoli esemplari di una stessa o di specie diverse, sequenze di DNA ecc.) di un certo campione sono più simili tra di loro, raggruppandoli a due a due sullo stesso ramo dell'albero.

Gli oggetti che sono più diversi fra loro vengono collocati in rami differenti.

```
> myostatin.dist
           baboon bovine chicken ovine
bovine       0.08
chicken      0.29   0.32
ovine        0.09   0.02    0.33
zebra_fish   0.67   0.64    0.70  0.63
```

	bab	bov	gal	ovi	zeb
bab	0	8	29	9	67
bov	8	0	32	2	64
gal	29	32	0	33	70
ovi	9	**2**	33	0	63
zeb	67	64	70	63	0

Matrice **D1** di distanze genetiche
(n° di nucleotidi differenti su 100)
fra coppie di specie

Il bovino e la pecora sono le specie tra le quali la D1 è <, quindi le mettiamo insieme

Quindi calcoliamo la distanza media tra bov-ovi e ciascuna delle altre specie

D2((bov,ovi),bab) = [D1(bov,bab)+D1(ovi,bab)]/2 = (8+9)/2 = 8,5
D2((bov,ovi),gal) = [D1(bov,gal)+D1(ovi,gal)]/2 = (32+33)/2 = 32,5
D2((bov,ovi),zeb) = [D1(bov,zeb)+D1(ovi,zeb)]/2 = (64+63)/2 = 63,5

	bab	bov-ovi	gal	zeb
bab	0	8,5	29	67
bov-ovi	**8,5**	0	32,5	63,5
gal	29	32,5	0	70
zeb	67	63,5	70	0

	(bov-ovi)bab	gal	zeb
(bov-ovi)bab	0	**30,75**	65,25
gal	30,75	0	70
zeb	65,25	70	0

D3(((bov,ovi),bab),gal) = [D2((bov,ovi),gal)+(D2(bab,gal)]/2 = (32,5+29)/2 = 30,75
D3(((bov,ovi),bab),zeb) = [D2((bov,ovi),zeb)+(D2(bab,zeb)]/2 = (63,5+67)/2 = 65,25

	(bov-ovi)bab	gal	zeb
(bov-ovi)bab	0	**30,75**	65,25
gal	30,75	0	70
zeb	65,25	70	0

D4((((bov,ovi),bab),gal),zeb) = [D3(((bov,ovi),bab),zeb)+D3(gal,zeb)]/2 = (65,25+70)/2 = 67,625

	((bov-ovi)bab)ga	zeb
((bov-ovi)bab)gal	0	67,625
zeb	67,625	0

```
> plot(myostatin.UPGMA, main = 'UPGMA distances')
```

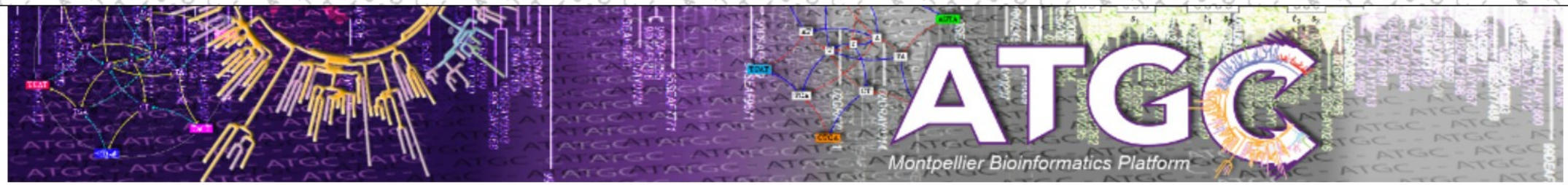


Scelta del modello di sostituzione nucleotidica

L'analisi filogenetica più semplice si basa sul calcolo delle distanze fra coppie di sequenze nucleotidiche, le quali diventano gli ingressi di una matrice che, tramite algoritmi di clusterizzazione, viene convertita in un albero filogenetico.
La distanza fra due sequenze è definita come numero atteso di sostituzioni nucleotidiche per sito (generalmente < 1) e può essere espressa come semplice proporzione dei siti variabili o distanza **p**.

```
> myostatin.dist
           baboon bovine chicken ovine
bovine       0.08
chicken      0.29   0.32
ovine        0.09   0.02    0.33
zebra_fish   0.67   0.64    0.70  0.63
```

www.atgc-montpellier.fr/phyml/download.php
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PhyML 3.0: new algorithms, methods and utilities

Please cite:
"New Algorithms and Methods to Estimate Maximum-Likelihood Phylogenies: Assessing the Performance of PhyML 3.0."
Guindon S., Dufayard J.F., Lefort V., Anisimova M., Hordijk W., Gascuel O.
Systematic Biology, 59(3):307-21, 2010.

Get the latest PhyML releases from GitHub.

Download PhyML binaries

Click **here** to [illegible] binaries
The binaries you have compiled on various Mac, Windows or LINUX systems are much welcome (**Stéphane Guindon**) and will be displayed on this web page.

Contact: **Webmaster**, LIRMM.

Il formato PHYLIP interleaved

```
5 100
baboon
atgcaaaaactgcaactctgtgtttatatttacctgtttatgctgattgttgctggtccagtggatctaaatgagaacagtgagcaaaaagaaaatgtgg
bovine
atgcaaaaactgcaaatctctgtttatatttacctatttatgctgattgttgctggcccagtggatctgaatgagaacagcgagcagaaggaaaatgtgg
chicken
atgcaaaagctagcagtctatgtttatatttacctgttcatgcagatcgcggttgatccggtggctctggatggcagtagtcagcccacagagaacgctg
ovine
atgcaaaaactgcaaatctttgtttatatttacctatttatgctgcttgttgctggcccagtggatctgaatgagaacagcgagcagaaggaaaatgtgg
zebrafish
atgcattttacacaggttttaatttctctaagtgtattaattgcatgtggtccagtgggttatggagatataacggcgcaccagcagccttccacagcca
```

```
> substitution_model_test = phymltest('myostatin_5_species_100_bp.phy', format
= "interleaved", itree = NULL, exclude = NULL, execname = '/home/piero/PhyML-
3.1/PhyML-3.1_linux64', append = TRUE)
```

```
oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo
                                  ---  PhyML 20120412  ---
                              http://www.atgc-montpellier.fr/phyml
                           Copyright CNRS - Universite Montpellier II
 oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo

. Sequence filename:            myostatin_5_species_100_bp.phy
. Data set:                 #1
. Tree topology search :        NNIs
. Initial tree:             BioNJ
. Model of nucleotides substitution:     GTR
. Number of taxa:           5
. Log-likelihood:           -402.43191
. Unconstrained likelihood:         -343.80642
. Parsimony:                100
. Tree size:                4.55423
. Nucleotides frequencies:
  - f(A)= 0.28600
  - f(C)= 0.17800
  - f(G)= 0.24000
  - f(T)= 0.29600
. GTR relative rate parameters :
  A <-> C   11.85779
  A <-> G   14.05957
  A <-> T    0.00993
  C <-> G    8.42143
  C <-> T   13.46606
  G <-> T    1.00000

. Instantaneous rate matrix :
  [A---------C---------G---------T------]
  -1.01259   0.38945   0.62260   0.00054
   0.62574  -1.73412   0.37293   0.73546
   0.74193   0.27659  -1.07313   0.05462
   0.00052   0.44227   0.04428  -0.48708
```

L’output del test

```
> substitution_model_test
          nb.free.para    loglik      AIC
JC69                 1 -414.3124 830.6248
JC69+I               2 -414.1910 832.3821
JC69+G               2 -413.8407 831.6814
JC69+I+G             3 -413.8313 833.6625
K80                  2 -411.0496 826.0993
K80+I                3 -408.3180 822.6359
K80+G                3 -408.8849 823.7699
K80+I+G              4 -407.9957 823.9915
F81                  4 -414.1411 836.2823
F81+I                5 -414.0996 838.1992
F81+G                5 -413.7568 837.5136
F81+I+G              6 -413.7486 839.4972
F84                  5 -409.8965 829.7931
F84+I                6 -407.9673 827.9346
F84+G                6 -408.0924 828.1848
F84+I+G              7 -407.5650 829.1301
HKY85                5 -409.8806 829.7612
HKY85+I              6 -407.9661 827.9323
HKY85+G              6 -408.0732 828.1464
HKY85+I+G            7 -407.5730 829.1460
TN93                 6 -409.8804 831.7607
TN93+I               7 -407.9637 829.9274
TN93+G               7 -408.0700 830.1400
TN93+I+G             8 -407.5648 831.1297
GTR                  9 -402.4319 822.8638
GTR+I               10 -401.2392 822.4784
GTR+G               10 -400.2603 820.5205
GTR+I+G             11 -400.2379 822.4758
>
```

ll file <myostatin_5_species_100_bp.phy_phyml_tree.txt>

```
(bovine:0.00000202,(ovine:0.00002963,zebrafish:9.99805710)0.000000:0.02049504,
(baboon:0.00008628,chicken:0.50886609)0.842000:0.09067509);
(baboon:0.01048862,(bovine:0.00000001,(zebrafish:1.37441488,ovine:0.00000508):0.02026739):0.07447686,chicken:0.35829670);
(baboon:0.00972686,chicken:0.36497752,(bovine:0.00000118,(ovine:0.00000029,zebrafish:1.54101437):0.02030498):0.07560156);
(baboon:0.00007584,chicken:0.44219573,(ovine:0.02056245,(bovine:0.00000001,zebrafish:3.54999086):0.00000001):0.08826723);
(baboon:0.00000910,chicken:0.45531080,(bovine:0.00000001,(zebrafish:4.16221975,ovine:0.00001874):0.02059383):0.08887950);
(bovine:0.00000099,(ovine:0.02028732,zebrafish:3.09237767):0.00003742,(baboon:0.01122146,chicken:0.36941408):0.07398107);
(zebrafish:8.82019763,(bovine:0.00000175,ovine:0.02039573):0.02277632,(chicken:0.39408351,baboon:0.01090682):0.05227845);
(bovine:0.00000099,(ovine:0.02044652,zebrafish:10.00000000):0.00002146,(baboon:0.00002629,chicken:0.50356626):0.08929782);
(chicken:0.43223177,baboon:0.00713230,(bovine:0.00000126,(zebrafish:8.89223736,ovine:0.01973263):0.00074093):0.08047846);
(baboon:0.01115834,chicken:0.36143405,(bovine:0.00000181,(zebrafish:1.42016394,ovine:0.00000268):0.02033915):0.07448850);
(bovine:0.00000091,(chicken:0.36579093,baboon:0.01074355):0.07520598,(zebrafish:1.52463974,ovine:0.00000001):0.02037077);
(bovine:0.00000117,(zebrafish:3.58786252,(chicken:0.44228493,baboon:0.00171289):0.08712725):0.00000872,ovine:0.02063859);
(baboon:0.00000824,chicken:0.46731497,(bovine:0.00000001,(zebrafish:4.82711651,ovine:0.00000756):0.02070614):0.08999625);
(bovine:0.00000101,(ovine:0.02042293,zebrafish:3.77229889):0.00003150,(baboon:0.01338516,chicken:0.37881952):0.07317730);
(zebrafish:8.98214792,(bovine:0.00000182,ovine:0.02045418):0.02675688,(chicken:0.39647730,baboon:0.01296603):0.04725382);
(bovine:0.00000204,(ovine:0.00002615,zebrafish:9.99812034):0.02054687,(baboon:0.00002661,chicken:0.50940308):0.09100487);
(chicken:0.44286719,zebrafish:9.08352240,(baboon:0.00942811,(bovine:0.00000132,ovine:0.02057252):0.07969971):0.00014779);
(bovine:0.00000099,(ovine:0.02039898,zebrafish:3.70535167):0.00003159,(baboon:0.01300483,chicken:0.37844433):0.07338886);
(zebrafish:8.98429384,(bovine:0.00000180,ovine:0.02042441):0.02673440,(chicken:0.39638088,baboon:0.01260347):0.04745109);
(bovine:0.00000202,(ovine:0.00002963,zebrafish:9.99805710):0.02049504,(baboon:0.00008628,chicken:0.50886609):0.09067509);
(chicken:0.44246753,zebrafish:9.08528623,(baboon:0.00898618,(bovine:0.00000130,ovine:0.02053515):0.07990025):0.00016885);
(bovine:0.00000099,(ovine:0.02039408,zebrafish:3.68224905):0.00003166,(baboon:0.01293915,chicken:0.37829051):0.07342438);
(zebrafish:8.98269671,(bovine:0.00000180,ovine:0.02044070):0.03061041,(chicken:0.39640554,baboon:0.01278578):0.04346390);
(bovine:0.00000205,(ovine:0.00003700,zebrafish:9.99683005):0.02045292,(baboon:0.00008704,chicken:0.50879952):0.09053445);
(chicken:0.44280285,zebrafish:9.08381557,(baboon:0.00936629,(bovine:0.00000130,ovine:0.02056699):0.07972586):0.00014726);
(bovine:0.00000001,(zebrafish:4.05071318,(baboon:0.00593667,chicken:0.39542302):0.08161689):0.00000004,ovine:0.02053827);
(zebrafish:4.34749500,(bovine:0.00000057,(chicken:0.40630281,baboon:0.00316606):0.08445039):0.02051856,ovine:0.00000251);
(zebrafish:10.00000000,(bovine:0.00000056,(baboon:0.00001018,chicken:0.54408882):0.09498755):0.02116598,ovine:0.00001086);
(baboon:0.00000993,chicken:0.53213267,(bovine:0.00000001,(zebrafish:9.76620951,ovine:0.00000001):0.02114857):0.09443561);
```

Scelta del modello di sostituzione nucleotidica

```
> r = '(baboon:0.00000993,chicken:0.53213267,(bovine:0.00000001,
(zebrafish:9.76620951,ovine:0.00000001):0.02114857):0.09443561);'
> albero = read.tree(text = r)
> albero

Phylogenetic tree with 5 tips and 3 internal nodes.

Tip labels:
  baboon, chicken, bovine, zebrafish, ovine

Unrooted; includes branch lengths.
> plot(albero, main = 'GTR model')
```

L’albero più verosimile

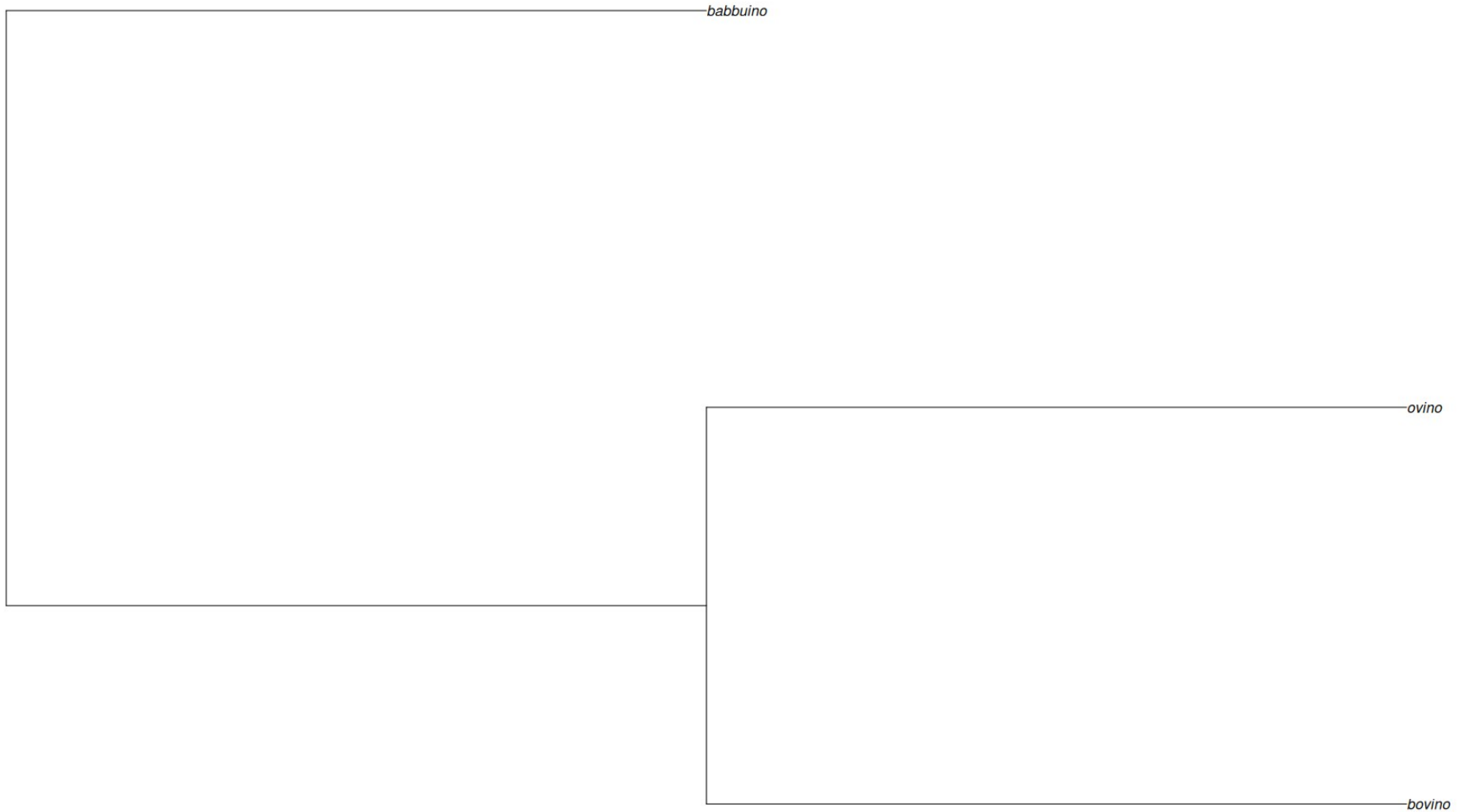


```
> alb = read.tree(text = "((bovino:1, ovino:1):1, babbuino:1);")
> alb

Phylogenetic tree with 3 tips and 2 internal nodes.

Tip labels:
  bovino, ovino, babbuino

Rooted; includes branch lengths.
> plot(alb)
```

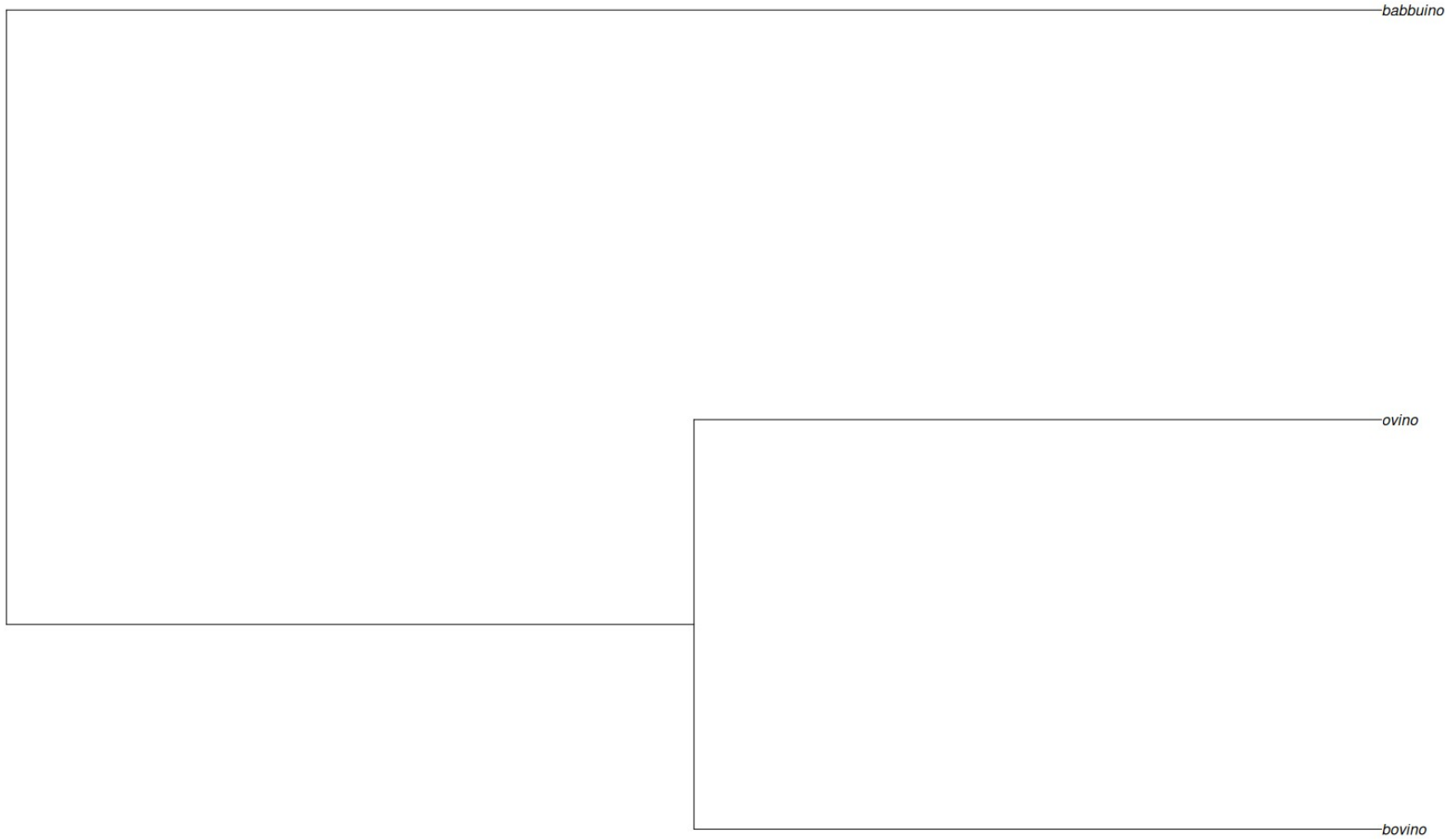


```
> alb = read.tree(text = "((bovino:1, ovino:1):1, babbuino:2);")
> alb

Phylogenetic tree with 3 tips and 2 internal nodes.

Tip labels:
  bovino, ovino, babbuino

Rooted; includes branch lengths.
> plot(alb)
```



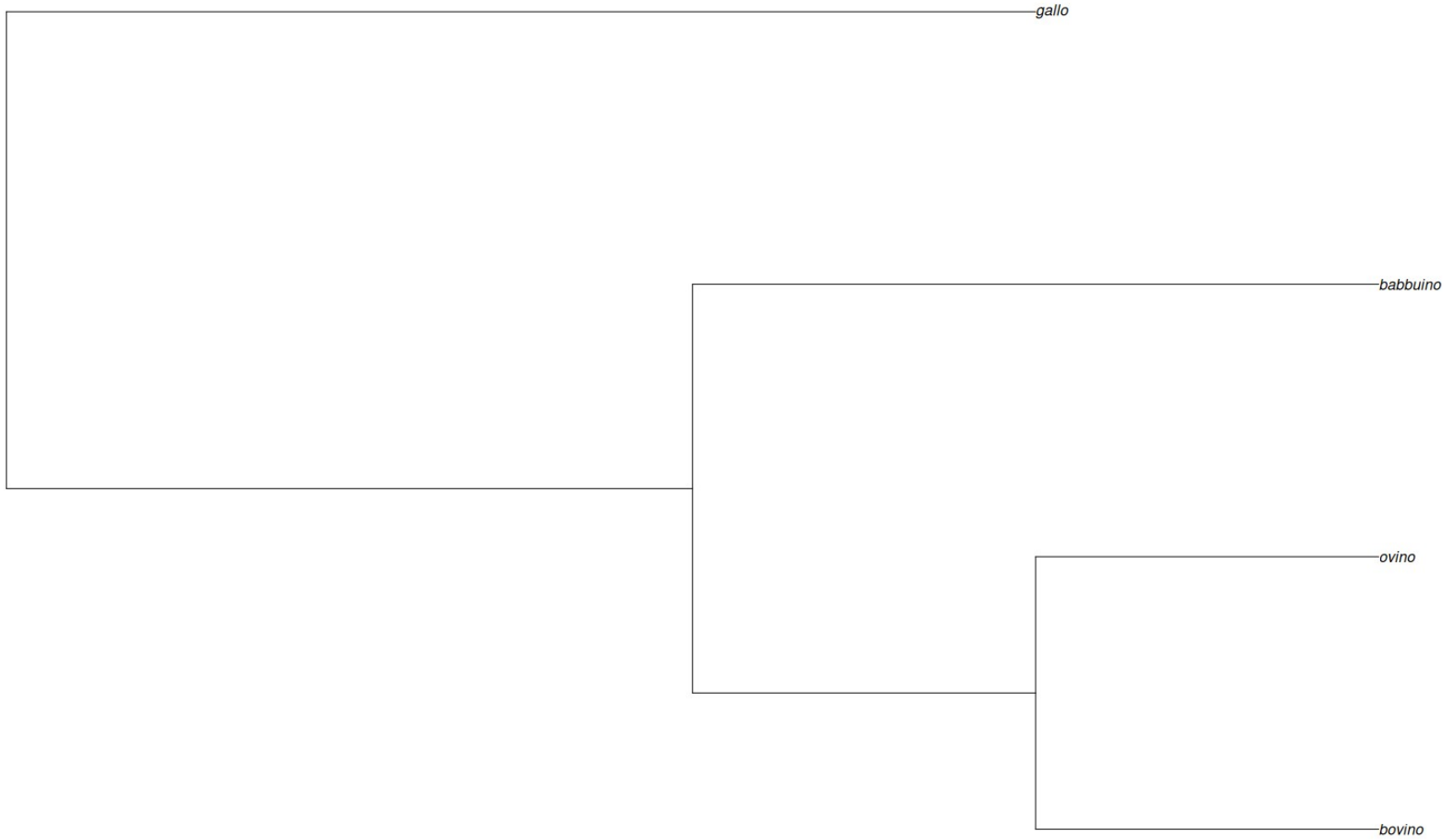
```
> alb = read.tree(text = "(((bovino:1, ovino:1):1, babbuino:2):2, gallo:3);")
> alb

Phylogenetic tree with 4 tips and 3 internal nodes.

Tip labels:
  bovino, ovino, babbuino, gallo

Rooted; includes branch lengths.

> plot(alb)
```



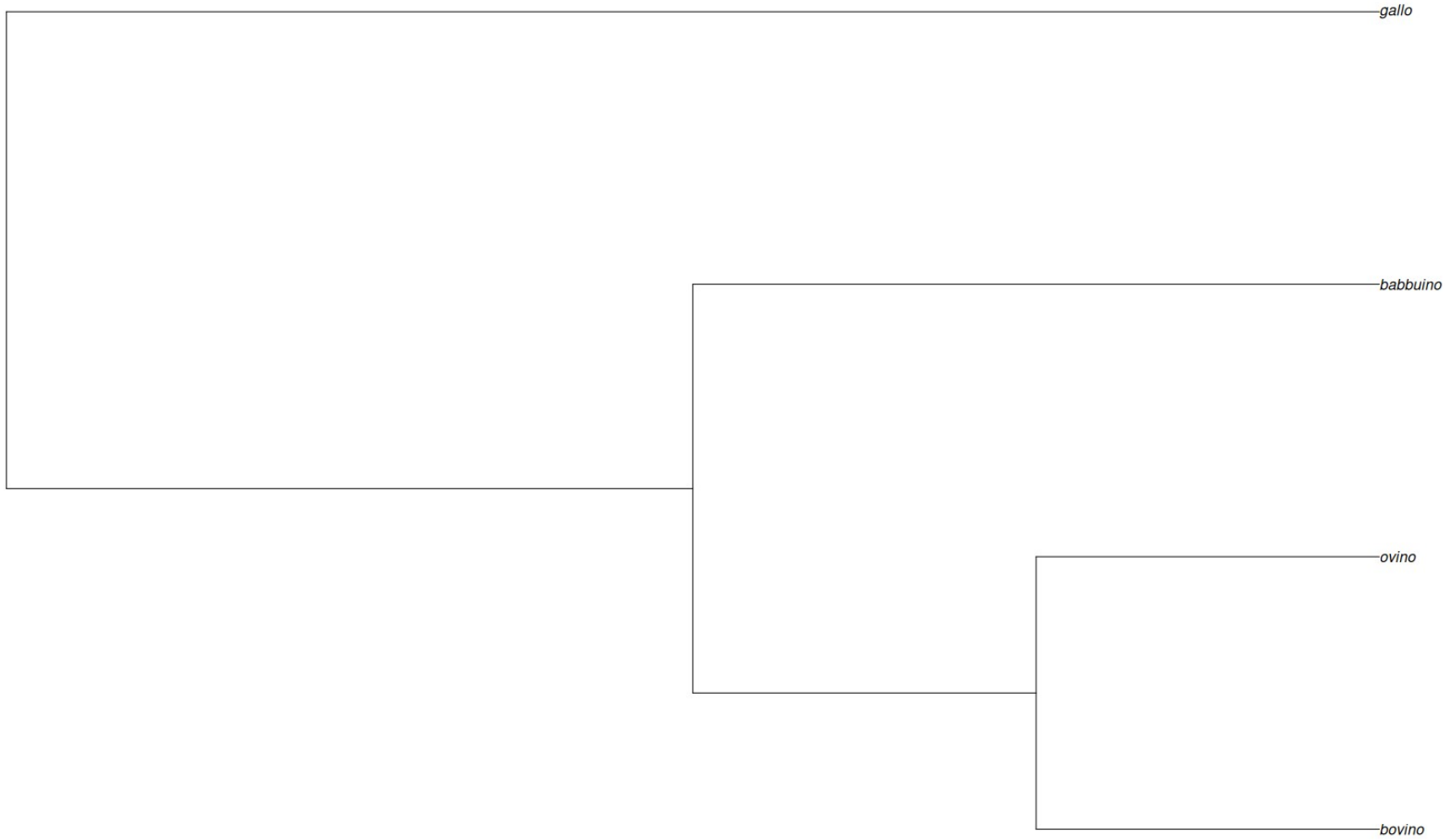
```
> alb = read.tree(text = "(((bovino:1, ovino:1):1, babbuino:2):2, gallo:4);")
> alb

Phylogenetic tree with 4 tips and 3 internal nodes.

Tip labels:
  bovino, ovino, babbuino, gallo

Rooted; includes branch lengths.

> plot(alb)
```



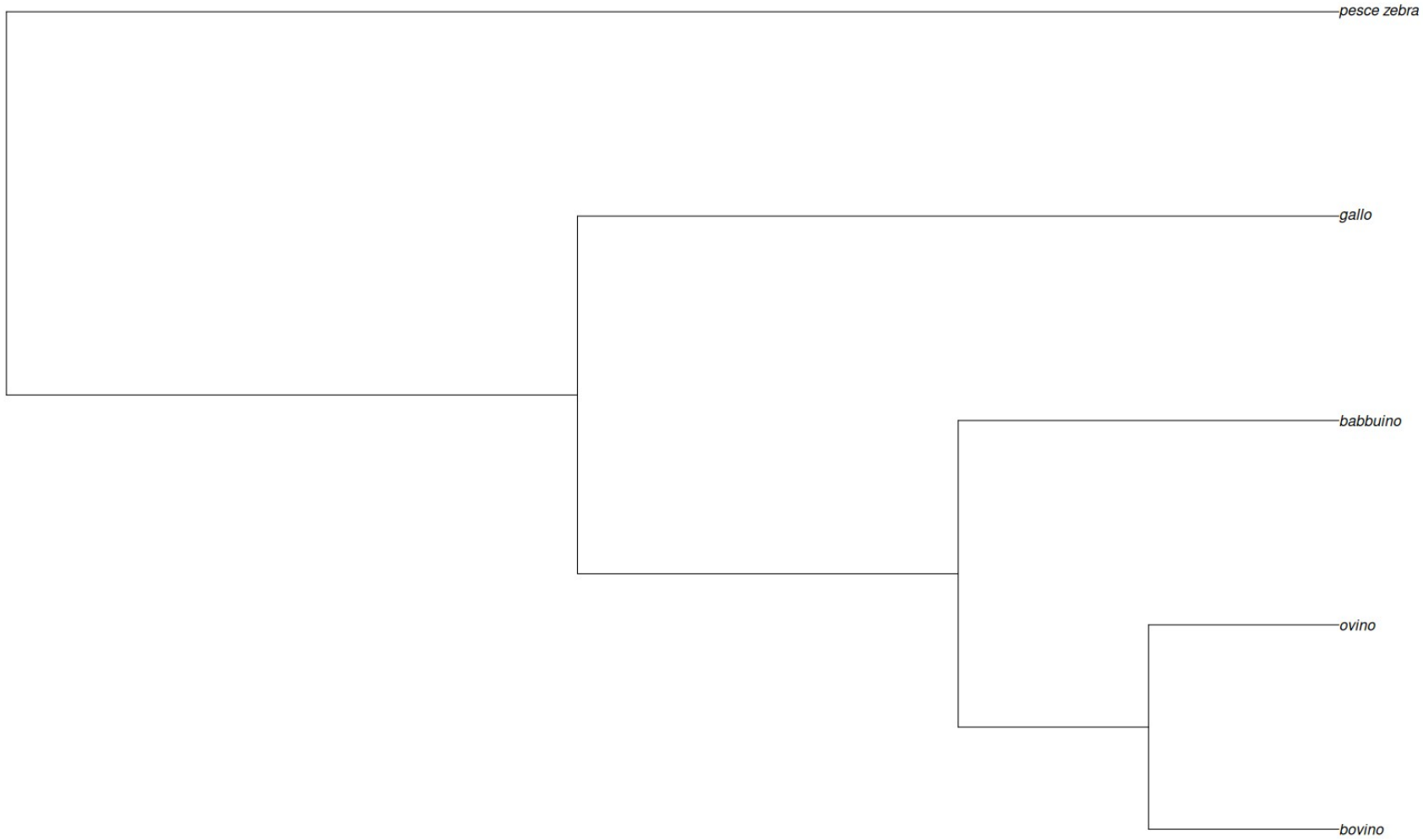
```
> alb = read.tree(text = "((((bovino:1, ovino:1):1, babbuino:2):2, gallo:4):3, pesce_zebra:7);")
> alb

Phylogenetic tree with 5 tips and 4 internal nodes.

Tip labels:
  bovino, ovino, babbuino, gallo, pesce_zebra

Rooted; includes branch lengths.

> plot(alb)
```

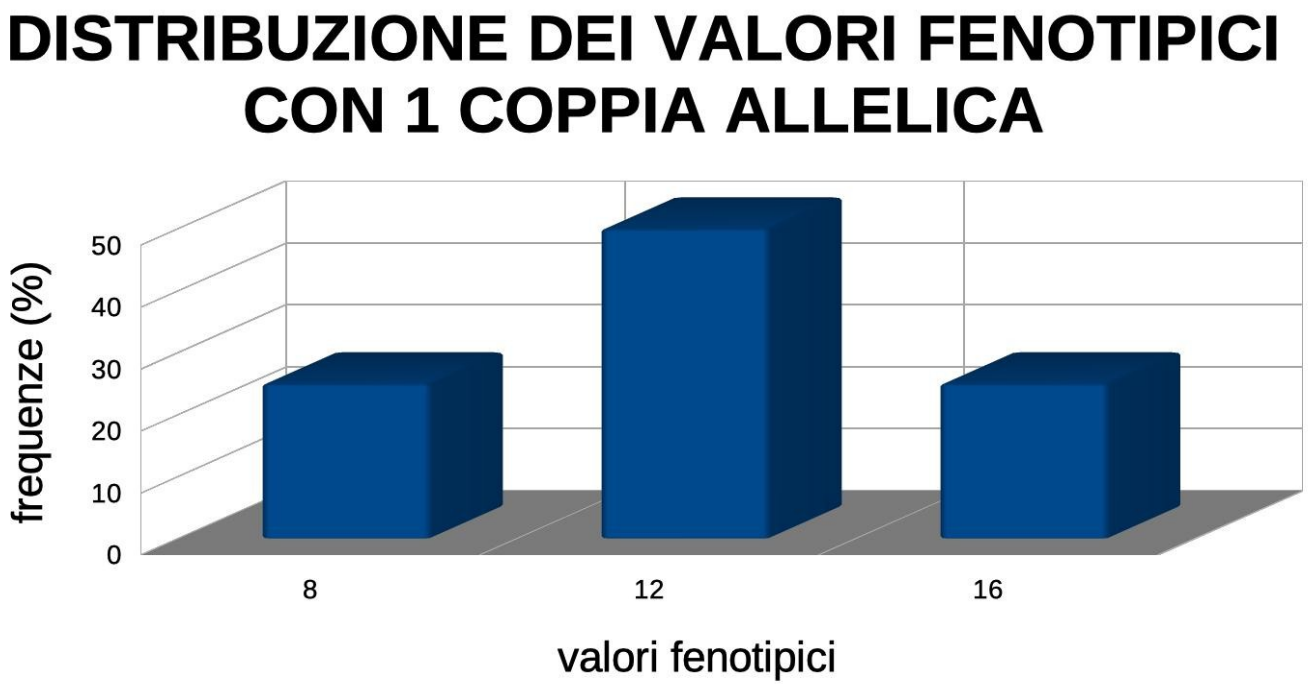


STATISTICA
L’effetto additivo dei fattori genetici ed ambientali

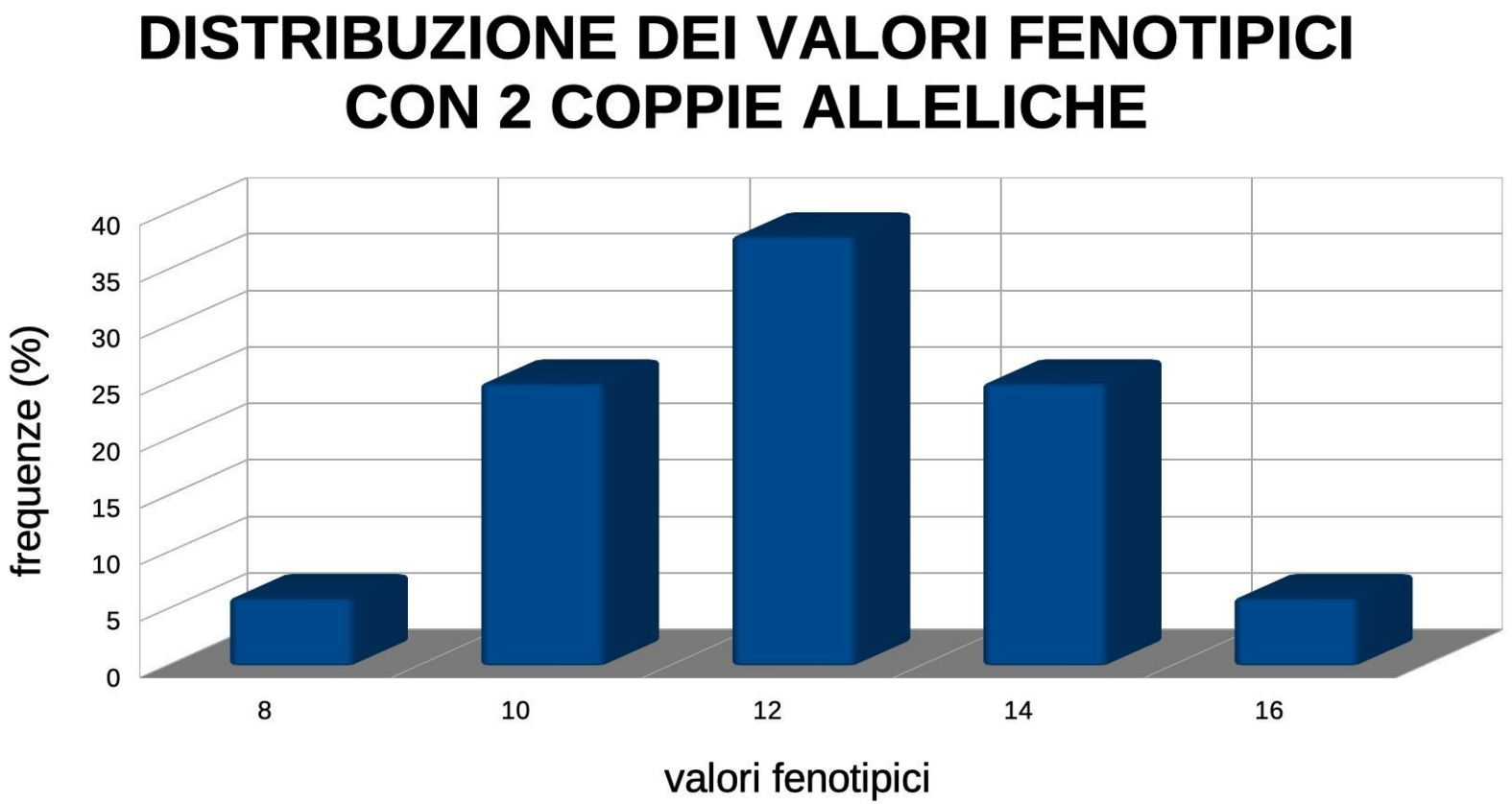
gene normale A	->	effetto fenotipico = 8	p = 0,5
allele mutante A’	->	effetto fenotipico = 4	q = 0,5
omozigoti AA	->	valore fenotipico = 16	F = 0,25
omozigoti A’A’	->	valore fenotipico = 8	F = 0,25
eterozigoti AA’	->	valore fenotipico = 12	F = 0,25
eterozigoti A’A	->	valore fenotipico = 12	F = 0,25

L’effetto additivo dei fattori genetici ed ambientali

CLASSE GENOTIPICA	VALORE FENOTIPICO	FREQUENZA
AA	16	1/4 = 25%
AA' - A'A	12	2/4 = 50%
A'A'	8	1/4 = 25%



Due loci



Valori e frequenze dei possibili genotipi nel caso in cui un carattere quantitativo è controllato da due*loci* («a» e «B»)

Tre loci

aumentando ulteriormente il numero di coppie alleliche interessate, aumenta in numero delle classi di valori ma diminuiscono le differenze fra i valori delle varie classi

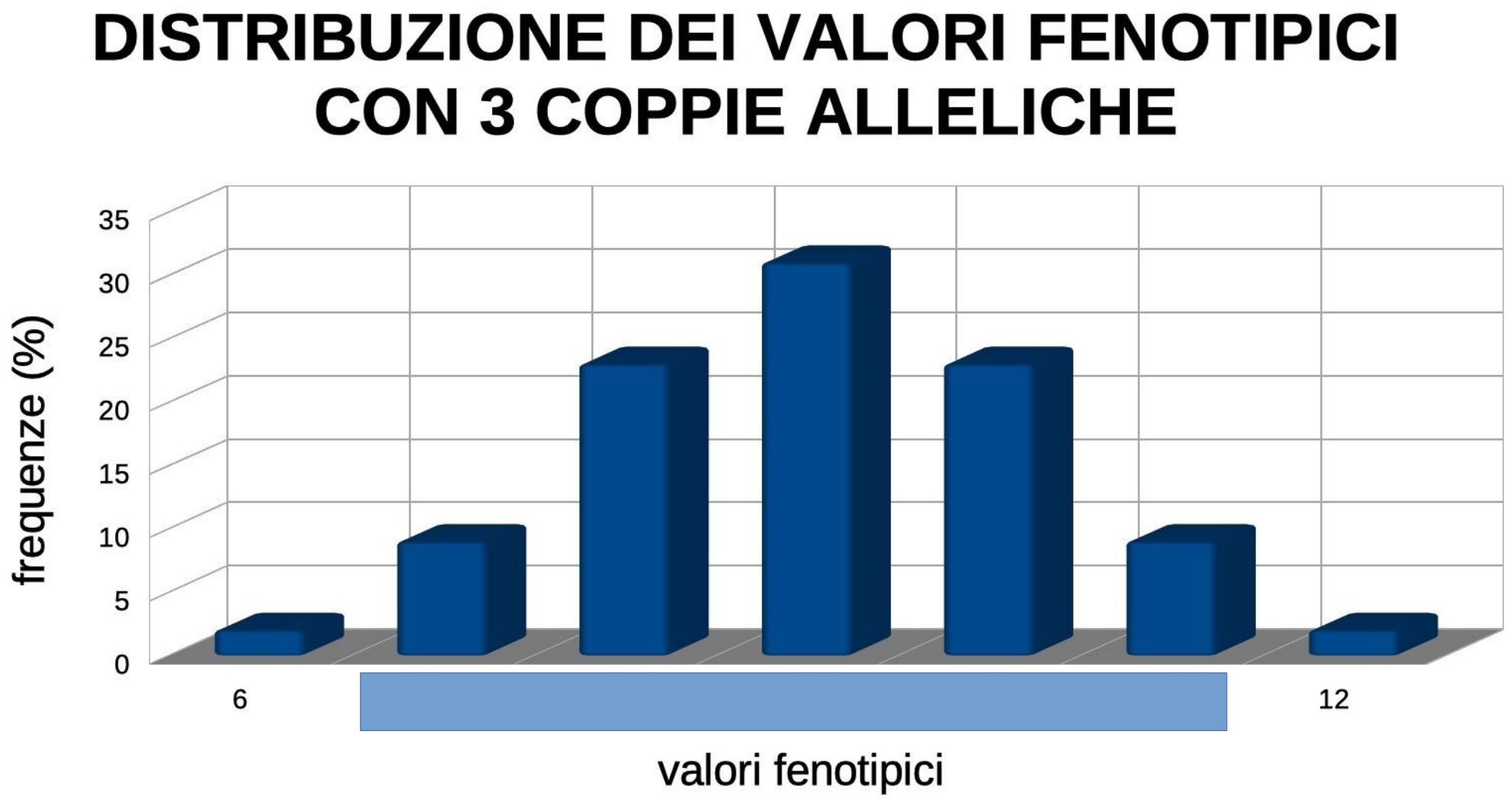
Se si considerano tre coppie di geni, a-a', B-B' e C-C', dove gli alleli «normali» a, B e C hanno effetto 2 e gli alleli «mutanti» a', B' e C' effetto 1, le classi genotipiche e i relativi valori fenotipici saranno:

Tre loci

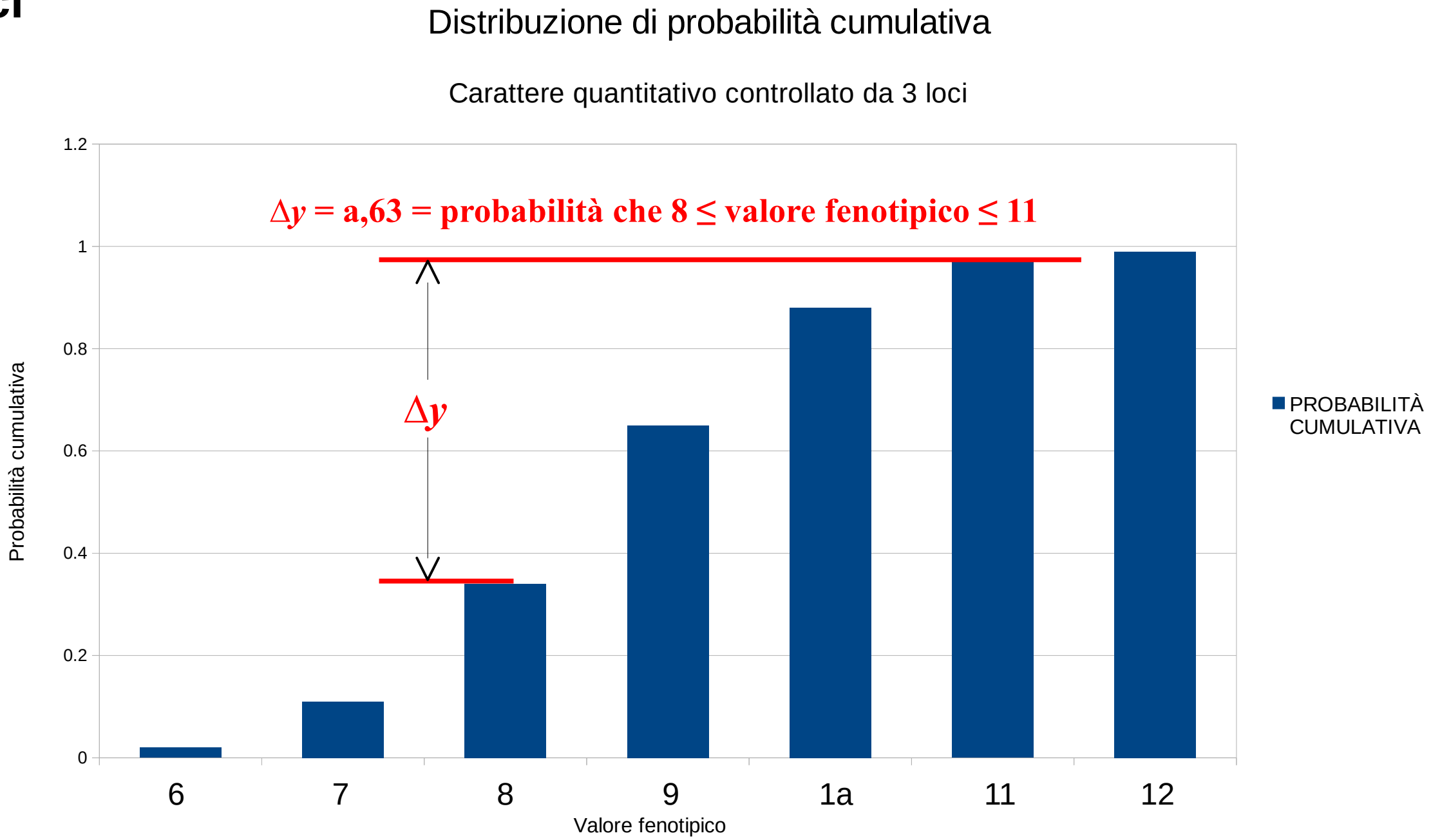
Quali sono i valori fenotipici intermedi?

CLaSSE GENOTIPICa	VaLORE FENOTIPICO	FREQUENZa
aaBBCC	12	1/64 = 2%
aaBBCC’ – aaBBC’C – aaBB’CC aaB’BCC – aa’BBCC – a’aBBCC		6/64 = 9%
aaBBC’C’ – aaBB’CC’ – aaB’BCC’ aa’BBCC’ – a’aBBCC’ – aaBB’C’C aaB’BC’C – aa’BBC’C – a’aBBC’C aaB’B’CC – aa’BB’CC – a’aBB’CC aa’B’BCC – a’aB’BCC – a’a’BBCC		15/64 = 23%
aaBB’C’C’ – aaB’BC’C’ – aaB’B’CC’ aaB’B’C’C – aa’BBC’C’ – aa’BB’CC’ aa’B’BCC’ – aa’BB’C’C – aa’B’BC’C aa’B’B’CC – a’aBBC’C’ – a’aBB’CC’ a’aB’BCC’ – a’aBB’C’C – a’aB’BC’C a’aB’B’CC – a’a’B’BCC – a’a’BB’CC, a’a’BBC’C – a’a’BBCC’		2a/64 = 31%
aaB’B’C’C’ – aa’BB’C’C’ – a’aBB’C’C’ a’a’BBC’C’ – aa’B’BC’C’ – a’aB’BC’C’ a’a’B’BCC’ – aa’B’B’CC’ – a’aB’B’CC’ a’a’B’BC’C – aa’B’B’C’C – a’aB’B’C’C a’a’B’B’CC – a’a’BB’CC’ – a’a’BB’C’C		15/64 = 23%
aa’B’B’C’C’ – a’aB’B’C’C’ – a’a’B’B’C’C a’a’B’B’CC’ – a’a’BB’C’C’ – a’a’B’BC’C’		6/64 = 9%
a’a’B’B’C’C’	6	1/64 = 2%

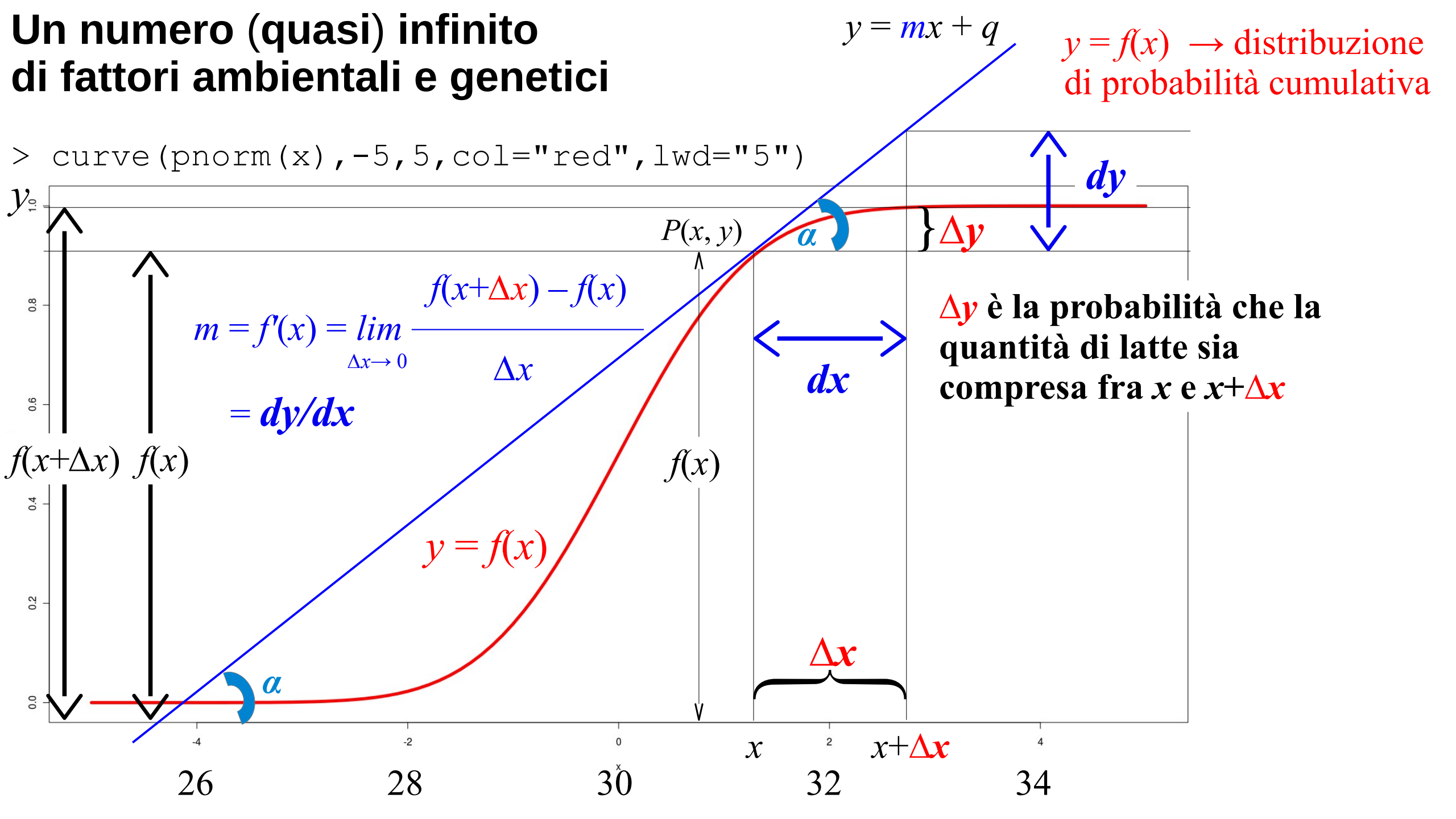
Tre loci


DISTRIBUZIONE DEI VALORI FENOTIPICI
CON 3 COPPIE ALLELICHE
frequenze (%)
35
30
25
20
15
10
5
0
6
12
valori fenotipici

Tre loci


Distribuzione di probabilità cumulativa
Carattere quantitativo controllato da 3 loci
Δy = a,63 = probabilità che 8 ≤ valore fenotipico ≤ 11
Δy
Probabilità cumulativa
1.2
1
0.8
0.6
0.4
0.2
0
6
7
8
9
1a
11
12
Valore fenotipico
PROBABILITÀ CUMULATIVA

Un numero (quasi) infinito di fattori ambientali e genetici

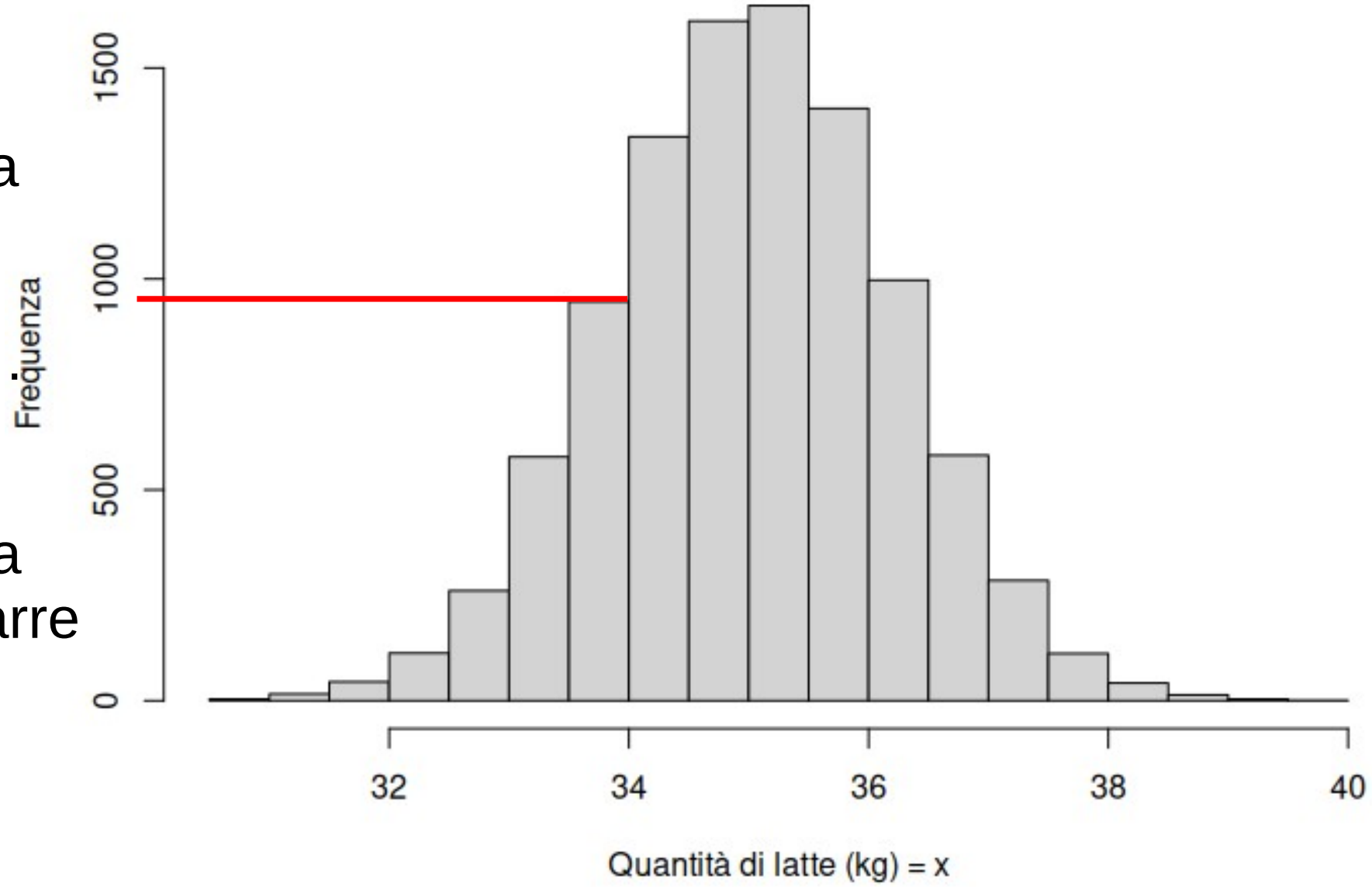


Un numero (quasi) infinito di fattori ambientali e genetici

Il N° di campioni compresi fra 33,5 e 34 è uguale a ≈ 95a

Quindi l'h di ogni barra rappresenta…

Se ci sono 1aaaa campioni, la somma delle h di tutte le barre sarà uguale a...

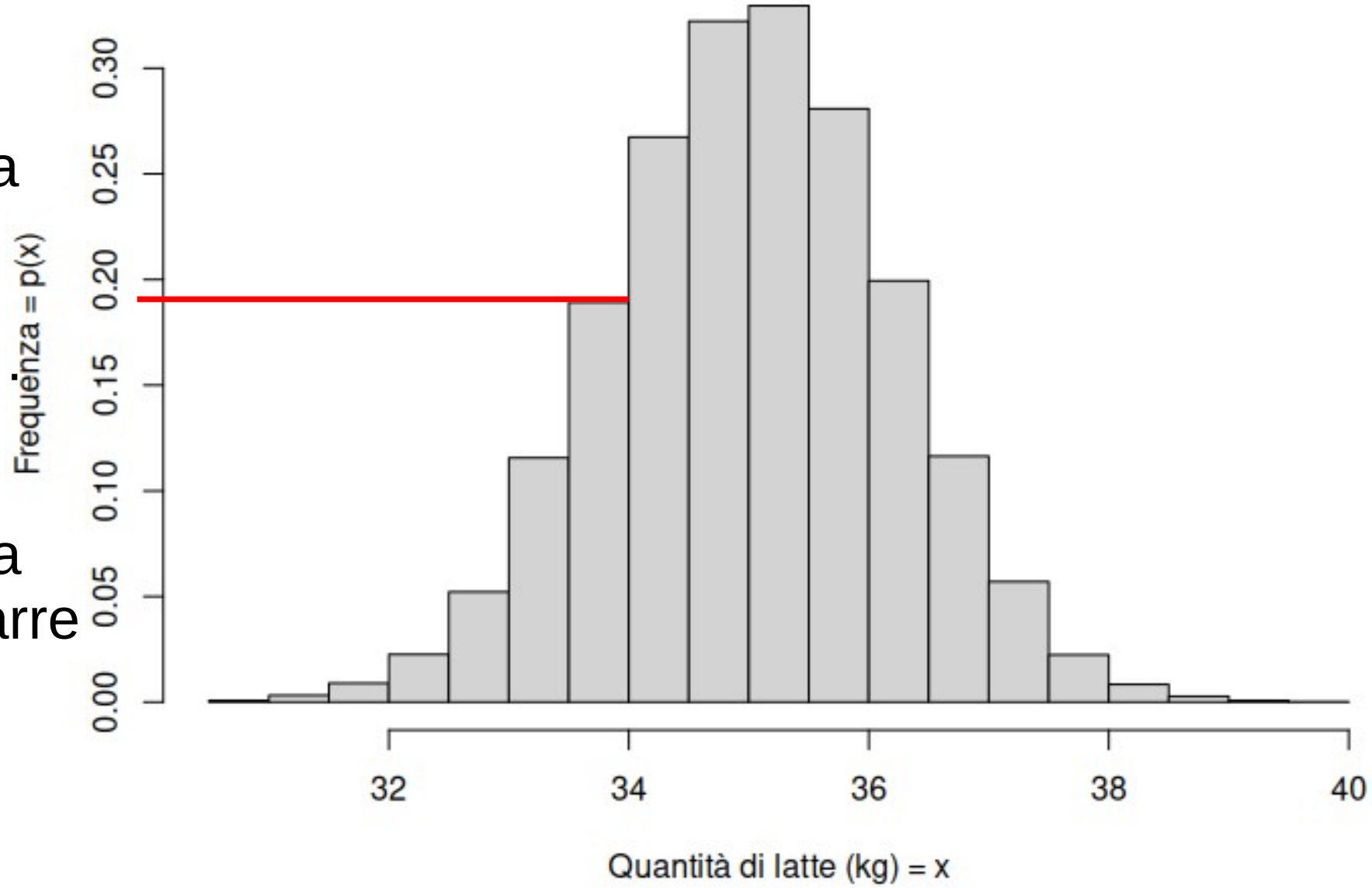


Un numero (quasi) infinito di fattori ambientali e genetici

Il N° di campioni compresi fra 33,5 e 34 è uguale a ≈ 95a

Quindi l'h di ogni barra rappresenta…

Se ci sono 1aaaa campioni, la somma delle h di tutte le barre sarà uguale a...

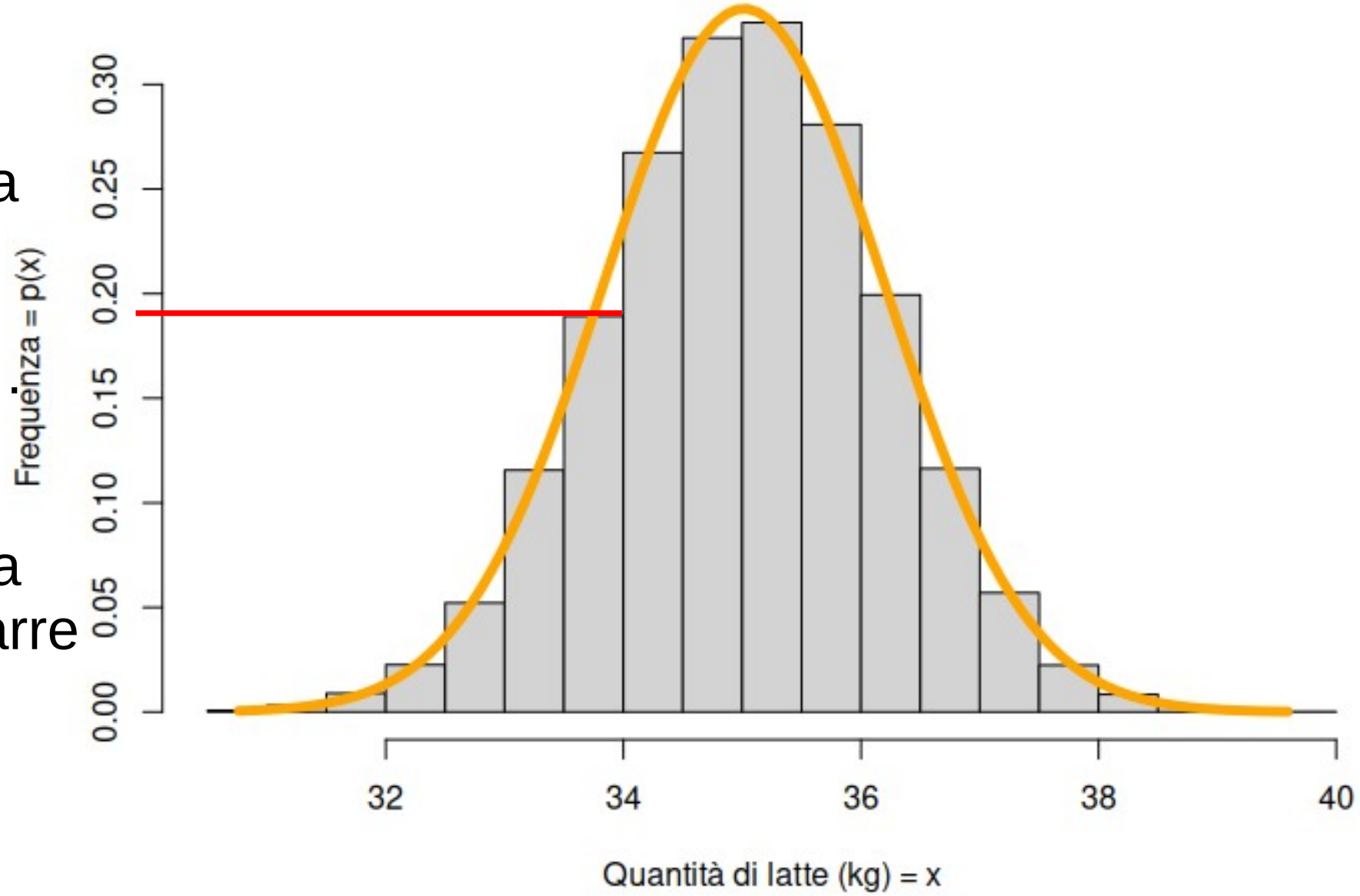


Un numero (quasi) infinito di fattori ambientali e genetici

Il N° di campioni compresi fra 33,5 e 34 è uguale a ≈ 95a

Quindi l'h di ogni barra rappresenta…

Se ci sono 1aaaa campioni, la somma delle h di tutte le barre sarà uguale a...



Un numero (quasi) infinito di fattori ambientali e genetici

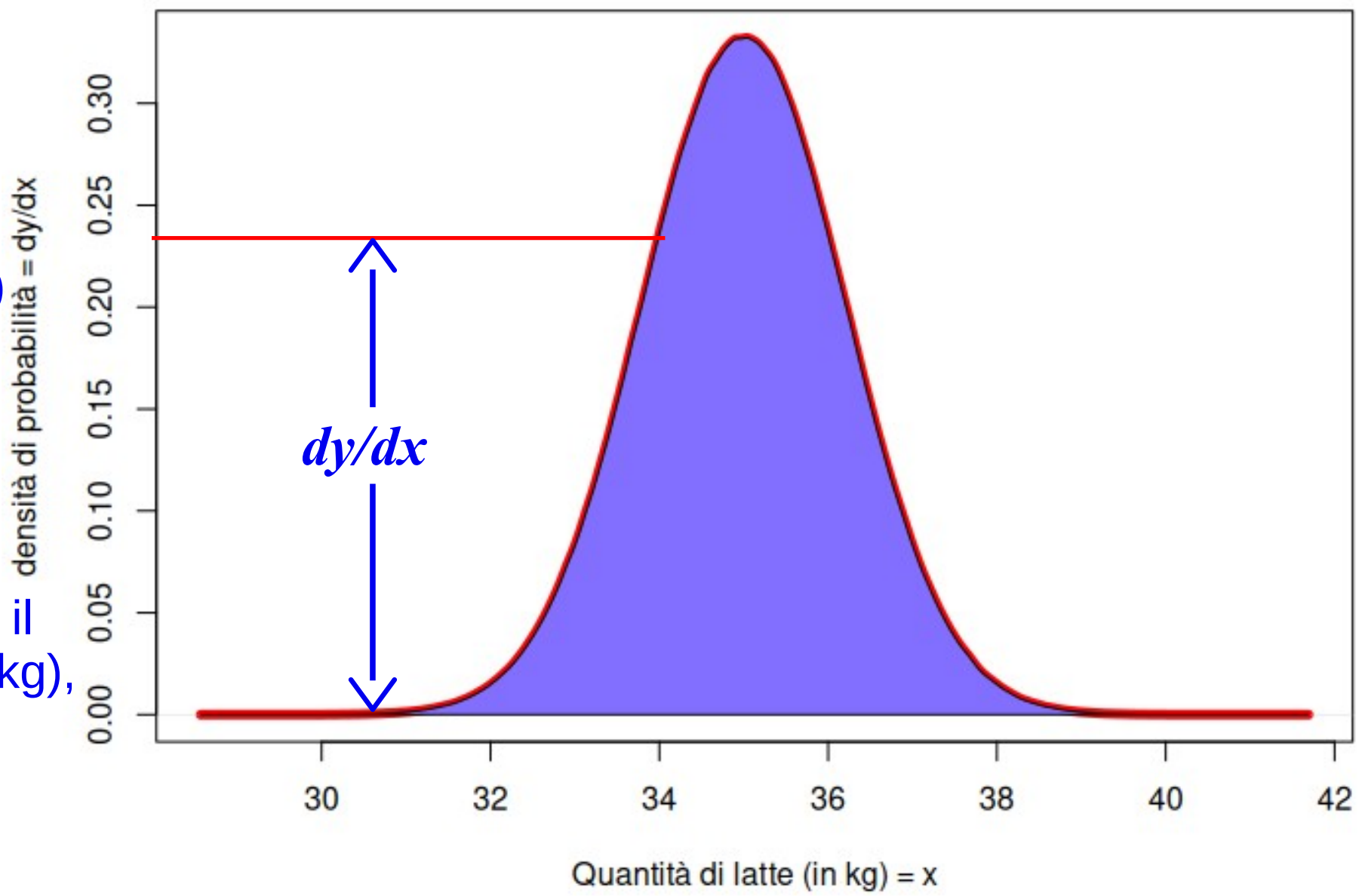
```
# per disegnare l'istogramma
> x = rnorm(1aaaa, mean = 35, sd = 1.2)
> hist(x, xlab = 'Quantità di latte (kg) = x', ylab = 'Frequenza')
> hist(x, prob = TRUE, xlab = 'Quantità di latte (kg) = x', ylab = 'Frequenza
= p(x)')
> x_values = seq(min(x), max(x), length = 1aa)
> fun = dnorm(x_values, mean = mean(x), sd = sd(x))
> lines(x_values, fun, col = 'orange', lwd = 5)

# per disegnare la densità
# attenzione: x ha un significato diverso
> y = rnorm(1aaaaaaa, mean = 35, sd = 1.2)
> den = density(y)
> plot(den, col = 'red', lwd = 5.a, xlab = 'Quantità di latte (in kg) = x',
ylab = 'densità di probabilità = dy/dx')
> polygon(den, col = 'slateblue1')
> value1 = 33
> value2 = 34
> l = min(which(den$x >= value1))
> h = max(which(den$x < value2))
> polygon(c(den$x[c(l, l:h, h)]), c(a, den$y[l:h], a), col = 'slateblue1')
```

Un numero (quasi) infinito di fattori ambientali e genetici

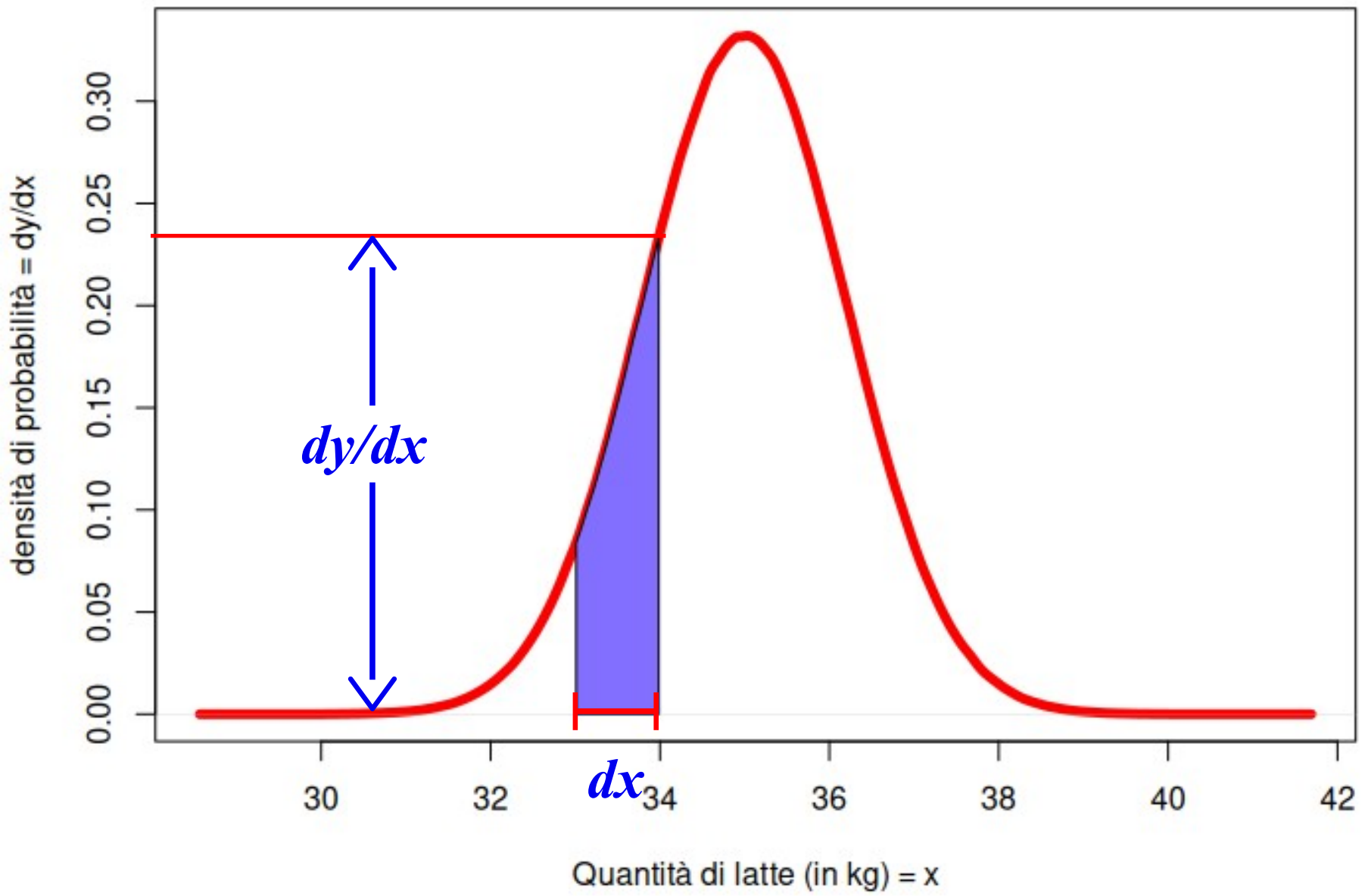
dy = probabilità che la quantità di latte sia compresa fra due valori, ***x*** e ***x+dx***, in cui ***dx*** → a (densità di probabilità)

L'area sotto la curva rappresenta la probabilità che x sia compreso tra il suo valore min (≈ 3a kg) e il suo valore max (≈ 4a kg), che equivale al 1aa% (evento certo).



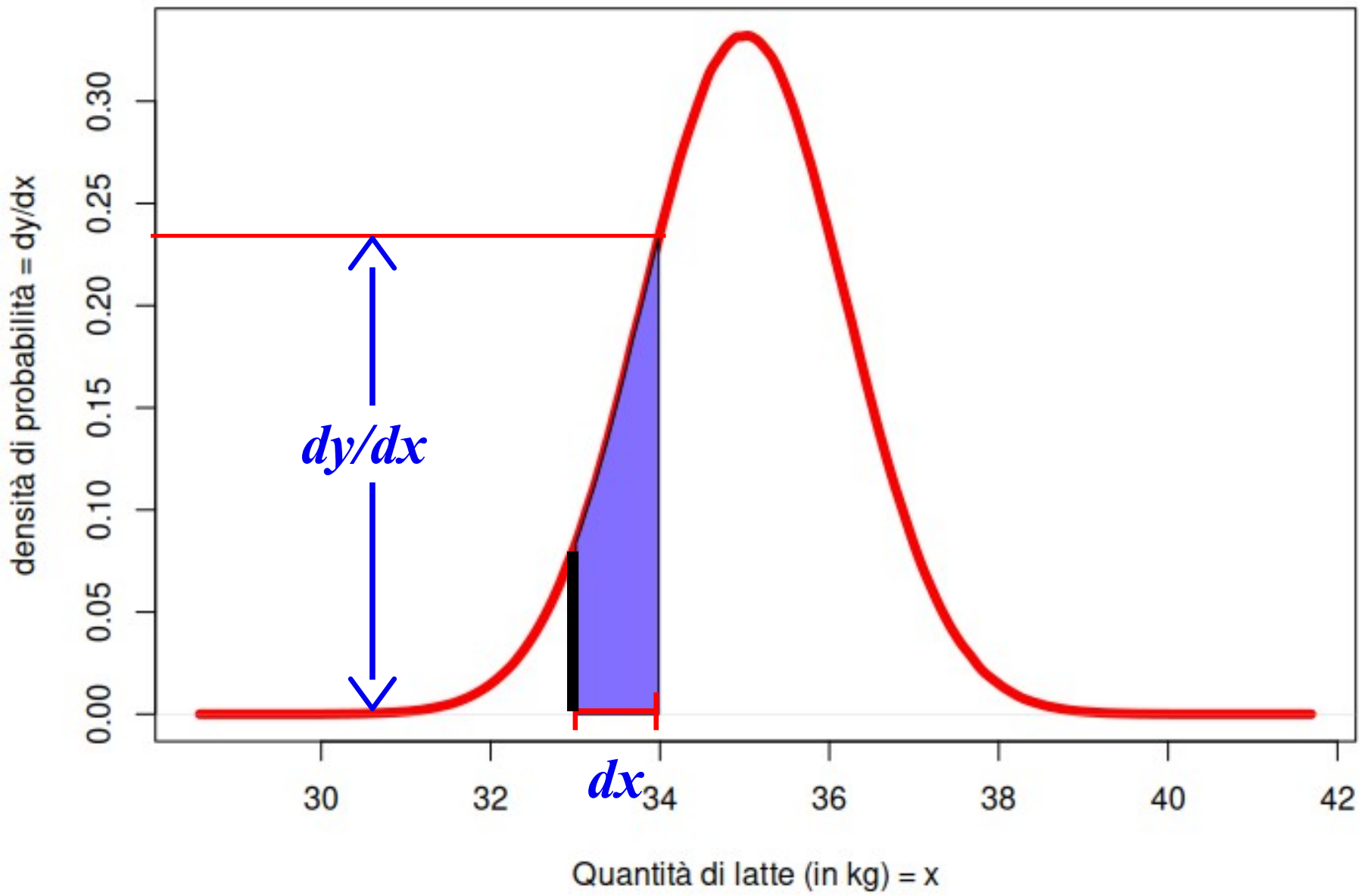
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



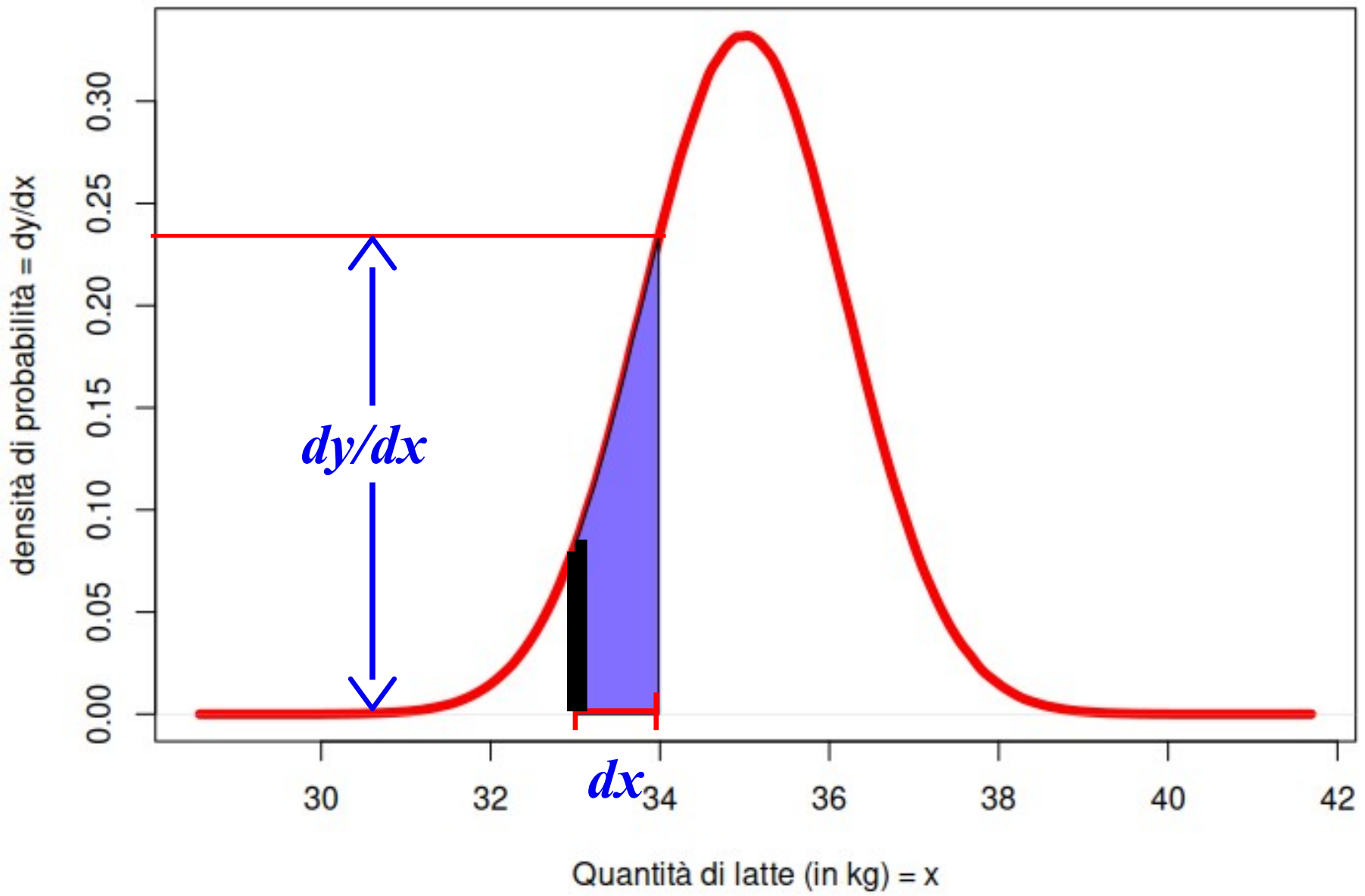
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



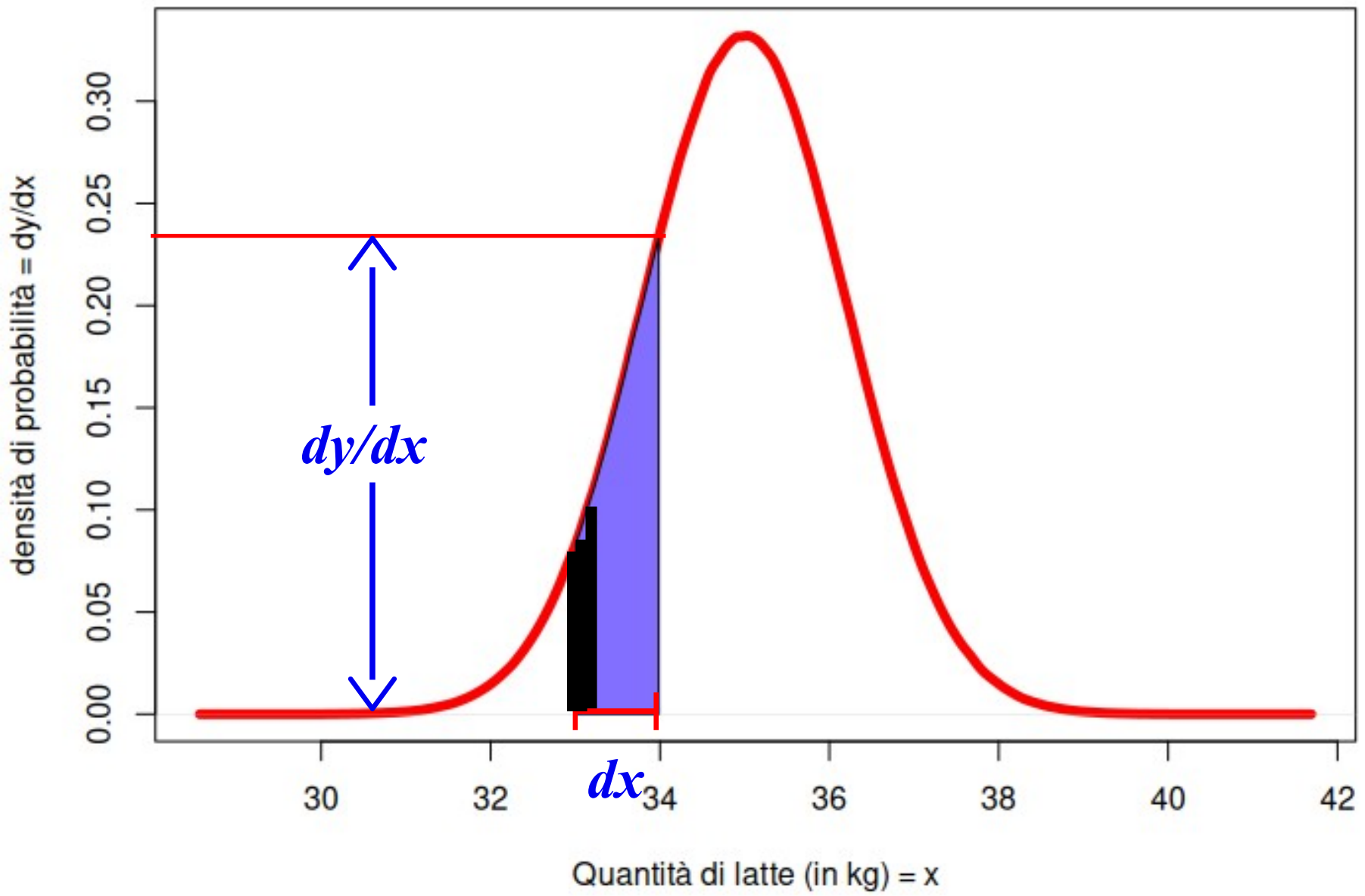
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



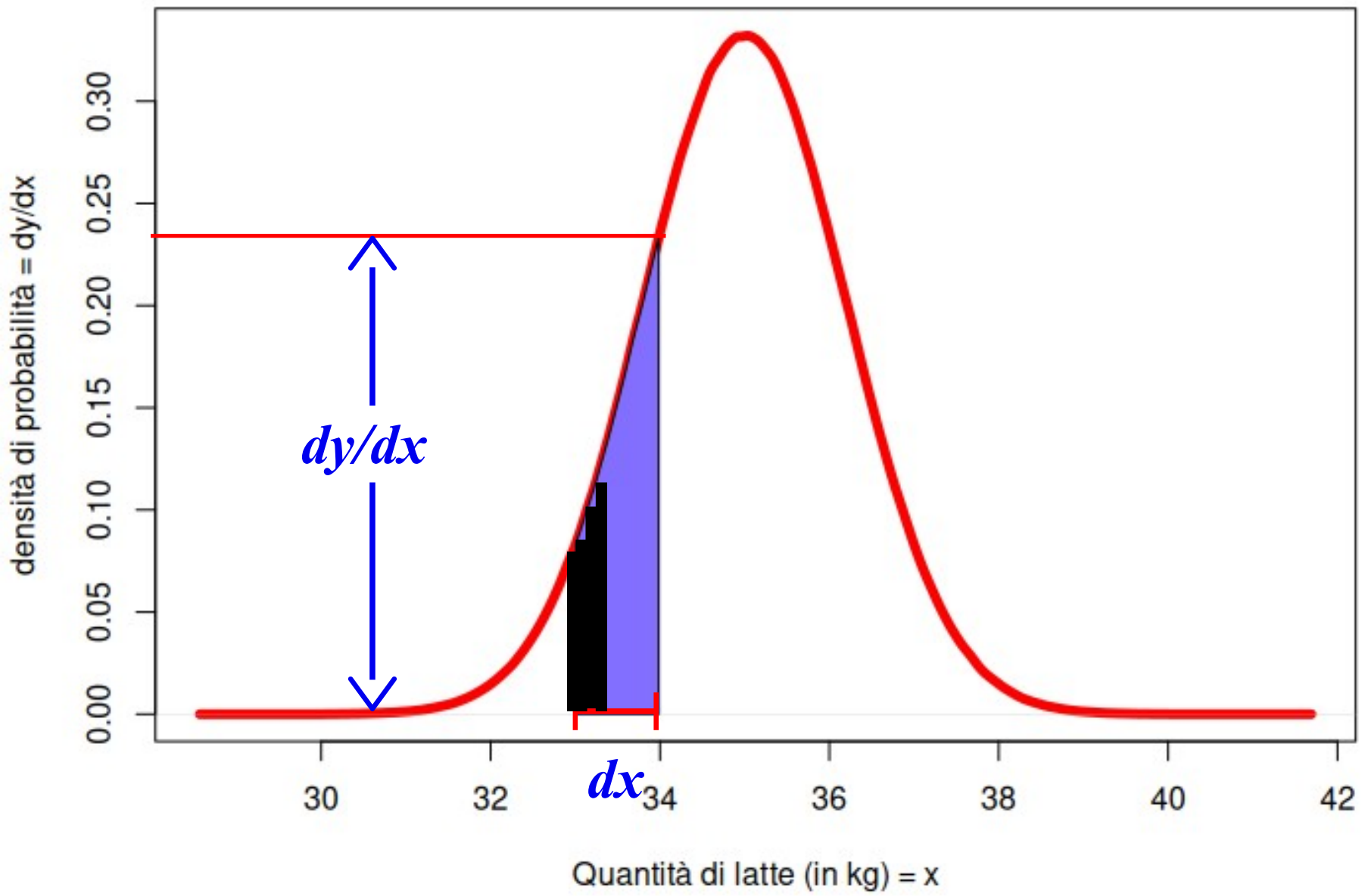
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



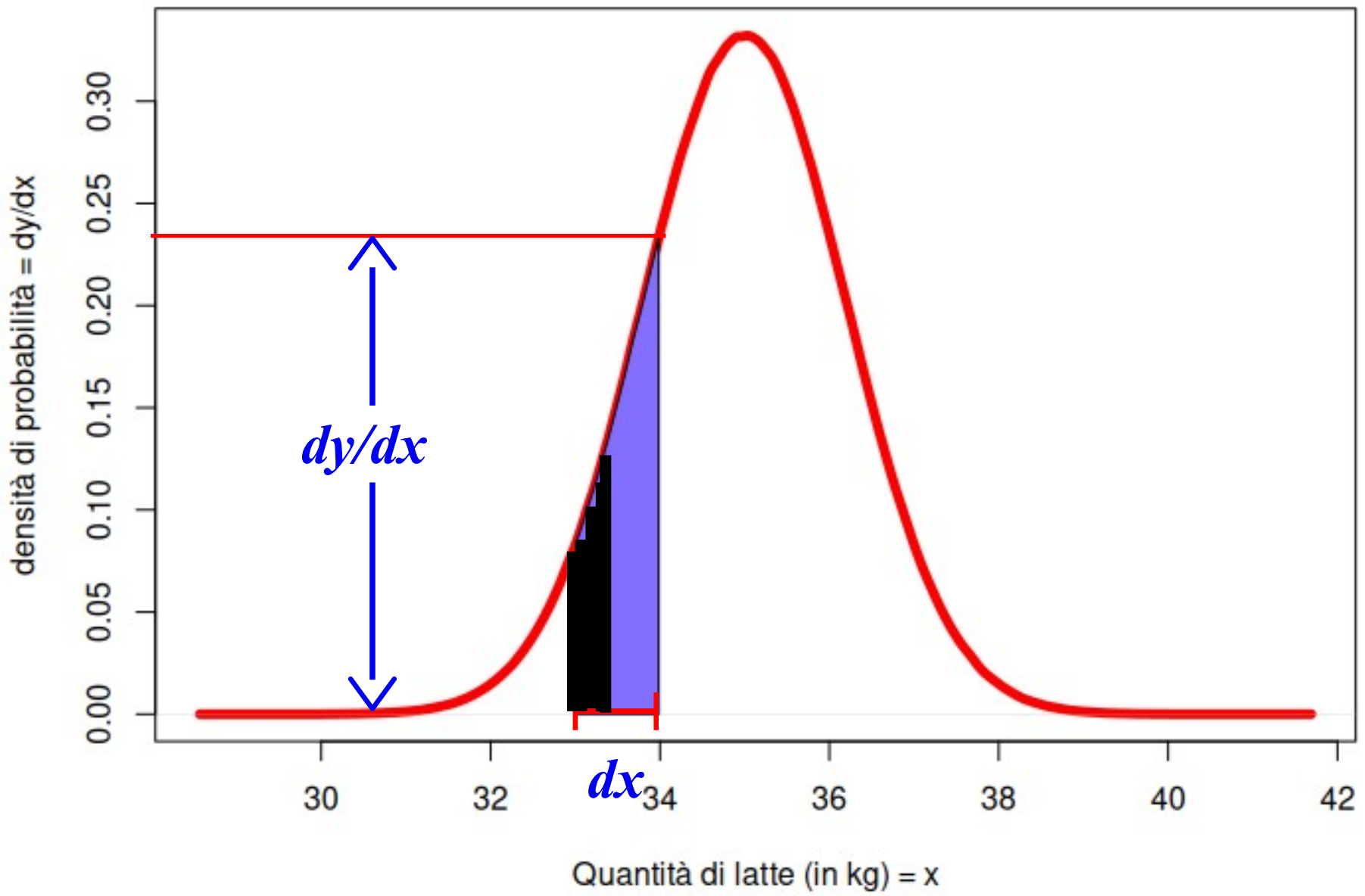
Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



Un numero (quasi) infinito di fattori ambientali e genetici

La probabilità che la quantità di latte sia compresa fra 33 e 34 Kg è data dall'area blu, che può essere calcolata utilizzando un'operazione che prende il nome di «**integrazione**».



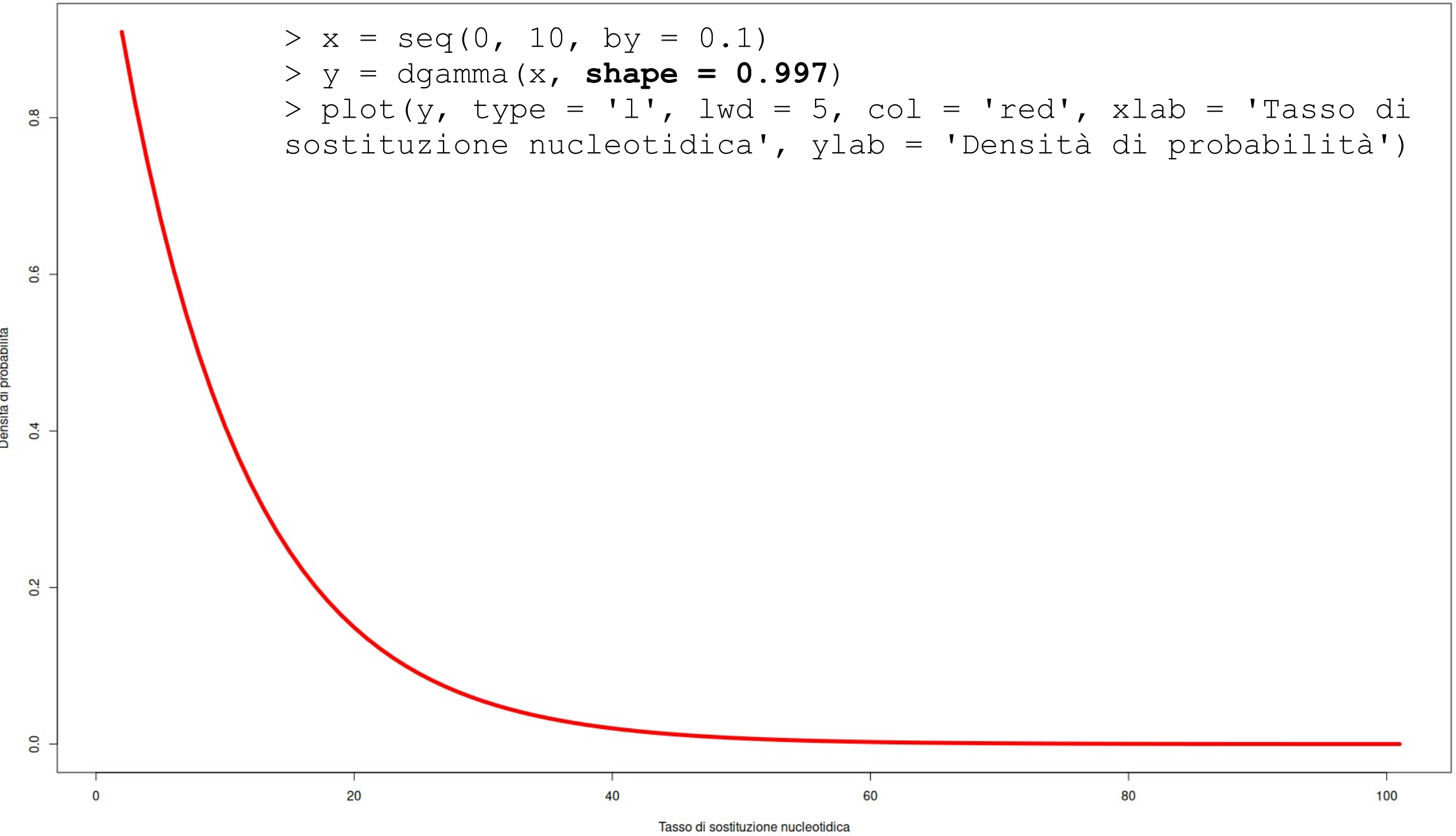
```
 oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo
                                      ---  PhyML 20120412  ---
                                   http://www.atgc-montpellier.fr/phyml
                                 Copyright CNRS - Universite Montpellier II
 oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo

. Sequence filename:           myostatin_5_species_100_bp.phy
. Data set:                 #1
. Tree topology search :       NNIs
. Initial tree:             BioNJ
. Model of nucleotides substitution:  GTR
. Number of taxa:              5
. Log-likelihood:              -400.23791
. Unconstrained likelihood:       -343.80642
. Parsimony:                   101
. Tree size:                   10.41394
. Discrete gamma model:        Yes
  - Number of categories:          4
  - Gamma shape parameter:         0.997
. Proportion of invariant:         0.023
. Nucleotides frequencies:
  - f(A)= 0.28600
  - f(C)= 0.17800
  - f(G)= 0.24000
  - f(T)= 0.29600
. GTR relative rate parameters :
  A <-> C   345.16529
  A <-> G   385.25348
  A <-> T    0.29877
  C <-> G   296.04247
  C <-> T   390.59090
  G <-> T    1.00000
```

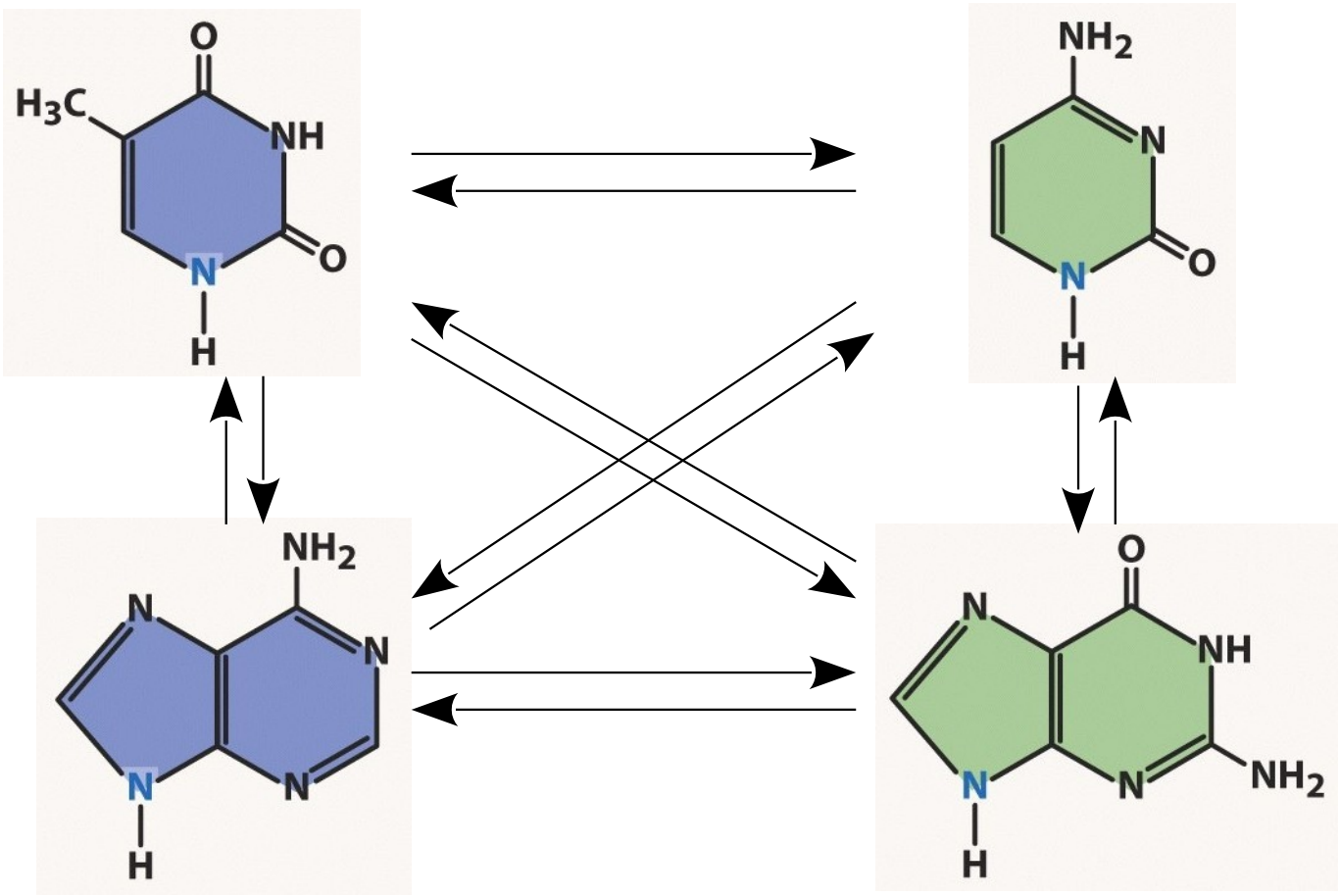
```
 oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo
                                      ---  PhyML 20120412  ---
                                 http://www.atgc-montpellier.fr/phyml
                               Copyright CNRS - Universite Montpellier II
 oooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooooo


. Instantaneous rate matrix :
  [A---------C---------G---------T------]
  -0.99555   0.39721   0.59777   0.00057
   0.63822  -1.84503   0.45935   0.74746
   0.71234   0.34068  -1.05494   0.00191
   0.00055   0.44949   0.00155  -0.45159
```

```
> x = seq(0, 10, by = 0.1)
> y = dgamma(x, shape = 0.997)
> plot(y, type = 'l', lwd = 5, col = 'red', xlab = 'Tasso di
sostituzione nucleotidica', ylab = 'Densità di probabilità')
```



Il modello JC69

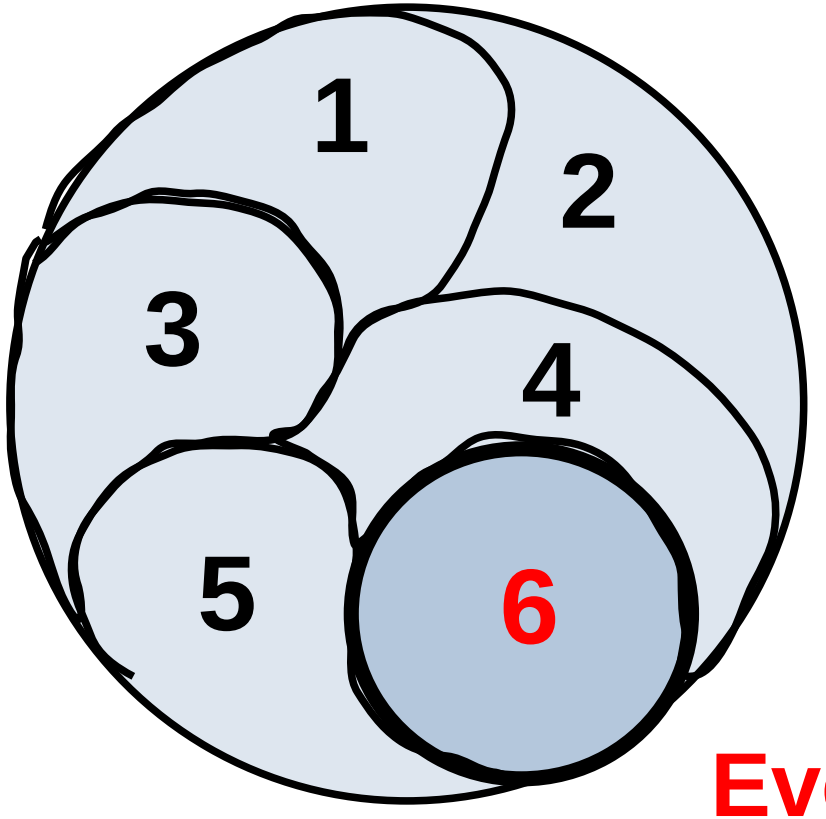


STATISTICA: concetti fondamentali

- **Esperimento**: processo che genera un'osservazione (per es., il lancio di un dado o di una moneta)
- **Evento**: ognuno dei possibili risultati di un esperimento (contrassegnati da lettere maiuscole)
- **Evento elementare**: evento che non può essere decomposto e che corrisponde ad un «punto campione» indivisibile

STATISTICA: spazio campione Ω

- **<u>Insieme di tutti i possibili risultati</u>** di un esperimento
  Es. lancio di un dado
- 6 risultati equiprobabili (se il dado non è truccato)



$\Omega = 1$

Evento certo

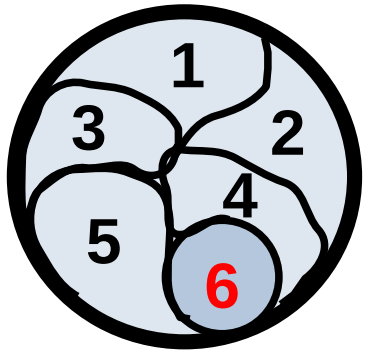
Evento elementare a

Cosa succede se lancio il dado una seconda volta?

- **Vinco se esce di nuovo il numero 6: che probabilità ho di vincere?**
- **Lo spazio campione Ω diventa uguale ad 1/6**
- **Poiché si tratta di EVENTI INDIPENDENTI, la P(6, 6) = 1/6 * 1/6 = 1/36**
- **Probabilità congiunta di eventi indipendenti = = prodotto delle probabilità dei singoli eventi**

$$\Omega = 1/6$$

Evento certo



Evento elementare B

STATISTICA: probabilità di due eventi indipendenti

- **a** = evento elementare esce 6 al primo lancio
- **B** = evento elementare esce 6 al secondo lancio

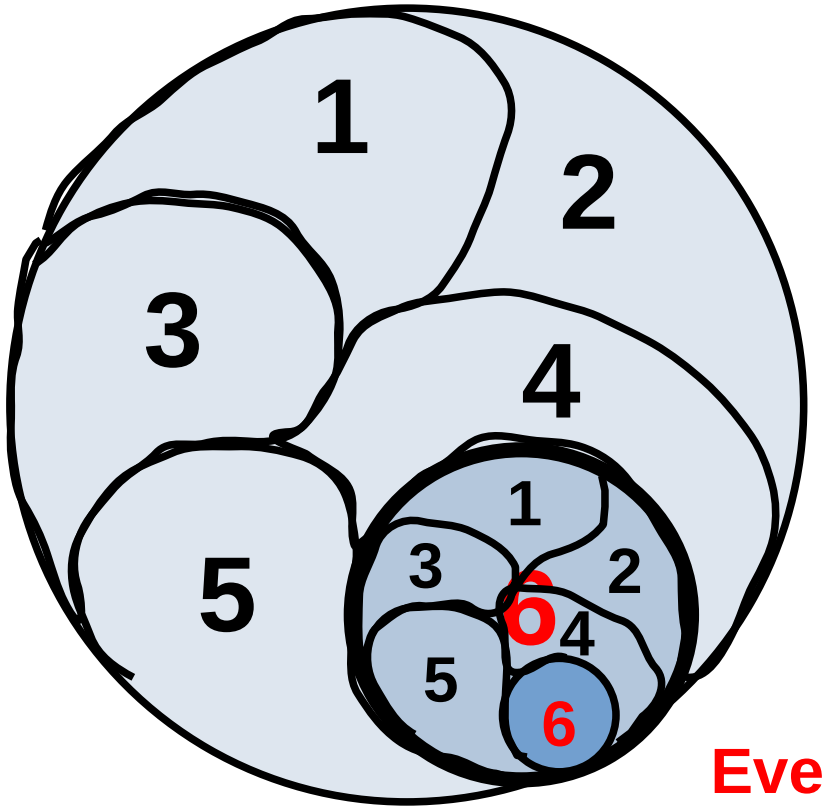
Quindi, noi vogliamo calcolare la Prob dell'intersezione fra il cerchio medio e il cerchio piccolo:

$P(a \cap B) = P(a)*P(B|a)$

Ma, poiché i due eventi sono indipendenti, $P(B|a) = P(B)$ e si applica la **regola del prodotto**

$$P(a \cap B) = P(a)*P(B)$$

$P(a \cap B) = 1/6*1/6 = 1/36$



$\Omega = 1/6$

Evento certo

Evento elementare a

Evento elementare B

STATISTICA: la probabilità condizionata

Da
$P(a \cap B) = P(a) * P(B|a)$
si ricava:

$P(B|a) = P(a \cap B) / P(a)$

La sostituzione nucleotidica è un processo di Markov

Nel corso del tempo sono stati elaborati vari modelli di sostituzione nucleotidica, più complessi, i quali prevedono che i quattro differenti nucleotidi, che possono essere presenti ad ogni sito, siano gli stati di una catena di Markov.

Ogni nucleotide può essere sostituito da ogni altro ed anche da un nucleotide identico. Le probabilità di tali cambiamenti si chiamano **probabilità di transizione** e sono gli ingressi di una matrice quadrata **P**(t) di dimensioni 4 X 4:

$$\mathbf{P} = \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix}$$

La sostituzione nucleotidica è un processo di Markov

I **processi di Markov**, chiamati così in onore del grande matematico russo Andrey Andreyevich Markov, sono catene di eventi che si presentano uno dopo l'altro in istanti successivi di tempo.
Nel nostro caso gli eventi sono le sostituzioni nucleotidiche casuali, in seguito alle quali lo stato X del carattere che si osserva ad un determinato sito nucleotidico, ossia la base azotata, può cambiare al trascorrere del tempo.

Esso dipende, però, soltanto dall'ultimo stato in cui il processo si trovava e non da quelli precedenti: la Prob che una certa base azotata venga sostituita da ciascuna delle altre tre dipende dalla base "attuale" e non da quelle che si trovavano in quella posizione della sequenza in passato.

$$P\{X_{t+1} = j \mid X_t = i, X_{t-1} = i\text{-}1, \ldots, X_0 = 0\} = P\{X_{t+1} = j \mid X_t = i\} = P_{ij}$$

per tutti i valori di j, i, X_{t-1}, …, X_0 e t. P_{ij} è costante nel tempo e non dipende da alcuno stato precedente; in altri termini, il processo non ha memoria degli stati precedenti a quello corrente, ad eccezione dell'ultimo (*proprietà di Markov*).

La sostituzione nucleotidica è un processo di Markov

$$P\{X_{t+1} = j \mid X_t = i, X_{t-1} = i\text{-}1, \ldots, X_0 = 0\} = P\{X_{t+1} = j \mid X_t = i\} = P_{ij}$$

per tutti i valori di j, i, X_{t-1}, …, X_0 e t.

P_{ij} = **probabilità di transizione** dallo stato i allo stato j (probabilità che il processo si sposti da uno stato ad un altro in un singolo passaggio) è costante nel tempo e non dipende da alcuno stato precedente; in altri termini, il processo non ha memoria degli stati precedenti a quello corrente, ad eccezione dell'ultimo (*proprietà di Markov*).

X_{t+1} = stato del processo nell'istante $t+1$

Ogni processo di Markov ha uno *spazio campione* $\boldsymbol{\Omega}$, definito come l'insieme, finito, dei possibili stati del processo (spazio degli stati):

$$\boldsymbol{\Omega} = \{1, 2, \ldots, n\}$$

con n stati possibili.

Stati transitori e stati definitivi

Gli **stati transitori**, dopo essere stati visitati, possono essere abbandonati mentre da quelli **definitivi** la catena, una volta entrata, non può più uscire. Gli stati definitivi, per cui $P_{ii} = 1$, si presentano nella matrice di transizione $\mathbf{P}$ come altrettanti 1 (uno) collocati nella diagonale principale, mentre in tutti gli altri ingressi ci sono solo zeri.
Per gli stati transitori $P_{ii} < 1$ cosicché almeno qualche altro ingresso della i-esima riga è > 0.
Per definizione, la catena deve trovarsi in uno degli stati possibili ad ogni *step*; per qualsiasi valore di i e j, $0 \leq P_{ij} \leq 1$ e

$$\sum_{j=1}^{n} P_{ij} = 1$$

per tutti gli i, ossia la somma delle probabilità che il processo si sposti dallo stato attuale i ad uno stato j qualsiasi (che può anche coincidere con i stesso) è uguale a 1.

La sostituzione nucleotidica è un processo di Markov

```
> mdat = matrix(c(2,1,3,4,1,3,1,5,7,1,1,1,4,2,1,3), nrow = 4, ncol = 4, byrow
= TRUE)
> mdat
     [,1] [,2] [,3] [,4]
[1,]    2    1    3    4
[2,]    1    3    1    5
[3,]    7    1    1    1
[4,]    4    2    1    3
>
```

Come passare da i a j in due passaggi

Le probabilità di transizione per $t = 1$ sono date dagli ingressi della matrice $\mathbf{P}$.

$$\mathbf{P} = \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix}$$

Per $t = 2$, invece, occorre tener conto del fatto che il processo può passare dallo stato i allo stato j attraversando un qualsivoglia stato intermedio k:

$$P_{ij}(2) = \sum_{k=1}^{n} P_{ik} P_{kj}$$

per tutti gli stati i, j.
Per ogni possibile stato intermedio k si calcola la probabilità congiunta che il processo passi da i a k e, successivamente, da k a j, quindi si sommano le singole probabilità congiunte, tante quanti sono i possibili stati k, ossia n

Come passare da i a j in due passaggi

Per calcolare la probabilità che il sistema passi dallo stato iniziale i = ① allo stato finale j = ③ passando per un qualsivoglia stato intermedio k occorre applicare la regola della **moltiplicazione di due matrici**: riga X colonna

$$\mathbf{P} = \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix}$$

i \ k	1	2	3	4
①	P_{11}	P_{12}	P_{13}	P_{14}
2	P_{21}	P_{22}	P_{23}	P_{24}
3	P_{31}	P_{32}	P_{33}	P_{34}
4	P_{41}	P_{42}	P_{43}	P_{44}

k \ j	1	2	③	4
1	P_{11}	P_{12}	P_{13}	P_{14}
2	P_{21}	P_{22}	P_{23}	P_{24}
3	P_{31}	P_{32}	P_{33}	P_{34}
4	P_{41}	P_{42}	P_{43}	P_{44}

Come passare da A a C in due passaggi

	A	T	**C**	G
	[,1]	[,2]	[,3]	[,4]
A[**1**,]	2	1	3	4
T[2,]	1	3	1	5
C[3,]	7	1	1	1
G[4,]	4	2	1	3

*

	A	T	**C**	G
	[,1]	[,2]	[,3]	[,4]
A[**1**,]	2	1	3	4
T[2,]	1	3	1	5
C[3,]	7	1	1	1
G[4,]	4	2	1	3

$= Q_{AC}$

2*3+1*1+3*1+4*1 = **14**

```
> mdat%*%mdat
     [,1] [,2] [,3] [,4]
[1,]   42   16   14   28
[2,]   32   21   12   35
[3,]   26   13   24   37
[4,]   29   17   18   36
>
```

Come passare da A a C in due passaggi

```
piero@piero-XPS-9320:~$ octave
QSocketNotifier: Can only be used with threads started with QThread
GNU Octave, version 8.3.0
Copyright (C) 1993-2023 The Octave Project Developers.
This is free software; see the source code for copying conditions.
There is ABSOLUTELY NO WARRANTY; not even for MERCHANTABILITY or
FITNESS FOR A PARTICULAR PURPOSE.  For details, type 'warranty'.

Octave was configured for "x86_64-pc-linux-gnu".

Additional information about Octave is available at https://www.octave.org.

Please contribute if you find this software useful.
For more information, visit https://www.octave.org/get-involved.html

Read https://www.octave.org/bugs.html to learn how to submit bug reports.
For information about changes from previous versions, type 'news'.

octave:1>
```

Come passare da A a C in due passaggi

```
octave:1> a = [2, 1, 3, 4; 1, 3, 1, 5; 7, 1, 1, 1; 4, 2, 1, 3]
a =

   2   1   3   4
   1   3   1   5
   7   1   1   1
   4   2   1   3

octave:2> a^2
ans =

   42   16   14   28
   32   21   12   35
   26   13   24   37
   29   17   18   36

octave:3>
```

Stati transitori e stati definitivi

Gli **stati transitori**, dopo essere stati visitati, possono essere abbandonati mentre da quelli **definitivi** la catena, una volta entrata, non può più uscire. Gli stati definitivi, per cui $P_{ii} = 1$, si presentano nella matrice di transizione $\mathbf{P}$ come altrettanti 1 (uno) collocati nella diagonale principale, mentre in tutti gli altri ingressi ci sono solo zeri.
Per gli stati transitori $P_{ii} < 1$ cosicché almeno qualche altro ingresso della i-esima riga è > 0.
Per definizione, la catena deve trovarsi in uno degli stati possibili ad ogni *step*; per qualsiasi valore di i e j, $0 \leq P_{ij} \leq 1$ e

$$\sum_{j=1}^{n} P_{ij} = 1$$

per tutti gli i, ossia la somma delle probabilità che il processo si sposti dallo stato attuale i ad uno stato j qualsiasi (che può anche coincidere con i stesso) è uguale a 1.

Il magico numero <e>

$e = 1 + 1/1! + 1/2! + 1/3! + 1/4! + \ldots \cong 2.718282$

$e^z = 1 + z/1! + z^2/2! + z^3/3! + z^4/4! + \ldots \cong 2.718282$

Per $z < 1$:

$e^z \cong 1 + z$

Es.

se z = 0.1

$e^{0.1} \cong 1 + 0.1 \cong 1.1$

```
> 10*9*8*7*6*5*4*3*2*1
[1] 3628800
> factorial(10)
[1] 3628800

> Euler = function(n,z)
+ {
+ result = NULL
+ numbers = seq(1:n)
+ for (i in numbers)
+ {
+ result[i] = (z^i)/factorial(i)
+ }
+ return(1 + sum(result))
+ }
> Euler(100,1)
[1] 2.718282
> Euler(100,10)
[1] 22026.47
> exp(10)
[1] 22026.47
```

```
> exp(0.1)
[1] 1.105171
> Euler(10,0.1)
[1] 1.105171
```

Da

$e^z = 1 + z/1! + z^2/2! + z^3/3! + z^4/4! + \ldots \cong 1 + z$

per $z < 1$ (al limite per z che tende a a)

Si ricava che:

$\boldsymbol{e^A} = 1 + \mathbf{A}/1! + \mathbf{A}^2/2! + \mathbf{A}^3/3! + \mathbf{A}^4/4! + \ldots \cong (\mathbf{I} + \mathbf{A}) = \mathbf{P}$

$$\mathbf{P} = \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix}$$

$$\mathbf{I} + \mathbf{A} = \mathbf{P}$$

$$\begin{matrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{matrix} + \begin{matrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{matrix} = \begin{matrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \\ p_{41} & p_{42} & p_{43} & p_{44} \end{matrix}$$